Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 1 — Algebraic expressions
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第1章 · 代数式
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C01-R01 · P1 · Chapter 1

For $x > 0$, $A = 6x^{3/2}$ and $B = \frac{3}{2}x^{-1/2}$.

(a) Express A/B as kx^n, where k and n are integers. (2)

(b) Given also that $AB = 81$, find

(i) the value of x, (2)

(ii) the exact value of $A + B$. (2)

(Total for Question 1 is 6 marks)

2.

A rectangular card has length $(\sqrt{11} + 2)$ cm and width $(\sqrt{11} - 2)$ cm.

(a) Find the exact area of the card. (2)

(b) A strip of width 1 cm is removed along the full length of the card. Find, in exact form,

(i) the perimeter of the remaining rectangle, (2)

(ii) the fraction of the original area that has been removed. (2)

(Total for Question 2 is 6 marks)

3.

The expression $E = (x + 3)^2 - (x - 1)(x + 5)$ is defined for all real x.

(a) Simplify E. (2)

(b) Given that $E = 3(x + 2)$, find x. (2)

(Total for Question 3 is 4 marks)

4.

Let $F = \dfrac{4x^2 - 25}{2x^2 + 3x - 20}$.

(a) Factorise the numerator and denominator completely. (3)

(b) Simplify F and state all real values of x for which the original expression is undefined. (3)

(Total for Question 4 is 6 marks)

Given that $a = \sqrt{6} + \sqrt{2}$ and $b = \sqrt{6} - \sqrt{2}$,

(a) find the exact values of

(i) $a^2 + b^2$, (2)

(ii) ab. (2)

(b) Hence simplify $\frac{a}{b} + \frac{b}{a}$. (2)

(Total for Question 5 is 6 marks)

For $t > 0$, $V = \dfrac{t^{5/3} + 2t^{2/3}}{t^{1/6}}$.

(a) Express V as the sum of two powers of t. (2)

(b) Given that $t = 9$, find

(i) the exact value of V, (2)

(ii) the exact value of $V/t^{1/2}$. (2)

(Total for Question 6 is 6 marks)

A square has area $(19 - 6\sqrt{10})\ \text{cm}^2$.

(a) Expand $(\sqrt{10} - 3)^2$. (2)

(b) Hence find the exact perimeter of the square. (2)

(Total for Question 7 is 4 marks)

8.

For $x > 0$, $p = \sqrt{x}$ and $q = 1/\sqrt{x}$. It is given that $p - q = 3$.

(a) Show that $p^2 + q^2 = 11$. (3)

(b) Find the exact value of $p + q$. (2)

(Total for Question 8 is 5 marks)

9. P1-C01-R09 · P1 · Chapter 1

A student writes $\sqrt{u+v} = \sqrt{u} + \sqrt{v}$ for positive real numbers u and v.

(a) By squaring the right-hand side, show that this statement is false for every $u > 0$, $v > 0$. (2)

(b) Simplify $\sqrt{48} + \sqrt{75} - \sqrt{27}$. (2)

(Total for Question 9 is 4 marks)

$N = 8^{2/3} \times 16^{-3/4}$.

(a) Find the exact value of N, showing your working. (2)

(b) Express $\dfrac{N}{\sqrt{2}-1}$ in the form $a + b\sqrt{2}$, where a, b are rational. (2)

(Total for Question 10 is 4 marks)

For $x \neq 0$, $E = \dfrac{x^{-2} + 3x^{-1}}{x^{-1}}$.

(a) Simplify E using positive powers of x only. (2)

(b) Find the value of x for which $E = 7$. (2)

(Total for Question 11 is 4 marks)

The dimensions of a box are x cm, $(x + 2)$ cm and $(x - 2)$ cm, where $x > 2$.

(a) Express the volume as a polynomial in x. (2)

(b) Find the total surface area of the box, giving your answer as a polynomial in x. (3)

(Total for Question 12 is 5 marks)

For $x > 0$, $A = \dfrac{\sqrt{50x^3}}{\sqrt{2x}}$.

(a) Show that $A = 5x$. (2)

(b) Given that $B = \sqrt{18x^2}$, express each of the following in simplest form:

(i) B/A, (2)

(ii) $(A + B)(A - B)$. (2)

(Total for Question 13 is 6 marks)

The expression $H = \dfrac{3}{\sqrt{7} - 2} - \dfrac{2}{\sqrt{7} + 2}$.

(a) Express H in the form $a + b\sqrt{7}$, where a, b are rational. (3)

(b) Find the exact value of $3H - \sqrt{7}$. (2)

(Total for Question 14 is 5 marks)

For $y > 0$, $K = \dfrac{(27y^6)^{2/3}}{3y^2}$.

(a) Simplify K. (2)

(b) Given that $K = 12$, find y. (2)

(Total for Question 15 is 4 marks)

A right-angled triangle has perpendicular sides $\sqrt{17} + 1$ cm and $\sqrt{17} - 1$ cm.

(a) Find the exact area of the triangle. (2)

(b) For the same triangle, find, in exact form,

(i) the length of the hypotenuse, (2)

(ii) the perimeter. (2)

(Total for Question 16 is 6 marks)

$f(x) = (2x - 3)(x + 4) - (x - 2)^2$.

(a) Express $f(x)$ in the form $ax^2 + bx + c$. (3)

(b) Find $f(-2)$ and $f(2)$, and hence find $f(2) - f(-2)$. (3)

(Total for Question 17 is 6 marks)

The positive numbers a and b satisfy $a^{1/2} = b^{1/3}$.

(a) Express a in terms of b. (2)

(b) Given also that $b = 64$, find

(i) the exact value of a, (2)

(ii) the exact value of $a^{-3/4}b^{1/2}$. (2)

(Total for Question 18 is 6 marks)

Let $R = \dfrac{x^2 - 6x + 9}{x^2 - 9}$.

(a) Simplify R and state the restrictions on x. (3)

(b) Find the exact value of R when $x = \sqrt{12}$. Give your answer as $a + b\sqrt{3}$. (3)

(Total for Question 19 is 6 marks)

A positive quantity y is related to $x > 0$ by $y = \dfrac{12x^{1/3}}{(3x)^{-2/3}}$.

(a) Express y in the form kx, giving the exact value of k. (3)

(b) Find the exact value of y when $x = 3$. (2)

(Total for Question 20 is 5 marks)

$P = (x - 1)^3 - (x + 1)(x^2 - x + 1)$.

(a) Expand and simplify P. (3)

(b) Find the value of P when $x = -\frac{1}{3}$. (2)

(Total for Question 21 is 5 marks)

$a = 2 + \sqrt{5}$ and $b = 2 - \sqrt{5}$.

(a) Find the exact values of

(i) $a + b$ and ab, (3)

(ii) $a^3 + b^3$, using $a^3 + b^3 = (a + b)^3 - 3ab(a + b)$. (2)

(b) Find $1/a + 1/b$. (2)

(Total for Question 22 is 7 marks)

The algebraic expression $T = 6u^2v - 9uv^2 + 12uv$ is defined for real u, v.

(a) Factorise T completely. (2)

(b) Given that $u = \sqrt{3}$ and $v = 2\sqrt{3}$, find the exact value of T. (3)

(Total for Question 23 is 5 marks)

For $x > 0$, $F = (x^{1/4} + x^{-1/4})^2$.

(a) Expand F as the sum of three terms. (2)

(b) Given that $x = 16$, find

(i) the exact value of F, (2)

(ii) the exact value of $(x^{1/4} - x^{-1/4})^2$. (2)

(Total for Question 24 is 6 marks)

The length L cm of a metal rod is $\dfrac{18}{3+\sqrt{3}}$.

(a) Write L in the form $a + b\sqrt{3}$, where a, b are integers. (2)

(b) The rod is cut into three equal pieces. Find the square of the length of one piece in the form $c + d\sqrt{3}$. (3)

(Total for Question 25 is 5 marks)

$E = (2x + 1)^2 - (x - 2)^2$.

(a) Factorise E completely without first expanding both squares. (2)

(b) Find the coefficient of x in the expansion of E. (2)

(Total for Question 26 is 4 marks)

For $a > 0$ and $b > 0$, $S = \sqrt{a^3 b}\,\sqrt{a b^3}$.

(a) Simplify S. (2)

(b) Given that $ab = 5$, find the exact value of $\dfrac{S + ab}{\sqrt{ab}}$, with a rational denominator. (3)

(Total for Question 27 is 5 marks)

The quantities $U = \dfrac{2x + 6}{x^2 + 5x + 6}$ and $V = \dfrac{1}{x + 2}$ are defined for $x \neq -3, -2$.

(a) Simplify U. (2)

(b) Express $U - V$ as a single fraction and find its value when $x = \sqrt{2}$. Give the value in the form $a + b\sqrt{2}$. (3)

(Total for Question 28 is 5 marks)

29.

A rectangular card has side lengths $(\sqrt{75} + \sqrt{12})$ cm and $(\sqrt{75} - \sqrt{12})$ cm. A square is cut from one corner. The square has the same perimeter as a rectangle of sides $\sqrt{3}$ cm and $2\sqrt{3}$ cm.

(a) Show that the area of the card before cutting is an integer. (2)

(b) For the card remaining after the cut, find

(i) its exact area. (2)

(ii) its exact perimeter, giving your answer in simplest surd form. (2)

(Total for Question 29 is 6 marks)

For $x > 0$, $F(x) = \dfrac{x^{3/2} - 9x^{1/2}}{x^{1/2}}$.

(a) Simplify $F(x)$ and state why $x = 0$ remains excluded. (2)

(b) A positive number a satisfies $F(a^2) = F(a) + 12$.

(i) Find a exactly. (2)

(ii) Hence find $F(1/a)$ exactly. (1)

(Total for Question 30 is 5 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C01-R01 · 6 marks

(a) · 2 marks

$4x^{2}$.

系数相除得4；相除时指数相减，$3/2-(-1/2)=2$。

M1	正确处理指数相减
A1	系数与指数均正确

(b) Given also that $AB=81$, find

(i) · 2 marks

$x=9$.

$AB=9x$，故$9x=81$。

M1	化积为9x
A1	得到x=9

(ii) · 2 marks

$\frac{325}{2}$.

$A=162$，$B=1/2$，相加。

M1	将x代回两个原式
A1	得到精确总和

Question 2

P1-C01-R02 · 6 marks

(a) · 2 marks

$7\ \text{cm}^2$.

用平方差：$(\sqrt{11})^2 - 2^2 = 7$。

M1	使用共轭乘积
A1	面积和单位

(b) A strip of width 1 cm is removed along the full length of the card. Find, in exact form,

(i) · 2 marks

$(4\sqrt{11} - 2)$ cm.

新宽为$\sqrt{11} - 3 > 0$；两边相加再乘2。

M1	正确写出新宽
A1	正确周长

(ii) · 2 marks

$\dfrac{\sqrt{11} + 2}{7}$.

去除面积等于原长乘1，再除原面积7。

M1	求去除面积
A1	正确比例

Question 3

P1-C01-R03 · 4 marks

(a) · 2 marks

$2x + 14$.

分别展开为$x^2 + 6x + 9$与$x^2 + 4x - 5$，相减。

M1	正确展开两项
A1	合并同类项

(b) · 2 marks

$x = 8$.

代入(a)得$2x + 14 = 3x + 6$。

M1	使用化简结果建立一次方程
A1	求得8

Question 4

P1-C01-R04 · 6 marks

(a) · 3 marks

$4x^2-25=(2x-5)(2x+5)$; $2x^2+3x-20=(2x-5)(x+4)$.

分子为平方差；分母分解时一次项来自$8x-5x$。

M1	分解分子
M1	分解分母
A1	两式均完全正确

(b) · 3 marks

$F=\dfrac{2x+5}{x+4}$, with $x\neq \frac{5}{2},-4$.

约去共同因子；原分母的两个零点都必须保留为限制，不能因约分而丢失。

M1	正确约分
A1	最简式
B1	保留两个原始限制

Question 5

P1-C01-R05 · 6 marks

(a) find the exact values of

(i) · 2 marks

16.

两个平方分别为$8+4\sqrt{3}$和$8-4\sqrt{3}$；相加根式项抵消。

M1	正确展开至少一个平方
A1	总和16

(ii) · 2 marks

4.

使用平方差$6-2$。

M1	平方差
A1	结果4

(b) · 2 marks

4.

通分为$(a^2+b^2)/(ab)=16/4$。

M1	正确通分并引用(a)
A1	结果4

Question 6

P1-C01-R06 · 6 marks

(a) · 2 marks

$t^{3/2} + 2t^{1/2}$.

两项分别减去指数$1/6$。

M1	对两项都减指数
A1	两个幂正确

(b) Given that $t = 9$, find

(i) · 2 marks

33.

$9^{1/2} = 3,\ 9^{3/2} = 27$。

M1	正确计算分数指数
A1	结果33

(ii) · 2 marks

11.

由(a)可得$V/t^{1/2} = t + 2,$ 代入9。

M1	化商为t+2
A1	结果11

Question 7

P1-C01-R07 · 4 marks

(a) · 2 marks

$19-6\sqrt{10}$.

平方含$10+9$及交叉项$-6\sqrt{10}$。

M1	写出交叉项
A1	正确展开

(b) · 2 marks

$4(\sqrt{10}-3)$ cm.

$\sqrt{10}>3$，故正边长为$\sqrt{10}-3$；周长为边长四倍。

M1	选取正边长
A1	正确周长及单位

Question 8

P1-C01-R08 · 5 marks

(a) · 3 marks

$p^2+q^2=11.$

$pq=1$，故$(p-q)^2=p^2+q^2-2=9$。

M1	识别pq=1
M1	展开差的平方
A1	推得11

(b) · 2 marks

$\sqrt{13}.$

$(p+q)^2=11+2=13$；p、q均为正，所以取正平方根。

M1	求和的平方
A1	选正根

Question 9

P1-C01-R09 · 4 marks

(a) · 2 marks

$(\sqrt{u}+\sqrt{v})^2 = u+v+2\sqrt{uv} > u+v.$

交叉项严格为正，两个非负数的平方不同，因此原数不同。

M1	正确展开平方
A1	用正交叉项得到严格结论

(b) · 2 marks

$6\sqrt{3}.$

各项提出完全平方因子，系数为4、5、3。

M1	将三项化为同类根式
A1	合并正确

Question 10

P1-C01-R10 · 4 marks

(a) · 2 marks

$\frac{1}{2}$.

$8^{2/3}=4$；$16^{-3/4}=1/8$。

M1	正确处理两个分数指数
A1	结果1/2

(b) · 2 marks

$\frac{1}{2}+\frac{1}{2}\sqrt{2}$.

代入N后，分子分母同乘$\sqrt{2}+1$；分母的共轭乘积为1。

M1	乘共轭式
A1	得到所要求形式

Question 11

P1-C01-R11 · 4 marks

(a) · 2 marks

$E = \dfrac{1}{x} + 3$.

逐项除以x^{-1}，指数相减。

M1	分别处理两项
A1	正确正指数形式

(b) · 2 marks

$x = \frac{1}{4}$.

$1/x = 4$，两边取倒数，所得值满足$x \neq 0$。

M1	使用化简式
A1	求得$1/4$

Question 12

P1-C01-R12 · 5 marks

(a) · 2 marks

$V = x^3 - 4x \text{ cm}^3$.

先将$(x+2)(x-2)$写成x^2-4再乘x。

M1	利用平方差
A1	正确多项式及单位

(b) · 3 marks

$(6x^2 - 8) \text{ cm}^2$.

三对面的面积和为$2[x(x+2)+x(x-2)+(x+2)(x-2)]$，展开并合并。

M1	正确计入三对面
M1	展开三个乘积
A1	多项式及单位

Question 13

P1-C01-R13 · 6 marks

(a) · 2 marks

$A = \sqrt{25x^2} = 5x.$

根号内的商为$25x^2$；由x正可取5x。

M1	合并商并化简根号内
A1	正确使用x为正

(b) Given that $B = \sqrt{18x^2}$, express each of the following in simplest form:

(i) · 2 marks

$\dfrac{3\sqrt{2}}{5}.$

$B = 3x\sqrt{2}$，与5x相除。

M1	化简B
A1	约去x

(ii) · 2 marks

$7x^2.$

用$A^2 - B^2 = 25x^2 - 18x^2$。

M1	使用平方差
A1	结果7x平方

Question 14

P1-C01-R14 · 5 marks

(a) · 3 marks

$\frac{10}{3}+\frac{1}{3}\sqrt{7}$.

两个分母的共轭乘积均为3；通分后分子$3(\sqrt{7}+2)-2(\sqrt{7}-2)=\sqrt{7}+10$。

M1	两个分式有理化
M1	正确相减分子
A1	所要求形式

(b) · 2 marks

10.

代入(a)，根式部分相消。

M1	正确代入化简式
A1	结果10

Question 15

P1-C01-R15 · 4 marks

(a) · 2 marks

$3y^2$.

分子为$9y^4$，除以$3y^2$。

M1	正确计算分子
A1	最简式

(b) · 2 marks

$y = 2$.

$y^2 = 4$，利用y正只保留2。

M1	解平方关系
A1	使用正值限制

Question 16

P1-C01-R16 · 6 marks

(a) · 2 marks

$8\ \text{cm}^2$.

面积为两直角边乘积的一半，即$(17-1)/2$。

M1	正确面积式
A1	结果8及单位

(b) For the same triangle, find, in exact form,

(i) · 2 marks

$6\ \text{cm}$.

两直角边平方和为$2(17+1)=36$。

M1	使用勾股并消去交叉项
A1	正根6

(ii) · 2 marks

$(2\sqrt{17}+6)\ \text{cm}$.

直角边之和为$2\sqrt{17}$，再加斜边6。

M1	正确合并边长
A1	周长表达式

Question 17

P1-C01-R17 · 6 marks

(a) · 3 marks

$f(x)=x^2+9x-16.$

首项乘积为$2x^2+5x-12$；减去x^2-4x+4。

M1	展开乘积和平方
M1	正确处理减号
A1	三个系数

(b) · 3 marks

$f(-2)=-30$, $f(2)=6$; difference 36.

代入得到$4-18-16$与$4+18-16$；按所给顺序相减。

M1	两次正确代入
A1	两函数值
A1	差为36

Question 18

P1-C01-R18 · 6 marks

(a) · 2 marks

$a = b^{2/3}$.

两边平方。

M1	使用指数幂运算
A1	正确指数

(b) Given also that $b = 64$, find

(i) · 2 marks

16.

64的立方根为4，再平方。

M1	正确开立方
A1	得到16

(ii) · 2 marks

1.

$16^{-3/4} = 1/8$，$64^{1/2} = 8$。

M1	正确负分数指数
A1	结果1

Question 19

P1-C01-R19 · 6 marks

(a) · 3 marks

$R = \dfrac{x-3}{x+3};\ x \neq \pm 3.$

分子$(x-3)^2$，分母$(x-3)(x+3)$；约分后保留原限制。

M1	分解并约分
A1	最简式
B1	两个限制

(b) · 3 marks

$7-4\sqrt{3}.$

$x=2\sqrt{3}$。将$(2\sqrt{3}-3)/(2\sqrt{3}+3)$有理化，得$(21-12\sqrt{3})/3$。

M1	正确代入并乘共轭
M1	展开平方及分母
A1	所要求形式

Question 20

P1-C01-R20 · 5 marks

(a) · 3 marks

$y = 12\,3^{2/3}x$.

除以负指数幂等于乘以正指数幂；x的指数相加为1。

M1	将除法改为乘法
M1	合并x的指数
A1	精确系数

(b) · 2 marks

$36\sqrt[3]{9}$.

代入得到$36\,3^{2/3}$，即所给根式。

M1	正确代入
A1	精确值

Question 21

P1-C01-R21 · 5 marks

(a) · 3 marks

$-3x^2+3x-2.$

第二个乘积为x^3+1，首项立方为x^3-3x^2+3x-1。

M1	展开立方
M1	展开第二乘积
A1	合并正确

(b) · 2 marks

$-\frac{10}{3}.$

代入得$-1/3-1-2$。

M1	正确处理负数平方
A1	精确结果

Question 22

P1-C01-R22 · 7 marks

(a) Find the exact values of

(i) · 3 marks

$a+b=4,\ ab=-1$.

和中根式相消；积用平方差$4-5$。

B1	和为4
M1	计算共轭乘积
A1	积为负1

(ii) · 2 marks

76.

使用题目给出的恒等式，得$64-3(-1)(4)=76$。

M1	正确代入三个已知量
A1	结果76

(b) · 2 marks

-4.

通分为$(a+b)/(ab)=4/(-1)$。

M1	通分
A1	结果负4

Question 23

P1-C01-R23 · 5 marks

(a) · 2 marks

$3uv(2u-3v+4)$.

三项最大公因式为3uv；依次除去得到括号中的三项。

M1	提取公因式3uv
A1	括号正确

(b) · 3 marks

$72-72\sqrt{3}$.

$3uv=18$，括号为$4-4\sqrt{3}$，相乘。

M1	代入化简后的两因子
M1	正确处理根式乘积
A1	精确结果

Question 24

P1-C01-R24 · 6 marks

(a) · 2 marks

$x^{1/2}+2+x^{-1/2}$.

交叉项为$2x^{0}=2$；两个平方的指数分别为$1/2$与$-1/2$。

M1	正确交叉项
A1	两平方项

(b) Given that $x=16$, find

(i) · 2 marks

$\frac{25}{4}$.

代入得到$4+2+1/4$。

M1	正确计算正负半次幂
A1	精确值

(ii) · 2 marks

$\frac{9}{4}$.

只改变交叉项的符号：$4-2+1/4=9/4$。

M1	使用差的平方
A1	结果$9/4$

Question 25

P1-C01-R25 · 5 marks

(a) · 2 marks

$L = 9 - 3\sqrt{3}$.

乘共轭后分母为6，分子为$18(3-\sqrt{3})$。

M1	使用共轭有理化
A1	正确整数系数

(b) · 3 marks

$(12 - 6\sqrt{3})\ \text{cm}^2$.

一段长$3-\sqrt{3}$；平方为$9+3-6\sqrt{3}$。

M1	先将全长除以3
M1	正确展开平方
A1	精确式及单位

Question 26

P1-C01-R26 · 4 marks

(a) · 2 marks

$E = (x+3)(3x-1)$.

平方差得$[(2x+1)-(x-2)][(2x+1)+(x-2)] = (x+3)(3x-1)$。

M1	使用平方差
A1	正确两因子

(b) · 2 marks

8.

直接展开得$3x^2+8x-3$，一次项系数8。

M1	正确展开或合并交叉项
A1	系数8

Question 27

P1-C01-R27 · 5 marks

(a) · 2 marks

a^2b^2.

合并根式得$\sqrt{a^4b^4}$，取非负平方根。

M1	合并根式
A1	正确平方根

(b) · 3 marks

$6\sqrt{5}$.

先得S=25，式子为$30/\sqrt{5}$；有理化得$6\sqrt{5}$。

M1	用ab求S
M1	分母有理化
A1	最简根式

Question 28

P1-C01-R28 · 5 marks

(a) · 2 marks

$U = \dfrac{2}{x+2}$.

分子$2(x+3)$，分母$(x+2)(x+3)$，约去共同因子。

M1	正确因式分解
A1	最简式

(b) · 3 marks

$U - V = \dfrac{1}{x+2}$; value $1 - \frac{1}{2}\sqrt{2}$.

先同分母相减；$1/(2+\sqrt{2}) = (2-\sqrt{2})/2$。

M1	正确单分式
M1	有理化
A1	指定形式

Question 29

P1-C01-R29 · 6 marks

(a) · 2 marks

$63\ \text{cm}^2$.

利用平方差：$(\sqrt{75}+\sqrt{12})(\sqrt{75}-\sqrt{12})=75-12=63$。

M1	平方差
A1	面积63

(b) For the card remaining after the cut, find

(i) · 2 marks

$225/4\ \text{cm}^2$.

小矩形周长为$6\sqrt{3}$，故正方形边长$3\sqrt{3}/2$，剪去面积$27/4$；剩余$63-27/4=225/4$。

M1	由周长求正方形边长
A1	剩余面积

(ii) · 2 marks

$20\sqrt{3}\ \text{cm}$.

从角上剪去正方形时，失去的两段外边各以等长切边补回，周长不变；$2(2\sqrt{75})=20\sqrt{3}$。

B1	说明周长不变
A1	最简根式

Question 30

P1-C01-R30 · 5 marks

(a) · 2 marks

$F(x) = x - 9,\ x > 0.$

分子为$\sqrt{x}(x-9)$，约去$\sqrt{x}$；原式在$x=0$分母为零，约分不能扩大定义域。

M1	提取并约去公因式
B1	原式定义域

(b) A positive number a satisfies $F(a^2) = F(a) + 12$.

(i) · 2 marks

$a = 4.$

$a^2 - 9 = a - 9 + 12$，即$(a-4)(a+3)=0$；由$a>0$取4。

M1	建立二次方程
A1	合法根

(ii) · 1 marks

$-35/4.$

代入已化简的式子：$F(1/4) = 1/4 - 9 = -35/4$。

B1	准确值

Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 2 — Quadratics
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第2章 · 二次函数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1.

P1-C02-R01 · P1 · Chapter 2

The curve C has equation $y = 3x^2 - 12x + 5$.

(a) Express y in the form $a(x - b)^2 + c$. (2)

(b) Hence find

(i) the coordinates of the minimum point of C, (1)

(ii) the exact values of x for which $y = 5$. (2)

(Total for Question 1 is 5 marks)

2. P1-C02-R02 · P1 · Chapter 2

The quadratic equation $x^2 - 2kx + k + 6 = 0$ has equal real roots.

(a) Find the possible values of k. (3)

(b) For each possible value of k, find

(i) the repeated root, (2)

(ii) the corresponding factorised quadratic expression. (2)

(Total for Question 2 is 7 marks)

3.

A rectangle has perimeter 28 cm. Its shorter side has length x cm, where $0 < x \leq 7$.

(a) Show that its area is $49 - (x - 7)^2$ cm^2. (3)

(b) The area is 40 cm^2. Find the dimensions of the rectangle. (3)

(Total for Question 3 is 6 marks)

$f(x) = 2x^2 + 8x + 11$.

(a) By completing the square, show that $f(x) > 0$ for every real x. (2)

(b) Find the exact minimum value of $\frac{1}{2}f(x) - 4$ and the value of x at which it occurs. (3)

(Total for Question 4 is 5 marks)

5.

The graph of $y = x^2 + px + q$ has minimum point $(3, -5)$.

(a) Find p and q. (3)

(b) Find the exact coordinates of the points where the graph meets the x-axis. (2)

(Total for Question 5 is 5 marks)

6. P1-C02-R06 · P1 · Chapter 2

The equation $(m-1)x^2 + 6x + 4 = 0$ has two distinct real roots, and $m \neq 1$.

(a) Show that $m < \frac{13}{4}$. (2)

(b) Find the value of x that solves the equation when $m = 1$, and explain why this case was excluded from part (a). (3)

(Total for Question 6 is 5 marks)

7.

A quadratic expression is $f(x) = x^2 - (k+2)x + 2k$, where k is real.

(a) Factorise $f(x)$ in terms of k. (2)

(b) Given that the graph of $y = f(x)$ has its minimum point on the x-axis, find

(i) the value of k, (2)

(ii) the coordinates of the minimum point. (1)

(Total for Question 7 is 5 marks)

8.

The equation $4x^2 - 4x - 7 = 0$ has roots $\alpha < \beta$.

(a) Find α and β in exact form. (3)

(b) Find the exact distance between the points $(\alpha, 0)$ and $(\beta, 0)$. (2)

(Total for Question 8 is 5 marks)

9.

A rectangular display has area $45\ \text{cm}^2$. Its length is 4 cm greater than its width.

(a) Form a quadratic equation for the width w cm. (2)

(b) Find the dimensions and the perimeter of the display. (3)

(Total for Question 9 is 5 marks)

$y = -2x^2 + 12x - 11$.

(a) Express y in the form $A - B(x - C)^2$. (2)

(b) Hence find

(i) the maximum value of y, (1)

(ii) the exact values of x for which $y = -1$. (2)

(Total for Question 10 is 5 marks)

11. P1-C02-R11 · P1 · Chapter 2

The line $y = c$ meets the curve $y = x^2 - 8x + 13$ at two distinct points.

(a) Find the range of possible values of c. (2)

(b) The two intersections have x-coordinates differing by 6. Find c. (3)

(Total for Question 11 is 5 marks)

$f(x) = x^2 + 6x + k$. The minimum value of $f(x)$ is twice $f(0)$.

(a) Find k. (3)

(b) For this value of k, solve $f(x) = 0$, giving exact answers. (2)

(Total for Question 12 is 5 marks)

13.

A ball is modelled by $h = 1 + 8t - 2t^2$, where h is its height in metres and $t \geq 0$ is time in seconds.

(a) Find the maximum height and the time at which it is reached. (3)

(b) Find the time when the ball first reaches the ground. Give your answer to 3 significant figures. (3)

(Total for Question 13 is 6 marks)

14.

The quadratic expression $x^2 + 2(k-1)x + k^2 + 3$ is defined for real x, k.

(a) Show that its discriminant is $-8(k+1)$. (3)

(b) Find all values of k for which the expression is strictly positive for every real x. (2)

(Total for Question 14 is 5 marks)

The equation $x^2 - 10x + 18 = 0$ has roots p and q, with $p < q$.

(a) Find p and q in exact form. (2)

(b) A new quadratic equation has roots $p + 2$ and $q + 2$. Find this equation in the form $x^2 + ax + b = 0$. (3)

(Total for Question 15 is 5 marks)

A rectangle is inscribed under the line $y = 12 - 2x$ in the first quadrant, with vertices $(0, 0)$, $(x, 0)$, (x, y) and $(0, y)$.

(a) Express its area A in terms of x and state the permitted interval for x. (3)

(b) Without differentiation, find the maximum possible area and the dimensions of the corresponding rectangle. (3)

(Total for Question 16 is 6 marks)

The curve $y = ax^2 + bx + 2$ passes through $(1, 1)$ and $(3, 11)$.

(a) Find a and b. (3)

(b) Find the exact minimum value of y. (2)

(Total for Question 17 is 5 marks)

18.

The equation $kx^2 + 4x + k = 0$ has no real roots. It is given that $k > 0$.

(a) Find the possible values of k. (3)

(b) For $k = 4$, write $kx^2 + 4x + k$ as a completed square and state its minimum value. (3)

(Total for Question 18 is 6 marks)

19.

Given that $x^2 + bx + 12 = (x - r)^2 + s$ for all real x, where $r = \frac{5}{2}$,

(a) find b and s. (3)

(b) Explain why the equation $x^2 + bx + 12 = 0$ has no real solutions. (2)

(Total for Question 19 is 5 marks)

A quadratic equation is $x^2 + px + q = 0$. One root is 2 and the other root is 3 greater.

(a) Find p and q. (3)

(b) For these values of p, q, find the value of c such that $x^2 + px + q = c$ has equal real roots. (2)

(Total for Question 20 is 5 marks)

21. P1-C02-R21 · P1 · Chapter 2

The graph of $y = x^2 - 4x + d$ meets the positive y-axis and has two distinct positive x-intercepts.

(a) Find the range of possible values of d. (3)

(b) Given that the smaller x-intercept is 1, find the other intercept and d. (3)

(Total for Question 21 is 6 marks)

A quadratic function has minimum value -8 at $x = -1$ and passes through $(1, 4)$.

(a) Find its equation in the form $y = ax^2 + bx + c$. (3)

(b) Find the exact values of x for which $y = 0$. (3)

(Total for Question 22 is 6 marks)

23. P1-C02-R23 · P1 · Chapter 2

For $x \geq 0$, $C = 2x^2 - 20x + 61$ is a model for the daily cost, in pounds, of producing a batch of size x hundred items.

(a) Find the minimum value of C and the batch size at which it occurs. (3)

(b) Find the two batch sizes for which $C = 29$. (3)

(Total for Question 23 is 6 marks)

The expression $f(x) = x^2 + 2ax + 9$ has minimum value 5.

(a) Find the possible values of a. (3)

(b) For each possible value of a, find

(i) the coordinates of the minimum point, (1)

(ii) the number of real solutions of $f(x) = 4$. Give a reason. (2)

(Total for Question 24 is 6 marks)

$x^2 - 6x + 7 = 2(x - 1)$.

(a) Solve the equation, giving exact answers. (3)

(b) Find the distance between the midpoint of the two roots on the number line and the origin. (2)

(Total for Question 25 is 5 marks)

26.

The quadratic equation $x^2 + (k - 4)x + 4 = 0$ has equal real roots.

(a) Find the two possible values of k. (3)

(b) For which of these values is the repeated root negative? State the repeated root. (2)

(Total for Question 26 is 5 marks)

A garden has width x m and length $(x + 6)$ m. A path of width 1 m runs inside all four edges. It is given that $x > 2$.

(a) Show that the remaining planted area is $(x^2 + 2x - 8)\ \text{m}^2$. (3)

(b) The planted area is $40\ \text{m}^2$. Find the exact outer dimensions of the garden. (3)

(Total for Question 27 is 6 marks)

The equation $3x^2 + 6x + c = 0$ has roots $-1 - r$ and $-1 + r$, where $r > 0$.

(a) Express c in terms of r. (2)

(b) Given that $c = -24$, find

(i) the value of r, (2)

(ii) the two roots of the equation. (1)

(Total for Question 28 is 5 marks)

The quadratic function f is defined by $f(x) = x^2 - 6x + c$, where c is a constant. The minimum value of f is -4.

(a) Find c and the coordinates of the minimum point. (2)

(b) The line $y = k$ meets $y = f(x)$ at points A and B. Their x-coordinates are positive and the x-coordinate of B is twice that of A.

(i) Find k and the coordinates of A, B. (2)

(ii) Find the area of triangle AMB, where M is the minimum point. (2)

(Total for Question 29 is 6 marks)

Consider the equation $(m+1)x^2 - 4x + 1 = 0$, where m is real.

(a) For $m \neq -1$, find the range of m for which the equation has two distinct real roots. (2)

(b) For one value of m, the equation has a repeated root.

(i) Find this value and the repeated root. (2)

(ii) For $m = -1$, solve the original equation and explain why this is not a repeated-root case. (2)

(Total for Question 30 is 6 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C02-R01 · 5 marks

(a) · 2 marks

$y = 3(x-2)^2 - 7$.

提出3后配方$x^2 - 4x = (x-2)^2 - 4$。

M1	正确配方
A1	三个常数

(b) Hence find

(i) · 1 marks

$(2, -7)$.

平方项最小为0，在x=2时取得。

B1	正确最小点

(ii) · 2 marks

$x = 0, 4$.

$3(x-2)^2 = 12$，取两个平方根得x−2=±2。

M1	建立平方关系
A1	两根

Question 2

P1-C02-R02 · 7 marks

(a) · 3 marks

$k = -2, 3$.

判别式为$4k^2 - 4(k+6) = 4(k-3)(k+2)$；令其为$0$。

M1	判别式等于零
M1	正确解参数方程
A1	两个k值

(b) For each possible value of k, find

(i) · 2 marks

$x = -2$ when $k = -2$; $x = 3$ when $k = 3$.

重根为$-b/(2a) = k$。

M1	求重根
A1	与两个k正确配对

(ii) · 2 marks

$(x+2)^2$ and $(x-3)^2$, respectively.

分别把重根写成对应平方因式；展开可复核常数项。

M1	根据重根写平方
A1	两式

Question 3

P1-C02-R03 · 6 marks

(a) · 3 marks

$A = x(14 - x) = 49 - (x - 7)^2$.

另一边为14−x；展开再配方。

M1	正确另一边
M1	面积关系
A1	配方结果

(b) · 3 marks

4 cm by 10 cm.

$(x - 7)^2 = 9$给x=4或10；短边限制只允许4。

M1	解平方关系
A1	按范围取4
A1	两边及单位

Question 4

P1-C02-R04 · 5 marks

(a) · 2 marks

$f(x) = 2(x+2)^2 + 3 \geq 3 > 0.$

平方非负，且常数下界为严格正数3。

M1	正确配方
A1	利用下界证明严格正

(b) · 3 marks

Minimum $-\frac{5}{2}$, at $x = -2$.

乘以正数1/2不改变最小值位置；最小值$3/2 - 4$。

M1	使用f的最小值
A1	新最小值
B1	取得位置

Question 5

P1-C02-R05 · 5 marks

(a) · 3 marks

$p = -6,\ q = 4.$

写成$y = (x-3)^2 - 5 = x^2 - 6x + 4$，比较系数。

M1	使用顶点式
A1	p
A1	q

(b) · 2 marks

$(3 - \sqrt{5}, 0)$ and $(3 + \sqrt{5}, 0).$

令y=0，得$(x-3)^2 = 5$；两根都应写成坐标。

M1	解配方等于零
A1	两个坐标

Question 6

P1-C02-R06 · 5 marks

(a) · 2 marks

$36-16(m-1)>0 \Rightarrow m<13/4$.

二次方程两不同实根要求判别式严格大于0。

M1	严格判别式条件
A1	参数上界

(b) · 3 marks

$x=-\frac{2}{3}$; the equation is linear, not quadratic.

m=1时二次项消失，只剩6x+4=0，所以不适用二次方程两根的讨论。

M1	解退化后方程
A1	正确根
B1	解释二次项消失

Question 7

P1-C02-R07 · 5 marks

(a) · 2 marks

$f(x) = (x-2)(x-k)$.

展开右式可得一次项$-(k+2)x$和常数$2k$。

M1	识别两个因式
A1	正确完整分解

(b) Given that the graph of $y = f(x)$ has its minimum point on the x-axis, find

(i) · 2 marks

$k = 2$.

最小点在x轴意味着两根相等，$2=k$。

M1	联系最小点与重根
A1	k为2

(ii) · 1 marks

$(2, 0)$.

此时$f(x) = (x-2)^2$。

B1	最小点坐标

Question 8

P1-C02-R08 · 5 marks

(a) · 3 marks

$\alpha = \frac{1}{2} - \sqrt{2}, \beta = \frac{1}{2} + \sqrt{2}.$

用二次公式，判别式16+112=128，$\sqrt{128} = 8\sqrt{2}$。

M1	使用二次公式
M1	化简根式
A1	排序正确的两根

(b) · 2 marks

$2\sqrt{2}.$

两点同在x轴，距离为β−α。

M1	使用有序根之差
A1	正确精确距离

Question 9

P1-C02-R09 · 5 marks

(a) · 2 marks

$w^2 + 4w - 45 = 0$.

长度w+4，面积关系$w(w+4)=45$。

M1	建立面积式
A1	标准二次方程

(b) · 3 marks

5 cm by 9 cm; perimeter 28 cm.

方程分解为(w+9)(w−5)=0；长度必须正，取w=5。

M1	正确解两代数根
A1	排除负宽并求长
A1	周长及单位

Question 10

P1-C02-R10 · 5 marks

(a) · 2 marks

$y = 7 - 2(x-3)^2$.

提出−2并配方。

M1	正确处理负系数配方
A1	结果

(b) Hence find

(i) · 1 marks

7.

被减去的平方项非负，在x=3为零。

B1	最大值7

(ii) · 2 marks

$x = 1, 5$.

$7 - 2(x-3)^2 = -1$，故平方为4。

M1	正确平方关系
A1	两解

Question 11

P1-C02-R11 · 5 marks

(a) · 2 marks

$c > -3$.

曲线为$(x-4)^2-3$，水平线高于最小值才有两个交点。

M1	求最小值或判别式
A1	严格范围

(b) · 3 marks

$c = 6$.

两横坐标为$4 \pm \sqrt{c+3}$，差为$2\sqrt{c+3} = 6$。

M1	建立根间距关系
M1	解平方关系
A1	c为6

Question 12

P1-C02-R12 · 5 marks

(a) · 3 marks

$k = -9$.

配方得最小值k−9；f(0)=k，因此k−9=2k。

M1	正确最小值
M1	建立条件方程
A1	k

(b) · 2 marks

$x = -3 \pm 3\sqrt{2}$.

此时$f = (x+3)^2 - 18$，解平方方程。

M1	正确代入k并配方
A1	两个根

Question 13

P1-C02-R13 · 6 marks

(a) · 3 marks

9 m at $t = 2$ s.

$h = 9 - 2(t-2)^2$，最大时平方项为零。

M1	配方
A1	最大高度
B1	对应时间

(b) · 3 marks

$t = 2 + \frac{3}{2}\sqrt{2} \approx 4.12$ s.

令h=0得$2t^2 - 8t - 1 = 0$，两根$2 \pm 3\sqrt{2}/2$；只保留非负时间。

M1	设高度为0并解二次方程
A1	选物理根
A1	按要求近似及单位

Question 14

P1-C02-R14 · 5 marks

(a) · 3 marks

$\Delta = 4(k-1)^2 - 4(k^2+3) = -8k - 8.$

展开平方后k^2项抵消。

M1	正确判别式
M1	展开化简
A1	目标式

(b) · 2 marks

$k > -1.$

二次项系数为正，恒严格正要求判别式负；$k=-1$时有零点不满足严格正。

M1	使用开口向上与判别式负
A1	严格范围

Question 15

P1-C02-R15 · 5 marks

(a) · 2 marks

$p = 5 - \sqrt{7}$, $q = 5 + \sqrt{7}$.

配方为$(x-5)^2=7$。

M1	配方
A1	有序两根

(b) · 3 marks

$x^2 - 14x + 42 = 0$.

新根为$7 \pm \sqrt{7}$，故$(x-7)^2 - 7 = 0$，展开。

M1	写出新根
M1	建立配方方程
A1	正确标准式

Question 16

P1-C02-R16 · 6 marks

(a) · 3 marks

$A = 12x - 2x^2$, $0 < x < 6$.

面积xy；两边长必须严格正，要求x>0与12−2x>0。

M1	正确面积式
A1	面积多项式
B1	开区间

(b) · 3 marks

18 square units; width 3, height 6.

$A = 18 - 2(x-3)^2$，顶点在x=3且位于允许范围。

M1	面积配方
A1	最大面积
A1	两个尺寸

Question 17

P1-C02-R17 · 5 marks

(a) · 3 marks

$a=2, b=-3.$

代入得a+b=−1及9a+3b=9；消元得a=2再求b。

M1	两个条件方程
M1	消元
A1	两系数

(b) · 2 marks

$\frac{7}{8}$.

$2x^2-3x+2=2(x-3/4)^2+7/8$。

M1	正确配方
A1	最小值7/8

Question 18

P1-C02-R18 · 6 marks

(a) · 3 marks

$k > 2$.

$16 - 4k^2 < 0$，结合k正得k>2。

M1	判别式小于零
M1	使用正值条件
A1	范围

(b) · 3 marks

$4(x + \frac{1}{2})^2 + 3$; minimum 3.

代入k后提出4配方，常数项4−1=3。

M1	正确配方
A1	完整式
B1	最小值

Question 19

P1-C02-R19 · 5 marks

(a) · 3 marks

$b = -5$, $s = \frac{23}{4}$.

比较一次项b=−2r；常数比较12=r²+s。

M1	比较系数
A1	b
A1	s

(b) · 2 marks

Its minimum value is $23/4 > 0$.

由(a)配方式，平方非负，整个式子至少23/4，不可能等于0。

M1	使用配方的非负平方
A1	正确的无根理由

Question 20

P1-C02-R20 · 5 marks

(a) · 3 marks

$p = -7,\ q = 10.$

另一根5；对应$(x-2)(x-5) = x^2 - 7x + 10$。

M1	得出第二根并写乘积
A1	p
A1	q

(b) · 2 marks

$c = -\frac{9}{4}.$

配方为$(x - 7/2)^2 - 9/4$，水平线仅在最小值处给重根。

M1	配方或判别式等零
A1	c

Question 21

P1-C02-R21 · 6 marks

(a) · 3 marks

$0 < d < 4$.

y轴正截距给d>0；判别式16−4d>0给d<4；两根$2 \pm \sqrt{4-d}$均正。

M1	正截距条件
M1	不同实根条件
A1	完整范围并保证正根

(b) · 3 marks

Other intercept $(3, 0)$; $d = 3$.

代入x=1得1−4+d=0；方程为(x−1)(x−3)=0。

M1	代入已知根
A1	d
A1	第二截距

Question 22

P1-C02-R22 · 6 marks

(a) · 3 marks

$y = 3x^2 + 6x - 5.$

顶点式$y = a(x+1)^2 - 8$；代入(1,4)得4a−8=4所以a=3。

M1	正确顶点式
M1	用点求a
A1	展开标准式

(b) · 3 marks

$x = -1 \pm \dfrac{2\sqrt{6}}{3}.$

由顶点式得$(x+1)^2=8/3$，开方并化简根式。

M1	正确平方关系
M1	化简根式
A1	两根

Question 23

P1-C02-R23 · 6 marks

(a) · 3 marks

£11, for 500 items.

$C = 2(x-5)^2 + 11$；x=5表示500件。

M1	配方
A1	最小费用
A1	正确解释百件单位

(b) · 3 marks

200 items and 800 items.

$2(x-5)^2 = 18$得x=2或8，再换算件数。

M1	解平方方程
A1	两个x值
A1	两个实际批量

Question 24

P1-C02-R24 · 6 marks

(a) · 3 marks

$a = \pm 2$.

$f = (x+a)^2 + 9 - a^2$；$9-a^2=5$。

M1	正确最小值
M1	参数平方关系
A1	两个值

(b) For each possible value of a, find

(i) · 1 marks

$(-2, 5)$ if $a = 2$; $(2, 5)$ if $a = -2$.

最小点横坐标为$-a$，纵坐标均5。

B1	两个正确配对的顶点

(ii) · 2 marks

No real solutions in either case.

f的最小值5大于4，水平线$y=4$不与曲线相交。

M1	比较最小值与4
A1	无实根结论

Question 25

P1-C02-R25 · 5 marks

(a) · 3 marks

$x = 4 \pm \sqrt{7}$.

整理成$x^2-8x+9=0$，配方$(x-4)^2=7$。

M1	正确移项
M1	配方或公式
A1	两根

(b) · 2 marks

4.

两根关于4对称，其中点为4，离原点距离4。

M1	求两根中点
A1	正确距离

Question 26

P1-C02-R26 · 5 marks

(a) · 3 marks

$k = 0, 8.$

判别式$(k-4)^2-16=0$，故$k-4=\pm 4$。

M1	判别式等零
M1	解平方方程
A1	两k值

(b) · 2 marks

$k = 8$, repeated root $x = -2.$

重根为$(4-k)/2$；$k=0$给2，$k=8$给-2。

M1	分别求重根
A1	选择k及根

Question 27

P1-C02-R27 · 6 marks

(a) · 3 marks

$A=(x-2)(x+4)=x^2+2x-8.$

内部两尺寸各减2，分别为x−2和x+4。

M1	正确内侧尺寸
M1	建立面积乘积
A1	展开结果

(b) · 3 marks

Width 6 m, length 12 m.

$x^2+2x-48=0$给$(x+8)(x-6)=0$；按$x>2$取6。

M1	正确二次方程
M1	因式分解并选根
A1	外尺寸

Question 28

P1-C02-R28 · 5 marks

(a) · 2 marks

$c = 3 - 3r^2$.

将方程写为$3(x+1)^2+c-3=0$；在根处平方为r^2。

M1	正确配方或因式乘积
A1	c表达式

(b) Given that $c = -24$, find

(i) · 2 marks

$r = 3$.

$-24=3-3r^2$给$r^2=9$，用r正取3。

M1	解平方关系
A1	r值

(ii) · 1 marks

$x = -4, 2$.

代入$-1\pm r$。

B1	两根

Question 29

P1-C02-R29 · 6 marks

(a) · 2 marks

$c = 5$; $(3, -4)$.

配方$f(x) = (x-3)^2 + c - 9$。最小值$c - 9 = -4$，所以$c = 5$。

M1	配方
A1	常数与顶点

(b) The line $y = k$ meets $y = f(x)$ at points A and B. Their x-coordinates are positive and the x-coordinate of B is twice that of A.

(i) · 2 marks

$k = -3$; $A(2, -3)$, $B(4, -3)$.

设两横坐标为$t, 2t$，关于对称轴$x = 3$，得$3t = 6$所以$t = 2$。代入$f(2) = f(4) = -3$。

M1	用对称中点与比例
A1	高度与两点

(ii) · 2 marks

1.

$AB = 2$，$M = (3, -4)$到水平底边$y = -3$的距离为1，面积$\frac{1}{2}(2)(1) = 1$。

M1	正确底与垂高
A1	面积

Question 30

P1-C02-R30 · 6 marks

(a) · 2 marks

$m < 3$, $m \neq -1$.

判别式$16 - 4(m+1) = 12 - 4m > 0$，另保留二次项系数不为零的条件。

M1	判别式严格为正
A1	完整参数范围

(b) For one value of m, the equation has a repeated root.

(i) · 2 marks

$m = 3$; $x = 1/2$.

判别式为零给出$m = 3$，原式为$(2x-1)^2 = 0$。

M1	判别式为零
A1	重根

(ii) · 2 marks

$x = 1/4$; the equation is linear.

原式化为$-4x + 1 = 0$，只有一个一次方程根；不能把“只有一个根”等同于二次重根。

B1	正确一次解
B1	区分一次与二次重根

Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 3 — Equations and inequalities Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第3章 · 方程与不等式
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C03-R01 · P1 · Chapter 3

The line L has equation $y = 2x + 1$ and the curve C has equation $y = x^2 - 4x + 6$.

(a) Find the exact coordinates of the intersections of L and C. (3)

(b) Find the set of values of x for which the curve lies strictly below the line. (2)

(Total for Question 1 is 5 marks)

2. P1-C03-R02 · P1 · Chapter 3

An integer n satisfies $3(2n - 1) \leq 4n + 9$ and $5 - n < 8$.

(a) Find all possible values of n. (3)

(b) Find the greatest possible value of n^2 and state the value of n that gives it. (2)

(Total for Question 2 is 5 marks)

3. P1-C03-R03 · P1 · Chapter 3

The equations $y = x + 4$ and $xy = 12$ are to be solved simultaneously.

(a) Find both pairs (x, y). (3)

(b) Only solutions with $x > 0$ and $y > 0$ are permitted in a rectangular area model. State the permitted solution and find $2x + 2y$. (3)

(Total for Question 3 is 6 marks)

$x^2 - 7x + 10 \leq 0$.

(a) Solve the inequality. (2)

(b) Given also that $2x - 1 > 5$, find

(i) the resulting interval for x, (2)

(ii) all integers in this interval. (1)

(Total for Question 4 is 5 marks)

5.

The equations $x + y = 7$ and $x^2 + y^2 = 29$ hold.

(a) Find all ordered pairs (x, y). (3)

(b) Find the possible values of $x - y$. (2)

(Total for Question 5 is 5 marks)

6.

A delivery charge is £4 plus £1.50 per kilometre. A second service charges £10 plus £0.75 per kilometre. Let the distance be $d > 0$ kilometres.

(a) Find the distance at which the two charges are equal. (2)

(b) A customer has a budget of £19 and wants the second service to be strictly cheaper. Find the permitted interval for d. (3)

(Total for Question 6 is 5 marks)

7. P1-C03-R07 · P1 · Chapter 3

The line $y = 3x - 2$ meets the curve $y = x^2 + k$ at the point $(1, 1)$.

(a) Find k and the coordinates of the other intersection. (3)

(b) Find the values of x for which $x^2 + k \geq 3x - 2$. (2)

(Total for Question 7 is 5 marks)

8.

For a real number x, $4 - 3x \geq 2x - 11$.

(a) Solve the inequality. (2)

(b) Find all values of x that also satisfy $x^2 > 4$. (3)

(Total for Question 8 is 5 marks)

9. P1-C03-R09 · P1 · Chapter 3

The simultaneous equations are $2x + y = 5$ and $x^2 + xy = 6$.

(a) Find all real solutions (x, y). (3)

(b) A student divides the second equation by x before solving. Explain why this division does not remove a solution in this question. (2)

(Total for Question 9 is 5 marks)

The inequalities $x^2 + 2x - 8 > 0$ and $x < 3$ must both hold.

(a) Find the complete solution set. (3)

(b) State the greatest integer that satisfies both inequalities. (2)

(Total for Question 10 is 5 marks)

A rectangle has perimeter 24 m and area at least $32\ \text{m}^2$. Its width is w m and is no greater than its length.

(a) Show that $w^2 - 12w + 32 \leq 0$. (2)

(b) Find the permitted interval for w. (3)

(Total for Question 11 is 5 marks)

12. P1-C03-R12 · P1 · Chapter 3

The equations $y = 2 - x$ and $y = x^2 - 4$ have two intersections.

(a) Find the coordinates of both intersections. (3)

(b) For each intersection, calculate $x + y$ and explain why the answers agree. (3)

(Total for Question 12 is 6 marks)

$2x^2 + 3x - 9 < 0$.

(a) Solve the inequality. (2)

(b) Find the smallest and largest integers satisfying the inequality. (2)

(Total for Question 13 is 4 marks)

14.

A line $y = x + c$ passes through an intersection of the curves $y = x^2$ and $y = 4x - 3$.

(a) Find the two possible intersections of the curves. (2)

(b) Find the two possible values of c. (2)

(Total for Question 14 is 4 marks)

15.

The equations are $x - y = 1$ and $xy = 6$.

(a) Find all possible ordered pairs (x, y). (3)

(b) Find the unique solution satisfying $x + y < 0$ and state the value of $x^2 - y^2$ for this solution. (3)

(Total for Question 15 is 6 marks)

The expression $Q(x) = 9 - (x - 1)^2$.

(a) Find all values of x for which $Q(x) \geq 0$. (2)

(b) Given also that x is an integer and $2x + 1 < 6$, find

(i) all permitted values of x, (2)

(ii) the greatest permitted value of $Q(x)$. (2)

(Total for Question 16 is 6 marks)

17.

A ticket office sells adult tickets at £8 and child tickets at £5. A group buys 14 tickets for a total of £94.

(a) Form and solve simultaneous equations to find the numbers of adult and child tickets. (3)

(b) The same group size is used for a second booking with a budget of at most £85. Find the greatest possible number of adult tickets. (3)

(Total for Question 17 is 6 marks)

18.

The equations $y = x^2 - 2x$ and $y = 4 - x$ are to be solved simultaneously.

(a) Find the exact x-coordinates of their intersections. (3)

(b) Hence solve $x^2 - 2x < 4 - x$. (2)

(Total for Question 18 is 5 marks)

19.

A real number x satisfies $5 - 2(x + 1) < 3x + 8$.

(a) Solve this inequality. (2)

(b) Find the values of x that also satisfy $(x + 1)(x - 4) \leq 0$. (2)

(Total for Question 19 is 4 marks)

The line $x + y = 6$ and the curve $x^2 + 2y = 20$ intersect at A and B.

(a) Find the exact coordinates of A and B. (3)

(b) Find the midpoint of AB. (2)

(Total for Question 20 is 5 marks)

21. P1-C03-R21 · P1 · Chapter 3

A rectangle has side lengths $(x - 1)$ cm and $(9 - x)$ cm, both positive.

(a) Find the interval in which x must lie. (2)

(b) The area must exceed $12\ \text{cm}^2$. Find the permitted interval for x, giving exact endpoints. (3)

(Total for Question 21 is 5 marks)

For real x, $x(x-5) \geq 6$.

(a) Solve the inequality. (2)

(b) A positive integer x must also satisfy $3x + 2 < 26$. Find all its possible values. (2)

(Total for Question 22 is 4 marks)

23.

The equations are $y = 3x$ and $x^2 + y^2 = 40$.

(a) Find all real solutions. (3)

(b) Find the distance between the two solution points. (3)

(Total for Question 23 is 6 marks)

A function is defined by $f(x) = x^2 - 3x - 10$.

(a) Solve $f(x) \leq 0$. (2)

(b) Find all real x for which both $f(x) \leq 0$ and $f(-x) \leq 0$. (3)

(Total for Question 24 is 5 marks)

The equations $x + 2y = 1$ and $x^2 - 4y^2 = 9$ hold.

(a) Without forming a quadratic equation, find $x - 2y$. (2)

(b) Hence find x and y. (3)

(Total for Question 25 is 5 marks)

For a school event, the profit P pounds is modelled by $P = -n^2 + 18n - 65$, where n is a non-negative integer number of tickets sold.

(a) Find all values of n for which the profit is positive. (3)

(b) Find the least number of tickets that gives a profit of at least £15. (3)

(Total for Question 26 is 6 marks)

27. P1-C03-R27 · P1 · Chapter 3

The inequalities $2x^2 - 5x - 3 \geq 0$ and $x^2 < 16$ must hold simultaneously.

(a) Solve the first inequality. (2)

(b) Find the complete simultaneous solution set. (3)

(Total for Question 27 is 5 marks)

The equations $x + y = k$ and $x^2 + y^2 = 10$ have a solution with $x = y$.

(a) Find all possible values of k. (3)

(b) For $k = 2$, find all real solutions (x, y). (3)

(Total for Question 28 is 6 marks)

A panel is made from a rectangle of length $(x + 5)$ cm and width $(x - 1)$ cm, where $x > 1$. Its area is at most $45\ \text{cm}^2$ and its perimeter is greater than 20 cm.

(a) Write each design requirement as an inequality in x. (2)

(b) Find

(i) the complete interval of permissible real values of x. (2)

(ii) all permissible integer values of x. (1)

(Total for Question 29 is 5 marks)

The line $y = tx + 1$, where t is a real constant, meets the curve $y = x^2 - 3x + 5$ at points A and B.

(a) Show that the x-coordinates of the intersections satisfy $x^2 - (t+3)x + 4 = 0$. (2)

(b) The x-coordinate of B is four times that of A.

(i) Find the two possible values of t. (2)

(ii) For $t < 0$, find the coordinates of A and B. (2)

(Total for Question 30 is 6 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C03-R01 · 5 marks

(a) · 3 marks

$(1, 3)$ and $(5, 11)$.

联立得$x^2-6x+5=0$，分解为$(x-1)(x-5)$；代入直线求y。

M1	消去y
M1	解两根
A1	两交点

(b) · 2 marks

$1 < x < 5$.

差为$(x-1)(x-5)$，开口向上，在两根之间负；严格不等号排除端点。

M1	建立曲线与直线差的不等式
A1	严格区间

Question 2

P1-C03-R02 · 5 marks

(a) · 3 marks

$n = -2, -1, 0, 1, 2, 3, 4, 5, 6.$

第一式给$n \leqslant 6$，第二式给$n > -3$，交集中取整数。

M1	解第一不等式
M1	解第二不等式
A1	全部整数

(b) · 2 marks

36, at $n = 6$.

可行整数中绝对值最大的是6；平方36。

M1	在可行整数中比较绝对值
A1	最大值与取值

Question 3

P1-C03-R03 · 6 marks

(a) · 3 marks

$(2, 6)$ and $(-6, -2)$.

代入得$x^2+4x-12=0$，即$(x+6)(x-2)=0$；回代求y。

M1	代入消元
M1	分解求两根
A1	两对解

(b) · 3 marks

$(2, 6)$; $2x + 2y = 16$.

排除两负的解；把正解代入周长式。

B1	正确选择
M1	代入所给式
A1	结果16

Question 4

P1-C03-R04 · 5 marks

(a) · 2 marks

$2 \le x \le 5$.

分解为(x−2)(x−5)，向上抛物线在两根间及根上非正。

M1	找两个根
A1	闭区间

(b) Given also that $2x - 1 > 5$, find

(i) · 2 marks

$3 < x \le 5$.

新条件给x>3，与(a)交集。

M1	解新不等式
A1	正确交集

(ii) · 1 marks

$4, 5$.

3被排除，5被包含。

B1	两个整数

Question 5

P1-C03-R05 · 5 marks

(a) · 3 marks

$(2,5)$ and $(5,2)$.

置y=7−x得$2x^2-14x+20=0$，约简后(x−2)(x−5)=0。

M1	正确消元
M1	化简并解方程
A1	两个有序对

(b) · 2 marks

$-3,3$.

两个有序解分别相减，不可只保留正差。

M1	用两组解
A1	两个差

Question 6

P1-C03-R06 · 5 marks

(a) · 2 marks

$d = 8$ km.

$4 + 1.5d = 10 + 0.75d$，移项得0.75d=6。

M1	建立等价方程
A1	距离8公里

(b) · 3 marks

$8 < d \leq 12$.

第二服务较便宜要求d>8；预算要求10+0.75d⩽19，即d⩽12；取交集。

M1	严格比较费用
M1	预算不等式
A1	完整区间

Question 7

P1-C03-R07 · 5 marks

(a) · 3 marks

$k = 0$; other intersection $(2, 4)$.

代入已知点得1=1+k；再解$x^2=3x-2$，得x=1,2。

M1	用点求k
M1	联立求另一根
A1	k和另一坐标

(b) · 2 marks

$x \leq 1$ or $x \geq 2$.

代入k后$(x-1)(x-2)\geqslant 0$；取外侧并包括端点。

M1	正确分解
A1	两个闭外区间

Question 8

P1-C03-R08 · 5 marks

(a) · 2 marks

$x \leq 3$.

移项得15≥5x，除以正数5。

M1	正确移项
A1	解集

(b) · 3 marks

$x < -2$ or $2 < x \leq 3$.

平方大于4给x<−2或x>2；与x≤3求交，平方的两个端点均不包含。

M1	正确两支平方不等式
M1	求交集
A1	完整端点

Question 9

P1-C03-R09 · 5 marks

(a) · 3 marks

$(2, 1)$ and $(3, -1)$.

y=5−2x，代入得−x²+5x=6，即x²−5x+6=0。

M1	正确代入
M1	解二次方程
A1	两解

(b) · 2 marks

$x = 0$ cannot satisfy $x^2 + xy = 6$, since its left side would be 0.

先核查可能被除去的x=0，发现不满足原式，因此此题除x合法。

M1	核查被除因子为零的情况
A1	与右边6矛盾并作结论

Question 10

P1-C03-R10 · 5 marks

(a) · 3 marks

$x < -4$ or $2 < x < 3$.

二次式分解(x+4)(x−2)，取外侧，再与x<3求交。

M1	正确根及符号
M1	求交
A1	两个严格区间

(b) · 2 marks

-5.

区间(2,3)没有整数；另一支x<−4的最大整数为−5。

M1	分别检查两个分支中的整数
A1	最大整数

Question 11

P1-C03-R11 · 5 marks

(a) · 2 marks

$w(12-w)\geq 32 \Rightarrow w^2-12w+32\leq 0.$

另一边为12−w；移项再乘−1需翻转不等号。

M1	正确面积不等式
A1	负号转换正确

(b) · 3 marks

$4\leq w\leq 6.$

二次不等式给4⩽w⩽8；宽不大于长给w⩽6；取交集。

M1	解二次不等式
M1	使用宽长关系
A1	最终区间

Question 12

P1-C03-R12 · 6 marks

(a) · 3 marks

$(-3, 5)$ and $(2, 0)$.

联立得$x^2+x-6=0=(x+3)(x-2)$。

M1	正确消元
M1	解两根
A1	两个坐标

(b) · 3 marks

$x + y = 2$ at both points; both lie on $y = 2 - x$.

直线方程等价于$x+y=2$，所以无论曲线上的哪一个交点都满足这个和。

B1	两个和均正确
M1	使用直线方程
A1	完整解释

Question 13

P1-C03-R13 · 4 marks

(a) · 2 marks

$-3 < x < \frac{3}{2}$.

分解为(2x−3)(x+3)，向上开口在两根之间负。

M1	正确因式分解或求根
A1	严格区间

(b) · 2 marks

Smallest -2; largest 1.

不包含−3，也不允许超过或等于1.5；可行整数−2、−1、0、1。

M1	正确处理两个端点
A1	最小最大整数

Question 14

P1-C03-R14 · 4 marks

(a) · 2 marks

$(1, 1)$ and $(3, 9)$.

$x^2-4x+3=0$，解x=1,3并计算y。

M1	联立并求根
A1	两点

(b) · 2 marks

$c = 0, 6$.

由c=y−x，分别用两个交点计算。

M1	使用c=y-x
A1	两值

Question 15

P1-C03-R15 · 6 marks

(a) · 3 marks

$(3, 2)$ and $(-2, -3)$.

x=y+1给y^2+y−6=0，分解后y=2或−3。

M1	正确代入
M1	两根及回代
A1	两对

(b) · 3 marks

$(-2, -3)$; $x^2 - y^2 = -5$.

两组和分别5和−5，故选负组；平方差为4−9。

M1	正确筛选
A1	解对
A1	平方差

Question 16

P1-C03-R16 · 6 marks

(a) · 2 marks

$-2 \leq x \leq 4$.

转成$(x-1)^2 \leqslant 9$，得$-3 \leqslant x-1 \leqslant 3$。

M1	正确平方范围
A1	闭区间

(b) Given also that x is an integer and $2x + 1 < 6$, find

(i) · 2 marks

$-2, -1, 0, 1, 2$.

新条件$x<2.5$，与(a)求交再取整数。

M1	联立并取整数
A1	全部值

(ii) · 2 marks

9, at $x = 1$.

配方最大9，且$x=1$在可行整数内。

M1	检查最大值位置可行
A1	最大值

Question 17

P1-C03-R17 · 6 marks

(a) · 3 marks

8 adult and 6 child tickets.

令成人a、儿童c，a+c=14，8a+5c=94；相减得3a=24。

M1	两个方程
M1	正确消元
A1	两个票数

(b) · 3 marks

5 adult tickets.

费用8a+5(14−a)=70+3a≤85给a≤5，结合整数与0≤a≤14。

M1	正确费用不等式
M1	解并考虑整数
A1	最大人数

Question 18

P1-C03-R18 · 5 marks

(a) · 3 marks

$x = \dfrac{1 \pm \sqrt{17}}{2}$.

联立得到$x^2-x-4=0$，用公式解。

M1	正确方程
M1	二次公式
A1	两个精确根

(b) · 2 marks

$\dfrac{1 - \sqrt{17}}{2} < x < \dfrac{1 + \sqrt{17}}{2}$.

二次差开口向上，严格为负的位置在两根之间，不包含端点。

M1	使用根和开口确定符号
A1	精确区间

Question 19

P1-C03-R19 · 4 marks

(a) · 2 marks

$x > -1$.

展开得3−2x<3x+8，故−5<5x。

M1	正确展开移项
A1	解集

(b) · 2 marks

$-1 < x \leq 4$.

二次条件给−1⩽x⩽4；首条件在−1严格，所以交集左开右闭。

M1	正确二次区间
A1	正确端点交集

Question 20

P1-C03-R20 · 5 marks

(a) · 3 marks

$(-2, 8)$ and $(4, 2)$.

y=6−x给$x^2-2x-8=0$，因式分解求x=−2,4并回代。

M1	代入消元
M1	求根
A1	两点

(b) · 2 marks

$(1, 5)$.

横纵坐标分别求平均。

M1	对两坐标取平均
A1	中点

Question 21

P1-C03-R21 · 5 marks

(a) · 2 marks

$1 < x < 9$.

两边正分别要求x>1与x<9。

M1	建立两条正值条件
A1	交集

(b) · 3 marks

$3 < x < 7$.

面积16−(x−5)²>12给(x−5)²<4，故3

M1	建立正确面积式
M1	解平方不等式
A1	正确区间

Question 22

P1-C03-R22 · 4 marks

(a) · 2 marks

$x \leq -1$ or $x \geq 6$.

移项后(x−6)(x+1)⩾0，取外侧含根。

M1	正确分解
A1	两外区间

(b) · 2 marks

$x = 6, 7$.

新条件x<8；只保留(a)中的正分支x⩾6并取整数。

M1	联立正值与线性限制
A1	全部整数

Question 23

P1-C03-R23 · 6 marks

(a) · 3 marks

$(2, 6)$ and $(-2, -6)$.

代入得$10x^2=40$，所以$x=\pm 2$，y与x同号。

M1	正确消元
M1	取正负平方根
A1	两对解

(b) · 3 marks

$4\sqrt{10}$.

横差4、纵差12，距离$\sqrt{(16+144)}=4\sqrt{10}$。

M1	使用两点距离
M1	化简根式
A1	精确距离

Question 24

P1-C03-R24 · 5 marks

(a) · 2 marks

$-2 \leq x \leq 5$.

分解为(x−5)(x+2)，开口向上。

M1	正确根与符号
A1	闭区间

(b) · 3 marks

$-2 \leq x \leq 2$.

将−x代入(a)的条件得−5≤x≤2；和原区间求交。

M1	正确处理替换后的不等号
M1	求交集
A1	最终区间

Question 25

P1-C03-R25 · 5 marks

(a) · 2 marks

$x - 2y = 9$.

平方差为(x−2y)(x+2y)=9，用给定第二因子1得到结论。

M1	正确平方差因式
A1	求得线性组合

(b) · 3 marks

$x = 5,\ y = -2$.

把x+2y=1与x−2y=9相加得2x=10，再回代。

M1	联立两线性组合
A1	x
A1	y

Question 26

P1-C03-R26 · 6 marks

(a) · 3 marks

$n = 6, 7, 8, 9, 10, 11, 12.$

正利润给$n^2-18n+65<0$，即$(n-5)(n-13)<0$；取5

M1	正确转换不等式
M1	求根与区间
A1	全部整数

(b) · 3 marks

8 tickets.

$P=16-(n-9)^2$，要求$(n-9)^2\leqslant 1$，n=8、9、10，最少8。

M1	正确利润门槛关系
M1	取可行整数
A1	最小值

Question 27

P1-C03-R27 · 5 marks

(a) · 2 marks

$x \le -\frac{1}{2}$ or $x \ge 3$.

因式为(2x+1)(x−3)，乘积非负取外侧。

M1	正确因式分解
A1	两外区间

(b) · 3 marks

$-4 < x \le -\frac{1}{2}$ or $3 \le x < 4$.

第二条件为−4

M1	正确平方范围
M1	对两个分支求交
A1	全部端点

Question 28

P1-C03-R28 · 6 marks

(a) · 3 marks

$k = \pm 2\sqrt{5}$.

代入x=y得$2x^2=10$，所以x=y=±√5且同号；k=2x。

M1	使用相等条件
M1	保留两个符号
A1	两k值

(b) · 3 marks

$(-1, 3)$ and $(3, -1)$.

y=2−x给$2x^2-4x-6=0$，解x=−1或3。

M1	正确消元
M1	解方程
A1	两对

Question 29

P1-C03-R29 · 5 marks

(a) · 2 marks

$x^2+4x-50\le 0$ and $4x+8>20$.

面积为$(x+5)(x-1)$，周长为$2[(x+5)+(x-1)]$；分别代入给定限制。

M1	正确面积不等式
B1	正确周长不等式

(b) Find

(i) · 2 marks

$3<x\le -2+3\sqrt{6}$.

面积方程根为$-2\pm 3\sqrt{6}$；与$x>3$及$x>1$取交集。

M1	解二次不等式
A1	取交集并保留端点

(ii) · 1 marks

$4, 5$.

$-2+3\sqrt{6}\approx 5.35$，所以整数仅为$4, 5$。

B1	整数解

Question 30

P1-C03-R30 · 6 marks

(a) · 2 marks

$x^2-(t+3)x+4=0$.

令两纵坐标相等，移项整理即可。

M1	联立曲线与直线
A1	正确方程

(b) The x-coordinate of B is four times that of A.

(i) · 2 marks

$t=2$ or $t=-8$.

设A横坐标为a，B为$4a$。将$x=a,4a$分别代入，联立可得$a^2=1$和$t+3=5a$；所以$t=2,-8$。可用展开$(x-a)(x-4a)$比较系数。

M1	利用两根比例建立因式
A1	两个参数值

(ii) · 2 marks

$A(-1,9)$, $B(-4,33)$.

$t=-8$对应$a=-1$。代入$y=-8x+1$求纵坐标。

M1	选负参数对应根
A1	两点坐标

Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 4 — Graphs and transformations Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第4章 · 图像与变换
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C04-R01 · P1 · Chapter 4

The curve C has equation $y = (x + 2)(x - 1)(x - 4)$.

(a) Sketch C, showing the coordinates of all intersections with the coordinate axes. (3)

(b) The curve is translated by $\begin{pmatrix} 3 \\ -2 \end{pmatrix}$. Find the coordinates of the image of $(0, 8)$ and write the equation of the translated curve. (3)

(Total for Question 1 is 6 marks)

The graph of $y = f(x)$ passes through $A = (-3, 2)$, $B = (1, -4)$ and $C = (4, 1)$.

(a) For the graph $y = 2f(x - 1) + 3$, find the images of

(i) A, (2)

(ii) B and C. (2)

(b) Describe a sequence of transformations taking $y = f(x)$ to $y = 2f(x - 1) + 3$. (3)

(Total for Question 2 is 7 marks)

The curve C is $y = \dfrac{4}{x}$.

(a) Sketch C, showing its asymptotes and the points on C with $x = \pm 2$. (3)

(b) The curve is translated 3 units left and 1 unit down. Find

(i) the equation of the new curve, (2)

(ii) its intersections with the coordinate axes. (2)

(Total for Question 3 is 7 marks)

The graph of $y = f(x)$ is shown in Figure 4. The marked points are $(-4, 0)$, $(-1, 3)$, $(2, -2)$ and $(5, 0)$; straight segments join consecutive points.

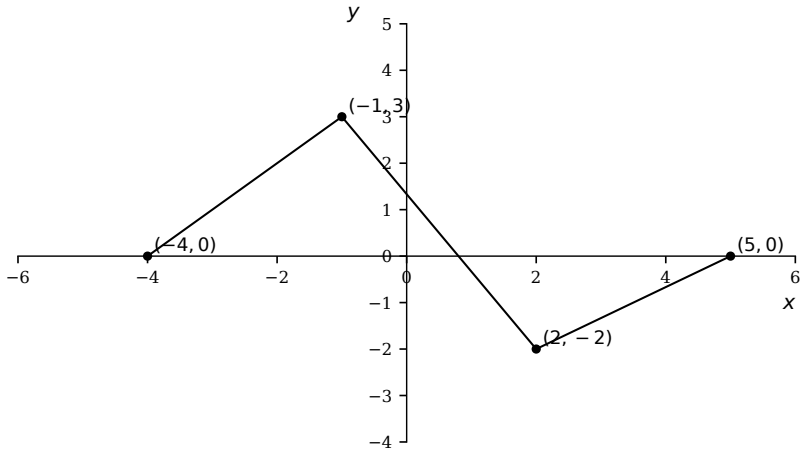


Figure 4

(a) Find the coordinates of the images of all four marked points under the transformation $y = -f(x)$. (2)

(b) Sketch $y = f(-x)$, labelling the images of the four marked points. (3)

(Total for Question 4 is 5 marks)

C is the curve $y = (x - 2)^2 + 1$.

(a) State the minimum point of C and explain why C does not intersect the x-axis. (3)

(b) C is stretched parallel to the y-axis by factor 3, then translated down 5. Find

(i) the equation and minimum point of the resulting curve, (3)

(ii) the exact x-intercepts. (2)

(Total for Question 5 is 8 marks)

The curve C is $y = \dfrac{6}{x^2}$.

(a) Sketch C, showing both asymptotes and the points at $x = \pm 1$. (3)

(b) After reflection in the x-axis and translation up 2, find the equation of the curve and its x-intercepts. (3)

(Total for Question 6 is 6 marks)

Point $P = (6, -3)$ lies on $y = f(x)$.

(a) Find the corresponding point on each graph:

(i) $y = f(2x)$, (2)

(ii) $y = \frac{1}{2}f(x)$. (2)

(b) Point $Q = (a, b)$ lies on $y = f(x)$. Give the corresponding point on $y = \frac{1}{2}f(2x)$. (2)

(Total for Question 7 is 6 marks)

8. P1-C04-R08 · P1 · Chapter 4

The curve C has equation $y = x^3$.

(a) Write the equation of the curve obtained by translating C so that $(0, 0)$ moves to $(2, -5)$. (2)

(b) Find the exact coordinates where the translated curve intersects the two coordinate axes. (3)

(Total for Question 8 is 5 marks)

9. P1-C04-R09 · P1 · Chapter 4

The graph of $y = f(x)$ has exactly three x-intercepts, at $x = -2, 1, 5$, and $f(0) = 4$.

(a) Find all x-intercepts of $y = f(3x)$. (2)

(b) For $y = f(x - 4)$, find all x-intercepts and the point corresponding to $(0, 4)$. (3)

(Total for Question 9 is 5 marks)

Curve A is $y = \dfrac{1}{x-2} + 3$ and curve B is $y = \dfrac{1}{x+1} - 2$.

(a) State the asymptotes of each curve. (2)

(b) Describe the single translation that maps A onto B. (2)

(Total for Question 10 is 4 marks)

A graph passes through $A = (2, 5)$ and $B = (-4, -1)$. A transformation maps every point (x, y) to $(x + 3, 2y)$.

(a) Find the images of A and B. (2)

(b) The original graph is $y = f(x)$. Write the equation of its image and state the original coordinates of a point whose image is $(7, 8)$. (3)

(Total for Question 11 is 5 marks)

C is the curve $y = (x + 1)^3$.

(a) State the coordinates where C meets the coordinate axes. (2)

(b) C is reflected in the y-axis. Find

(i) the equation of the reflected curve, (2)

(ii) the image of each intercept from part (a). (2)

(Total for Question 12 is 6 marks)

13.

The graph of $y = f(x)$ has a maximum point $(2, 6)$ and a minimum point $(5, -3)$.

(a) Find the maximum and minimum points of $y = f(x) + 4$. (2)

(b) Find the maximum and minimum points of $y = -2f(x)$. (3)

(Total for Question 13 is 5 marks)

A curve is obtained from $y = 1/x$ by a translation. Its asymptotes are $x = -4$ and $y = 2$.

(a) Write the equation of the curve. (2)

(b) Find its exact intersections with the coordinate axes. (3)

(Total for Question 14 is 5 marks)

$f(x) = x(x - 3)(x + 2)$.

(a) Find the roots of $f(x) = 0$ and sketch $y = f(x)$. (3)

(b) A second curve has equation $y = f(x + 1)$. Find its roots and the coordinates of the point corresponding to $(0, 0)$. (3)

(Total for Question 15 is 6 marks)

The graph of $y = f(x)$ contains the points $(-2, 7)$ and $(3, -1)$.

(a) For the graph $y = f(2x - 4)$, find

(i) the corresponding two points, (2)

(ii) the point on $y = f(x)$ corresponding to $(5, 9)$ on the new graph. (2)

(b) Describe a sequence of two transformations taking $y = f(x)$ to $y = f(2x - 4)$. (2)

(Total for Question 16 is 6 marks)

C is the curve $y = \dfrac{2}{x^2} + 1$.

(a) State its asymptotes and explain why it has no x-intercepts. (3)

(b) Find the coordinates where C meets the horizontal line $y = 9$. (2)

(Total for Question 17 is 5 marks)

A point $P = (a, b)$ on $y = f(x)$ is mapped to $(4, -6)$ on $y = -3f(x + 2)$.

(a) Find a and b. (2)

(b) Find the image of P on $y = f(x) - 5$. (2)

(Total for Question 18 is 4 marks)

19.

A parabola has equation $y = (x - 4)^2 - 9$.

(a) Find its minimum point and its x-intercepts. (3)

(b) The parabola is translated so that its minimum point moves to the origin. Find the translation vector and the equation of the resulting curve. (2)

(Total for Question 19 is 5 marks)

The curve C is $y = x^3 - 8$.

(a) Find the coordinates of its intersections with the axes. (2)

(b) Curve D is obtained by reflecting C in the x-axis. Find the equation of D and state where D meets C. (3)

(Total for Question 20 is 5 marks)

Figure 21 shows $y = f(x)$ with marked points $A = (-3, -1)$, $B = (0, 2)$ and $C = (4, 0)$. Straight segments join A to B and B to C.

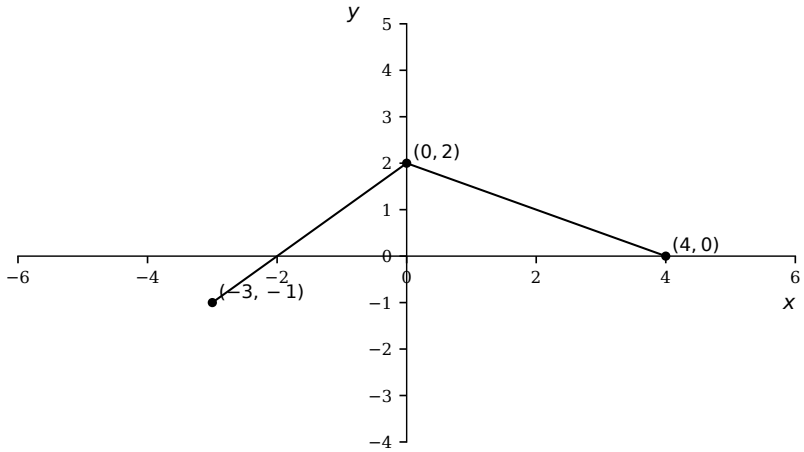


Figure 21

(a) Sketch $y = 2f(x)$, marking the three corresponding points. (3)

(b) Sketch $y = f(2x)$ on separate axes, marking the three corresponding points. (3)

(Total for Question 21 is 6 marks)

The curve $y = f(x)$ is reflected in the y-axis and then translated right by 3.

(a) Write the equation of the resulting curve in terms of f. (3)

(b) If the order of these transformations is reversed, find the resulting equation and the image of the point $(2, 5)$. (3)

(Total for Question 22 is 6 marks)

The curve C is $y = \dfrac{a}{x-1} + 2$, where a is a non-zero constant. It passes through $(3, 5)$.

(a) Find a. (2)

(b) For this value of a, find

(i) the intersections of C with the axes, (2)

(ii) the translation that maps C onto $y = 6/x$. (2)

(Total for Question 23 is 6 marks)

$f(x) = (x - 1)(x + 2)^2$.

(a) Sketch $y = f(x)$, showing all axis intersections and distinguishing a crossing from a touching point. (3)

(b) For the graph $y = f(x) - f(0)$, find the y-intercept and state the vertical translation from the original graph. (2)

(Total for Question 24 is 5 marks)

The curve C has equation $y = \dfrac{3}{(x+2)^2}$.

(a) State its asymptotes and its intersection with the y-axis. (2)

(b) The curve is translated down by 3. Find the equation of its image and all x-intercepts. (3)

(Total for Question 25 is 5 marks)

A curve $y = f(x)$ passes through $(2, 0)$ and $(0, -6)$. Its image under a translation is $y = f(x - 3) + 4$.

(a) Find the images of the two given points. (2)

(b) Explain why the image of $(2, 0)$ need not be an x-intercept of the new curve, and state the equation that determines its actual x-intercepts. (2)

(Total for Question 26 is 4 marks)

The curve $y = f(x)$ is stretched parallel to the x-axis by factor 3 and then reflected in the x-axis.

(a) Write the equation of the resulting curve. (2)

(b) Point (a, b) on the original curve maps to $(-6, 5)$. Find

(i) a and b, (2)

(ii) the image of (a, b) under translation by $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$. (2)

(Total for Question 27 is 6 marks)

The curve C is $y = x^2$ and the curve D is $y = (x - 2)^2 + 4$.

(a) Describe the single translation taking C onto D. (2)

(b) Find the coordinates where C and D intersect, and explain why there is only one intersection. (3)

(Total for Question 28 is 5 marks)

Figure 29 shows the graph of $y = f(x)$. It consists of straight line segments joining $A(-4, 1)$, $B(-1, -2)$ and $C(3, 2)$, in that order.

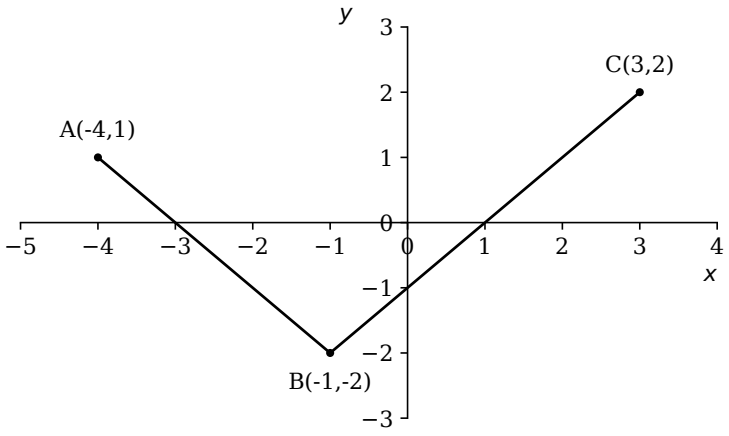


Figure 29

(a) Sketch the graph of $y = -2f(x + 1)$, giving the coordinates of the images of A, B, C. (3)

(b) The graph of $y = f(x)$ crosses the x-axis at two points.

(i) Find their coordinates. (2)

(ii) Hence solve $f(x + 1) = 0$. (1)

(Total for Question 29 is 6 marks)

The curve C has equation $y = x(x - 2)(x + 3)$.

(a) Sketch C, showing all axis intercepts and the behaviour of the curve for large positive and large negative x. Do not calculate turning points. (3)

(b) The curve is translated 2 units to the right to form D.

(i) Find an equation for D in factorised form. (2)

(ii) Find the interval on which D lies below the x-axis and has positive x. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C04-R01 · 6 marks

(a) · 3 marks

x-intercepts $(-2,0),(1,0),(4,0)$; y-intercept $(0,8)$; cubic rises to the right.

三个不同的一次因子给三个穿越点；代入0得8；首项x^3决定左下右上。

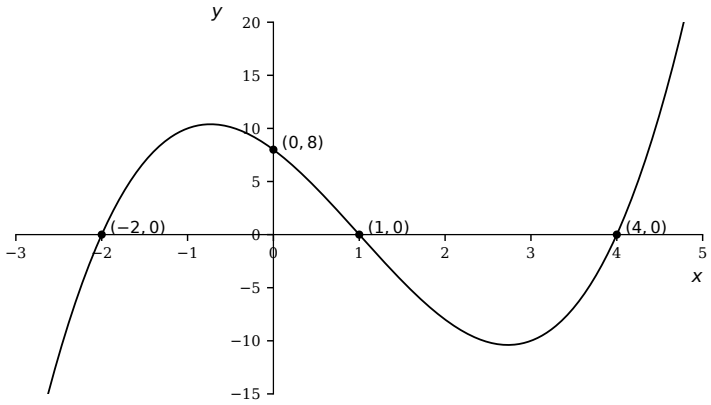
M1	三个横截距
B1	纵截距
A1	正确三次曲线走向

(b) · 3 marks

Image $(3,6)$; $y=(x-1)(x-4)(x-7)-2$.

点坐标加平移向量；方程以x−3替换x再减2。

B1	平移后坐标
M1	正确替换自变量
A1	完整方程



Suggested sketch

Question 2

P1-C04-R02 · 7 marks

(a) For the graph $y = 2f(x-1)+3$, find the images of

(i) · 2 marks

$(-2, 7)$.

x坐标加1，y坐标乘2再加3。

M1	正确变换横纵坐标
A1	点坐标

(ii) · 2 marks

$(2, -5)$ and $(5, 5)$.

两点都应用映射$(x, y) \mapsto (x+1, 2y+3)$，顺序保持。

M1	两点使用一致映射
A1	两个像点

(b) · 3 marks

Translate right 1, stretch parallel to the y-axis by factor 2, then translate up 3.

横移只改x位置，纵向伸缩先于向上平移；如先上移3再伸缩会误成上移6。

M1	正确横移
M1	正确纵向伸缩
A1	上移和顺序

Question 3

P1-C04-R03 · 7 marks

(a) · 3 marks

Asymptotes $x = 0, y = 0$; points $(2, 2), (-2, -2)$; branches in quadrants I and III.

正系数倒数曲线在第一、第三象限，不能画过渐近线；直接代入±2。

B1	两渐近线
B1	两点
A1	两支走向

(b) The curve is translated 3 units left and 1 unit down. Find

(i) · 2 marks

$y = \dfrac{4}{x + 3} - 1.$

向左3用x+3替换，再减1。

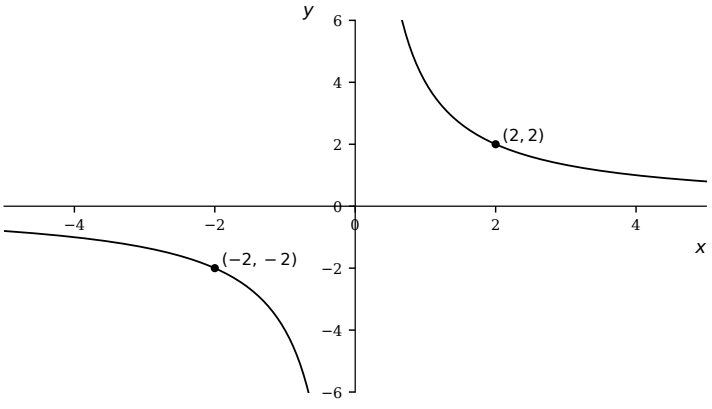
M1	正确水平替换
A1	完整方程

(ii) · 2 marks

$(1, 0)$ and $(0, \frac{1}{3})$.

y=0时x+3=4；x=0时y=4/3−1。

M1	分别令x或y为零
A1	两个截距



Suggested sketch

Question 4

P1-C04-R04 · 5 marks

(a) · 2 marks

$(-4, 0), (-1, -3), (2, 2), (5, 0)$.

关于x轴反射，横坐标不变，纵坐标变号。

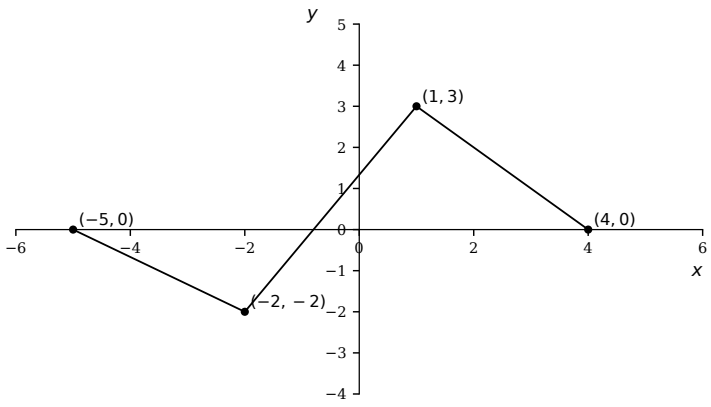
M1	正确映射规律
A1	四点正确

(b) · 3 marks

$(-5, 0), (-2, -2), (1, 3), (4, 0)$, joined in this left-to-right order.

关于y轴反射，横坐标变号，并重新按从左到右顺序连线。

M1	四横坐标反号
A1	四像点
A1	正确连接顺序



Suggested sketch

Question 5

P1-C04-R05 · 8 marks

(a) · 3 marks

Minimum $(2, 1)$; $y \geq 1 > 0$.

平方项非负，最低纵坐标1，故没有横截距。

B1	顶点
M1	用平方非负
A1	无交点理由

(b) C is stretched parallel to the y-axis by factor 3, then translated down 5. Find

(i) · 3 marks

$y = 3(x-2)^2 - 2$; minimum $(2, -2)$.

原整体y值乘3后减5，常数3−5=−2。

M1	变换整个原式
A1	方程
B1	顶点

(ii) · 2 marks

$x = 2 \pm \dfrac{\sqrt{6}}{3}$.

令y=0得$(x-2)^2=2/3$。

M1	解平方方程
A1	两精确根

Question 6

P1-C04-R06 · 6 marks

(a) · 3 marks

$x = 0, y = 0$; $(-1, 6), (1, 6)$; two positive branches symmetric about the y-axis.

x^2正使两支均在x轴上方，且左右对称；趋近x=0时向上无界。

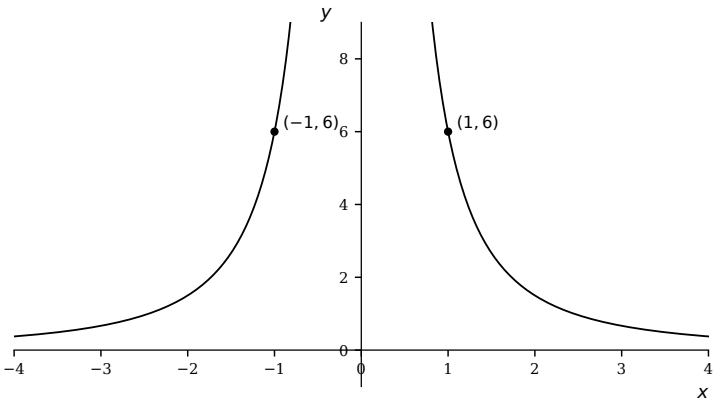
B1	渐近线
B1	两点
A1	正确形状与对称

(b) · 3 marks

$y = 2 - 6/x^2$; $(\pm\sqrt{3}, 0)$.

先反射变为$-6/x^2$，再上移2；令其为0解$x^2=3$。

M1	正确组合变换
A1	方程
A1	两截距



Suggested sketch

Question 7

P1-C04-R07 · 6 marks

(a) Find the corresponding point on each graph:

(i) · 2 marks

$(3, -3)$.

要使函数输入仍为6，需2x=6；y不变。

M1	正确处理水平因子
A1	点坐标

(ii) · 2 marks

$(6, -\frac{3}{2})$.

x不变，y乘1/2。

M1	纵坐标减半
A1	点坐标

(b) · 2 marks

$(\frac{a}{2}, \frac{b}{2})$.

两个独立映射分别使横纵坐标减半。

M1	正确横坐标
A1	正确纵坐标

Question 8

P1-C04-R08 · 5 marks

(a) · 2 marks

$y = (x-2)^3 - 5$.

平移向量$(2,-5)$，在函数中用$x-2$替换并减5。

M1	正确水平平移
A1	完整式

(b) · 3 marks

$(2+\sqrt[3]{5}, 0)$ and $(0, -13)$.

令$y=0$得到$(x-2)^3=5$；令$x=0$得到$-8-5$。

M1	解立方关系
A1	横截距
A1	纵截距

Question 9

P1-C04-R09 · 5 marks

(a) · 2 marks

$(-\frac{2}{3},0),(\frac{1}{3},0),(\frac{5}{3},0)$.

输入3x分别等于原三个零点，所以每个横坐标除以3。

M1	正确水平缩放
A1	全部截距

(b) · 3 marks

$(2,0),(5,0),(9,0)$; corresponding point $(4,4)$.

所有点横坐标加4，纵坐标保持。

M1	统一横移
A1	三个截距
B1	对应点

Question 10

P1-C04-R10 · 4 marks

(a) · 2 marks

A: $x = 2, y = 3$; B: $x = -1, y = -2$.

分母为零给竖直渐近线，外加常数给水平渐近线。

B1	A的两条渐近线
B1	B的两条渐近线

(b) · 2 marks

Translation by $\begin{pmatrix} -3 \\ -5 \end{pmatrix}$.

竖直渐近线从2移到-1，水平线从3移到-2；同样移动整个曲线。

M1	正确水平位移
A1	完整向量

Question 11

P1-C04-R11 · 5 marks

(a) · 2 marks

$A \mapsto (5, 10)$; $B \mapsto (-1, -2)$.

逐坐标应用给定映射。

M1	对两点使用映射
A1	两像点

(b) · 3 marks

$y = 2f(x - 3)$; original point $(4, 4)$.

方程水平位移用x−3；逆推点坐标时横坐标减3，纵坐标除2。

M1	方程水平处理正确
A1	完整方程
A1	逆推点

Question 12

P1-C04-R12 · 6 marks

(a) · 2 marks

$(-1, 0)$ and $(0, 1)$.

分别令y=0和x=0。

B1	横截距
B1	纵截距

(b) C is reflected in the y-axis. Find

(i) · 2 marks

$y = (1 - x)^3$.

用−x替换原式中的x。

M1	正确反射替换
A1	方程

(ii) · 2 marks

$(1, 0)$ and $(0, 1)$.

横坐标反号；纵轴上的点不动。

M1	正确坐标规律
A1	两像点

Question 13

P1-C04-R13 · 5 marks

(a) · 2 marks

Maximum $(2, 10)$; minimum $(5, 1)$.

纵向平移不改变横坐标和极值类型。

M1	两纵坐标加4
A1	两极值点

(b) · 3 marks

Maximum $(5, 6)$; minimum $(2, -12)$.

纵坐标乘-2，负因子会反射并互换最大和最小类型。

M1	坐标乘负因子
M1	交换极值类型
A1	两个正确分类的点

Question 14

P1-C04-R14 · 5 marks

(a) · 2 marks

$y = \dfrac{1}{x+4} + 2.$

水平平移−4、竖直平移2。

M1	正确平移参数
A1	方程

(b) · 3 marks

$(-\frac{9}{2}, 0)$ and $(0, \frac{9}{4})$.

横截距解1/(x+4)=−2；纵截距代入0。

M1	分别设轴条件
A1	横截距
A1	纵截距

Question 15

P1-C04-R15 · 6 marks

(a) · 3 marks

Roots $-2, 0, 3$; positive-leading cubic crossing all three.

每个一次因子各给一零点，首项系数为正；原点同时为纵截距。

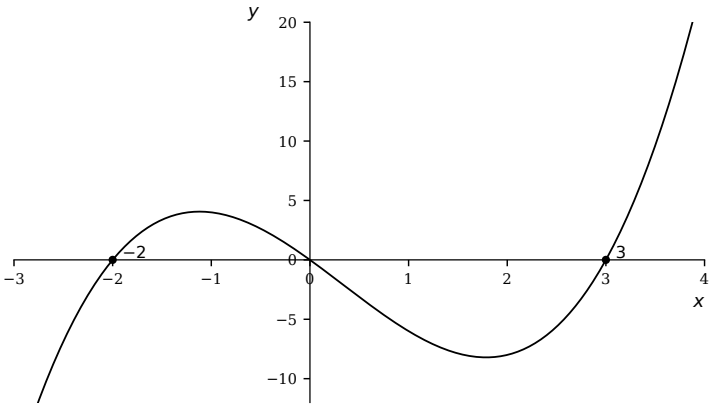
B1	三根
M1	三个穿越点
A1	正确端部走向

(b) · 3 marks

Roots $-3, -1, 2$; image $(-1, 0)$.

x+1等于原根，所有横坐标减1。

M1	正确水平映射
A1	三根
B1	原点的像



Suggested sketch

Question 16

P1-C04-R16 · 6 marks

(a) For the graph $y = f(2x - 4)$, find

(i) · 2 marks

$(1, 7)$ and $(\frac{7}{2}, -1)$.

新横坐标满足2x−4等于原横坐标，所以x=(原x+4)/2。

M1	正确输入对应关系
A1	两坐标

(ii) · 2 marks

$(6, 9)$.

新x=5时原输入$2\times5-4=6$，纵坐标不变。

M1	正确反向对应
A1	点坐标

(b) · 2 marks

Stretch parallel to the x-axis by factor $1/2$, then translate right 2.

由横坐标映射x旧/2+2判断顺序；先压缩后向右移2。

M1	正确水平因子
A1	平移及顺序

Question 17

P1-C04-R17 · 5 marks

(a) · 3 marks

$x = 0$, $y = 1$; $y > 1$ for all permitted x.

反平方项严格正，所以整条曲线在y=1之上；x=0无定义。

B1	两渐近线
M1	利用正反平方项
A1	无横截距理由

(b) · 2 marks

$(-\frac{1}{2}, 9)$ and $(\frac{1}{2}, 9)$.

$2/x^2=8$，故$x^2=1/4$，保留两个符号。

M1	正确平方关系
A1	两坐标

Question 18

P1-C04-R18 · 4 marks

(a) · 2 marks

$a = 6, b = 2.$

新横坐标a−2=4，纵坐标−3b=−6。

M1	正确两坐标关系
A1	两值

(b) · 2 marks

$(6, -3).$

回到原点P=(6,2)，仅把纵坐标减5。

M1	用原坐标而非已变换坐标
A1	像点

Question 19

P1-C04-R19 · 5 marks

(a) · 3 marks

Minimum $(4,-9)$; intercepts $(1,0),(7,0)$.

顶点由配方式直接读出；令平方等于9求截距。

B1	顶点
M1	解平方方程
A1	两截距

(b) · 2 marks

$\binom{-4}{9}$; $y=x^2$.

把顶点(4,−9)移到(0,0)，故左移4上移9，消去配方中的两平移。

M1	正确平移向量
A1	新方程

Question 20

P1-C04-R20 · 5 marks

(a) · 2 marks

$(2, 0)$ and $(0, -8)$.

令$x^3=8$求横截距；$x=0$得纵截距。

M1	解立方关系
A1	两坐标

(b) · 3 marks

D: $y = 8 - x^3$; they meet at $(2, 0)$.

反射使y变号；交点满足$x^3-8=8-x^3$，得$x^3=8$。

M1	正确反射方程
M1	联立两曲线
A1	交点

Question 21

P1-C04-R21 · 6 marks

(a) · 3 marks

$(-3,-2),(0,4),(4,0)$, joined by straight segments.

纵坐标乘2、横坐标不变，保持线段连接关系。

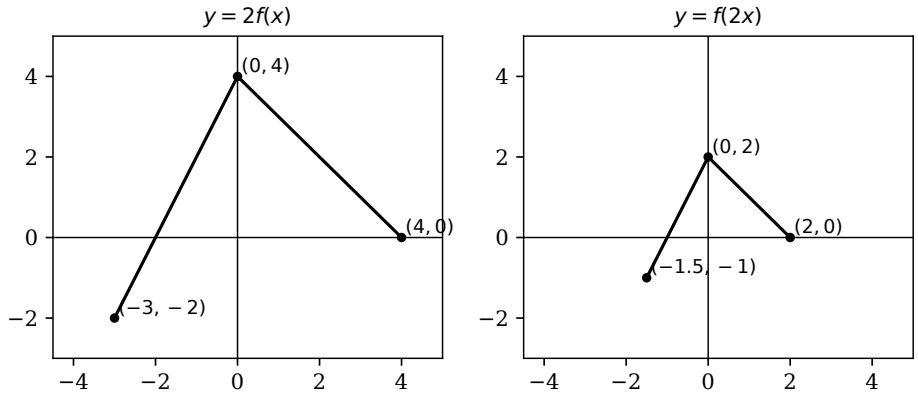
M1	正确纵向伸缩
A1	三点
A1	连线

(b) · 3 marks

$(-\frac{3}{2},-1),(0,2),(2,0)$, joined by straight segments.

横坐标除2、纵坐标不变；与(a)的纵向变换不同。

M1	正确水平伸缩
A1	三点
A1	连线



Suggested sketch

Question 22

P1-C04-R22 · 6 marks

(a) · 3 marks

$y = f(3 - x)$.

反射后为f(−x)，再右移3得f(−(x−3))。

M1	正确反射
M1	对反射结果平移
A1	最终式

(b) · 3 marks

$y = f(-x - 3)$; image $(-5, 5)$.

先右移把(2,5)变(5,5)，再反射成(−5,5)；函数为f(−x−3)。

M1	正确处理逆序函数
A1	方程
A1	点坐标

Question 23

P1-C04-R23 · 6 marks

(a) · 2 marks

$a = 6$.

代入5=a/2+2给a=6。

M1	正确代入已知点
A1	参数

(b) For this value of a, find

(i) · 2 marks

$(-2, 0)$ and $(0, -4)$.

y=0给x−1=−3；x=0给−6+2。

M1	分别设轴条件
A1	两截距

(ii) · 2 marks

$\begin{pmatrix} -1 \\ -2 \end{pmatrix}$.

把渐近线x=1,y=2分别移到坐标轴，需左1下2。

M1	使用两渐近线位移
A1	向量

Question 24

P1-C04-R24 · 5 marks

(a) · 3 marks

Touches at $(-2, 0)$, crosses at $(1, 0)$; y-intercept $(0, -4)$; rises to the right.

平方因子在−2不改变符号，简单因子在1改变符号；首项正，代0得−4。

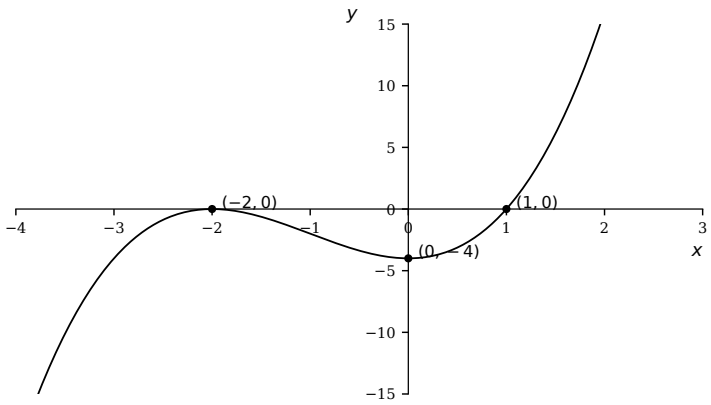
M1	两横截距及不同接触方式
B1	纵截距
A1	整体走向

(b) · 2 marks

Intercept $(0, 0)$; translation up 4.

f(0)=−4，减f(0)即加4。

M1	正确使用f(0)符号
A1	截距与方向



Suggested sketch

Question 25

P1-C04-R25 · 5 marks

(a) · 2 marks

$x=-2, y=0$; $(0, \frac{3}{4})$.

分母为零的竖线是x=−2，反平方远处趋于0；代0求截距。

B1	两渐近线
A1	纵截距

(b) · 3 marks

$y=3/(x+2)^2-3$; $(-3,0), (-1,0)$.

令$3/(x+2)^2=3$得$(x+2)^2=1$。

M1	正确新方程
M1	解平方关系
A1	两截距

Question 26

P1-C04-R26 · 4 marks

(a) · 2 marks

$(5, 4)$ and $(3, -2)$.

点映射为(x+3,y+4)。

M1	正确映射
A1	两点

(b) · 2 marks

The image has $y = 4 \neq 0$; actual intercepts satisfy $f(x - 3) = -4$.

上移会把原横截距离开x轴；新截距应令整个新函数值为0。

M1	用像点纵坐标解释
A1	正确截距方程

Question 27

P1-C04-R27 · 6 marks

(a) · 2 marks

$y = -f(x/3)$.

水平伸长3意味着函数输入除3；再使函数值变号。

M1	正确水平替换
A1	反射符号

(b) Point (a, b) on the original curve maps to $(-6, 5)$. Find

(i) · 2 marks

$a = -2$, $b = -5$.

新坐标为(3a,−b)，与给定像点比较。

M1	正确两坐标关系
A1	a与b

(ii) · 2 marks

$(0, -6)$.

用原点(−2,−5)加向量(2,−1)。

M1	使用原点坐标
A1	新像点

Question 28

P1-C04-R28 · 5 marks

(a) · 2 marks

$\binom{2}{4}$.

配方形式显示顶点由$(0,0)$到$(2,4)$。

M1	正确水平平移
A1	完整向量

(b) · 3 marks

$(2, 4)$; subtraction reduces to $4x = 8$, which has one solution.

联立$x^2=x^2-4x+8$，二次项抵消成一次方程；回代得4。

M1	联立并消去平方项
A1	交点
B1	只一个解的理由

Question 29

P1-C04-R29 · 6 marks

(a) · 3 marks

$(-5,-2),(-2,4),(2,-4)$.

输入$x+1$使横坐标减1；输出乘-2使纵坐标乘-2，按原顺序连接。

M1	水平平移正确
M1	纵向反射伸缩正确
A1	三点与连接

(b) The graph of $y=f(x)$ crosses the x-axis at two points.

(i) · 2 marks

$(-3,0),(1,0)$.

AB直线为$y=-x-3$，BC为$y=x-1$，分别令$y=0$。

M1	利用线段求零点
A1	两零点

(ii) · 1 marks

$x=-4,0$.

输入$x+1=-3$或1，解得$x=-4,0$。

B1	变换后零点

Question 30

P1-C04-R30 · 7 marks

(a) · 3 marks

Zeros $-3, 0, 2$; left end down, right end up.

三个一次因式均为单根，曲线在$-3, 0, 2$穿过x轴。首项x^3决定左下右上；无需超出题目要求求转折点。

B1	三个截距
B1	端点趋势
B1	连贯三次图像

(b) The curve is translated 2 units to the right to form D.

(i) · 2 marks

$y = (x-2)(x-4)(x+1)$.

所有x用$x-2$代入，保留因式结构。

M1	正确输入替换
A1	因式形式

(ii) · 2 marks

$2 < x < 4$.

D零点为$-1, 2, 4$；结合三次曲线符号和$x > 0$得到唯一所求区间。

M1	图像符号
A1	与正横坐标取交集

Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 5 — Straight line graphs
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第5章 · 直线
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C05-R01 · P1 · Chapter 5

$A = (-2, 5)$ and $B = (4, -1)$.

(a) Find an equation of the line AB. (3)

(b) Find an equation of the perpendicular bisector of AB. (3)

(Total for Question 1 is 6 marks)

2. P1-C05-R02 · P1 · Chapter 5

The line L has equation $3x + 2y = 12$.

(a) Find

(i) the coordinates where L meets the coordinate axes, (2)

(ii) the area of the triangle bounded by L and the coordinate axes. (2)

(b) Find an equation of the line through $(1, -2)$ perpendicular to L. (3)

(Total for Question 2 is 7 marks)

3.

Three points are $A = (1, 2)$, $B = (4, 8)$ and $C = (7, k)$.

(a) Find k if A, B, C are collinear. (2)

(b) For this value of k, find

(i) the ratio $AB : BC$, (2)

(ii) the midpoint of AC. (2)

(Total for Question 3 is 6 marks)

4.

The line L passes through $(2, 7)$ and is parallel to $4x - y = 9$.

(a) Find an equation of L. (2)

(b) Find where L intersects $x + 2y = 19$. (3)

(Total for Question 4 is 5 marks)

5.

$A = (-3, 1)$, $B = (5, 1)$ and $C = (1, 7)$ are vertices of a triangle.

(a) Find the area of triangle ABC. (2)

(b) Find an equation of the median from C to AB and an equation of the altitude from C to AB. (3)

(Total for Question 5 is 5 marks)

6.

P1-C05-R06 · P1 · Chapter 5

A straight-line calibration relates temperature T °C to a sensor reading R. Readings $R = 2$ and $R = 8$ correspond to $T = 15$ and $T = 51$, respectively.

(a) Find T in terms of R. (3)

(b) Use the model to find the reading for $T = 33$ °C, and state the temperature represented by the intercept. (3)

(Total for Question 6 is 6 marks)

$P = (a, 3)$ and $Q = (5, 11)$. The gradient of PQ is 2.

(a) Find a. (2)

(b) Find the exact length of PQ. (2)

(Total for Question 7 is 4 marks)

8. P1-C05-R08 · P1 · Chapter 5

Line L_1 is $y = mx + 4$ and line L_2 is $2y + 3x = 10$. They are perpendicular.

(a) Find m. (2)

(b) Find their point of intersection. (3)

(Total for Question 8 is 5 marks)

9. P1-C05-R09 · P1 · Chapter 5

The vertices $A = (0, 0)$, $B = (6, 2)$ and $C = (8, 8)$ are consecutive vertices of a parallelogram $ABCD$.

(a) Find the coordinates of D. (2)

(b) Find the point of intersection of its diagonals, and verify that both diagonals have this midpoint. (3)

(Total for Question 9 is 5 marks)

The line L passes through $A = (-1, 4)$ and $B = (3, -2)$.

(a) Find the equation of L in the form $ax + by = c$, where a, b, c are integers. (3)

(b) Point P lies on L and has equal coordinates. Find

(i) the coordinates of P, (2)

(ii) the exact distance AP. (2)

(Total for Question 10 is 7 marks)

A triangle has vertices $A = (2, 1)$, $B = (8, 3)$ and $C = (4, 7)$.

(a) Determine whether $\angle ACB = 90°$. Show your working. (3)

(b) Find an equation of the line through C perpendicular to AB. (3)

(Total for Question 11 is 6 marks)

$M = (3, -2)$ is the midpoint of AB, where $A = (-1, 5)$.

(a) Find B. (2)

(b) Find an equation of the perpendicular bisector of AB. (3)

(Total for Question 12 is 5 marks)

13.

P1-C05-R13 · P1 · Chapter 5

A line through $(0, 6)$ meets the positive x-axis at A. The triangle formed with the axes has area 15 square units.

(a) Find the coordinates of A. (2)

(b) Find the equation of the line and its gradient. (2)

(Total for Question 13 is 4 marks)

The line L has equation $(k+1)x + 2y = 8$, where k is real.

(a) Find k if L is parallel to $y = 3x - 1$. (2)

(b) For this value of k, find

(i) the y-intercept of L, (1)

(ii) where L meets $y = 0$. (2)

(Total for Question 14 is 5 marks)

15. P1-C05-R15 · P1 · Chapter 5

Two roads are modelled by $y = \frac{1}{2}x + 1$ and $y = -2x + 11$. Coordinates are measured in kilometres.

(a) Find the junction and show that the roads meet at right angles. (4)

(b) A depot is at $(0, 1)$ on the first road. Find the exact distance from the depot to the junction. (2)

(Total for Question 15 is 6 marks)

16. P1-C05-R16 · P1 · Chapter 5

$A = (1, -3)$ and $B = (7, 5)$. Point P lies on AB with $AP : PB = 1 : 2$.

(a) Find the coordinates of P. (2)

(b) Find an equation of the line through P parallel to $y = 2x + 7$. (3)

(Total for Question 16 is 5 marks)

17.

A point $P = (t, 2t + 1)$ is equidistant from $A = (0, 0)$ and $B = (4, 0)$.

(a) Find t. (3)

(b) Find P and its exact distance from A. (3)

(Total for Question 17 is 6 marks)

The line L passes through $(2, 3)$ and has x-intercept 6.

(a) Find an equation of L. (3)

(b) Find the y-intercept and the area of the triangle enclosed by L and the axes. (3)

(Total for Question 18 is 6 marks)

19.

The vertices of a rectangle are $A = (1, 1)$, $B = (5, 3)$ and $C = (2, 9)$ in consecutive order.

(a) Verify that AB and BC are perpendicular. (3)

(b) Find the fourth vertex D and the exact area of the rectangle. (3)

(Total for Question 19 is 6 marks)

Lines L_1 and L_2 are $2x - y = 3$ and $x + 2y = 4$.

(a) Find their intersection P. (2)

(b) Find the equation of the line through P and the midpoint of $A = (-2, 4)$ and $B = (6, 2)$. (3)

(Total for Question 20 is 5 marks)

A straight line passes through $(a, 0)$ and $(0, 2a)$, where $a > 0$.

(a) Find its gradient and write its equation in terms of a. (2)

(b) The line also passes through $(3, 4)$. Find a and the area of the triangle it forms with the axes. (4)

(Total for Question 21 is 6 marks)

$A = (-4, 2)$ and $B = (2, 6)$. Point $C = (5, k)$ lies on the line through B perpendicular to AB.

(a) Find k. (3)

(b) Find the exact length BC. (2)

(Total for Question 22 is 5 marks)

Point $P = (4, 7)$ is reflected in the horizontal line $y = 2$ to form Q.

(a) Find Q and an equation of the line PQ. (3)

(b) Find the point where the line through Q parallel to $y = 3x + 1$ meets the x-axis. (3)

(Total for Question 23 is 6 marks)

A machine costs £4200 when new. Its value is modelled by a straight line and is £1800 after 6 years.

(a) Find the model $V = at + b$, where t is age in years. (3)

(b) Find the age at which the model predicts zero value, and explain why the model should not be used for larger ages. (3)

(Total for Question 24 is 6 marks)

Line L is $y = -\frac{2}{3}x + 4$. Point $P = (3, 1)$ does not lie on L.

(a) Find an equation of the perpendicular from P to L. (2)

(b) Find the foot F of this perpendicular and hence the exact shortest distance from P to L. (4)

(Total for Question 25 is 6 marks)

The line joining $A = (1, 5)$ and $B = (7, 1)$ intersects the line $y = x$ at P.

(a) Find P. (3)

(b) Find the ratio $AP : PB$. (3)

(Total for Question 26 is 6 marks)

The line L_k has equation $y = kx + 2 - k$, where k is real.

(a) Show that every line L_k passes through the same point, and find this point. (3)

(b) Find the member of this family perpendicular to $y = 4x - 7$. (3)

(Total for Question 27 is 6 marks)

A triangle has $A = (0, 4)$, $B = (8, 4)$ and $C = (3, -2)$.

(a) Find equations of the median from C and the perpendicular bisector of AB. (4)

(b) Find where the median meets the x-axis. (2)

(Total for Question 28 is 6 marks)

29.

Points $A(-2, 5)$ and $B(6, 1)$ are the ends of a straight path. A lamp is to be placed at a point P on the line $y = x + 1$ so that $PA = PB$.

(a) Find an equation for the perpendicular bisector of AB. (2)

(b) Find

(i) the coordinates of P. (2)

(ii) the exact common distance $PA = PB$. (2)

(Total for Question 29 is 6 marks)

A straight-line model relates temperature $T\,^\circ\mathrm{C}$ to resistance R ohms. The line passes through $(R, T) = (12, 5)$ and $(20, 45)$.

(a) Find the model in the form $T = aR + b$. (2)

(b) A second sensor uses the model $T = 4R - 30$.

(i) Find the resistance at which the two sensors give the same temperature. (2)

(ii) The first model was calibrated only for $12 \le R \le 20$. State why the agreement does not establish accuracy at the resistance found in (b)(i). (1)

(Total for Question 30 is 5 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C05-R01 · 6 marks

(a) · 3 marks

$y = -x + 3$.

斜率(−1−5)/(4+2)=−1；代一点得截距3。

M1	正确斜率
M1	用点求截距
A1	直线方程

(b) · 3 marks

$y = x + 1$.

中点(1,2)，垂直斜率1；点斜式y−2=x−1。

M1	中点
M1	垂直斜率
A1	方程

Question 2

P1-C05-R02 · 7 marks

(a) Find

(i) · 2 marks

$(4, 0)$ and $(0, 6)$.

分别令y=0与x=0。

B1	横截距
B1	纵截距

(ii) · 2 marks

12 square units.

截距形成直角三角形，面积为4×6/2。

M1	正确底高面积式
A1	面积

(b) · 3 marks

$y = \frac{2}{3}x - \frac{8}{3}$.

L斜率−3/2，故垂线斜率2/3，过点求截距。

M1	负倒数斜率
M1	点斜式
A1	最终方程

Question 3

P1-C05-R03 · 6 marks

(a) · 2 marks

$k = 14$.

AB斜率2；BC斜率(k−8)/3必须等于2。

M1	用斜率相等
A1	k值

(b) For this value of k, find

(i) · 2 marks

$1 : 1$.

两段横增量都3且在同一直线上，或两长度均$3\sqrt{5}$。

M1	比较坐标增量或长度
A1	比值

(ii) · 2 marks

$(4, 8)$.

两坐标平均，恰好为B。

M1	正确平均坐标
A1	中点

Question 4

P1-C05-R04 · 5 marks

(a) · 2 marks

$y = 4x - 1$.

给定线斜率4，平行线同斜率，代(2,7)得截距−1。

M1	正确平行斜率
A1	方程

(b) · 3 marks

$(\frac{7}{3}, \frac{25}{3})$.

代y=4x−1得9x−2=19，x=7/3；回代得25/3。

M1	正确联立消元
A1	x
A1	y

Question 5

P1-C05-R05 · 5 marks

(a) · 2 marks

24 square units.

AB水平，底长8；C到AB竖直距离6，面积24。

M1	正确底与垂直高
A1	面积

(b) · 3 marks

Both are $x = 1$.

AB中点(1,1)，与C有相同横坐标；AB水平，其垂线也为过C的竖直线。

M1	求AB中点
A1	中线方程
A1	高线方程及方向

Question 6

P1-C05-R06 · 6 marks

(a) · 3 marks

$T = 6R + 3$.

斜率(51−15)/(8−2)=6，代入一对数值求截距3。

M1	用两读数求斜率
M1	求截距
A1	模型

(b) · 3 marks

$R = 5$; intercept represents 3 °C at $R = 0$.

33=6R+3得R=5；截距指读数零时的模型温度。

M1	反求读数
A1	R值
B1	截距解释

Question 7

P1-C05-R07 · 4 marks

(a) · 2 marks

$a = 1$.

8/(5−a)=2给5−a=4。

M1	正确斜率关系
A1	a

(b) · 2 marks

$4\sqrt{5}$.

坐标差4与8，勾股长度$\sqrt{80}$。

M1	正确平方距离
A1	最简根式

Question 8

P1-C05-R08 · 5 marks

(a) · 2 marks

$m = \frac{2}{3}$.

L2斜率−3/2，垂直斜率负倒数为2/3。

M1	正确L2斜率
A1	m

(b) · 3 marks

$(\frac{6}{13}, \frac{56}{13})$.

2x/3+4=5−3x/2，乘6得13x=6；回代。

M1	正确联立
M1	解精确横坐标
A1	交点

Question 9

P1-C05-R09 · 5 marks

(a) · 2 marks

$(2, 6)$.

平行四边形对边相同位移，D=A+(C−B)。

M1	使用相等位移
A1	D坐标

(b) · 3 marks

$(4, 4)$; midpoint of AC and of BD both $(4, 4)$.

AC平均为(4,4)；BD平均为((6+2)/2,(2+6)/2)。

M1	求一条对角线中点
M1	独立求另一中点
A1	一致结论

Question 10

P1-C05-R10 · 7 marks

(a) · 3 marks

$3x + 2y = 5$.

斜率$-6/4=-3/2$；用点斜式消去分母。

M1	正确斜率
M1	使用已知点
A1	整数系数式

(b) Point P lies on L and has equal coordinates. Find

(i) · 2 marks

$(1, 1)$.

令y=x代入3x+2x=5。

M1	联立y=x
A1	P

(ii) · 2 marks

$\sqrt{13}$.

A到P坐标差2、-3，长度$\sqrt{13}$。

M1	两点距离
A1	根式

Question 11

P1-C05-R11 · 6 marks

(a) · 3 marks

It is not right-angled at C: $m_{CA} = 3$ and $m_{CB} = -1$, with product $-3 \neq -1$.

分别计算两条过C边斜率，乘积不为−1，所以不垂直。

M1	正确两个斜率
M1	应用垂直条件
A1	明确否定结论

(b) · 3 marks

$y = -3x + 19$.

AB斜率2/6=1/3，故所求斜率−3，过(4,7)。

M1	AB斜率
M1	垂直斜率及过点
A1	方程

Question 12

P1-C05-R12 · 5 marks

(a) · 2 marks

$B = (7, -9)$.

由(A+B)/2=M，得B=2M−A。

M1	中点方程
A1	B

(b) · 3 marks

$y + 2 = \frac{4}{7}(x - 3)$.

AB斜率−14/8=−7/4；垂直平分线过M且斜率4/7。

M1	AB斜率
M1	垂线斜率并用中点
A1	方程

Question 13

P1-C05-R13 · 4 marks

(a) · 2 marks

$A = (5, 0)$.

面积$(1/2)\times$OA$\times 6$=15给OA=5；正x轴确定符号。

M1	面积关系
A1	A坐标

(b) · 2 marks

$y = -\frac{6}{5}x + 6$; gradient $-6/5$.

连接截距$(0,6)$和$(5,0)$，斜率$-6/5$。

M1	正确斜率
A1	完整方程

Question 14

P1-C05-R14 · 5 marks

(a) · 2 marks

$k = -7$.

L斜率−(k+1)/2=3，求k。

M1	正确参数斜率
A1	k

(b) For this value of k, find

(i) · 1 marks

$(0, 4)$.

令x=0，2y=8。

B1	截距

(ii) · 2 marks

$(-\frac{4}{3}, 0)$.

L为−6x+2y=8，令y=0。

M1	正确代入参数及y=0
A1	横截距

Question 15

P1-C05-R15 · 6 marks

(a) · 4 marks

Junction $(4,3)$; gradient product $(1/2)(-2)=-1$.

联立得2.5x=10；两个非竖直斜率乘积−1意味着垂直。

M1	联立
A1	交点
M1	梯度乘积
A1	垂直结论

(b) · 2 marks

$2\sqrt{5}$ km.

坐标差4和2，距离$\sqrt{20}$。

M1	正确两点距离
A1	精确距离与单位

Question 16

P1-C05-R16 · 5 marks

(a) · 2 marks

$(3, -\frac{1}{3})$.

从A走全位移的1/3，即(1,−3)+(2,8/3)。

M1	正确使用三分之一位移
A1	两坐标

(b) · 3 marks

$y = 2x - \frac{19}{3}$.

斜率2，代P求截距−1/3−6。

M1	采用相同斜率
M1	使用P
A1	方程

Question 17

P1-C05-R17 · 6 marks

(a) · 3 marks

$t = 2$.

PA²=t²+(2t+1)²；PB²=(t−4)²+(2t+1)²；相等后得t²=(t−4)²。

M1	建立等距离平方
M1	正确消去共同项
A1	t

(b) · 3 marks

$P = (2, 5)$; distance $\sqrt{29}$.

把t=2代回坐标，再用勾股求长度。

M1	回代求点
A1	点坐标
A1	距离

Question 18

P1-C05-R18 · 6 marks

(a) · 3 marks

$y = -\frac{3}{4}x + \frac{9}{2}$.

另一点为(6,0)，斜率−3/4，代入求截距9/2。

M1	正确识别截距点
M1	求斜率
A1	方程

(b) · 3 marks

$(0, \frac{9}{2})$; area $\frac{27}{2}$ square units.

截距9/2，三角形面积6×(9/2)/2。

B1	纵截距
M1	底高面积式
A1	面积

Question 19

P1-C05-R19 · 6 marks

(a) · 3 marks

$m_{AB} = 1/2$, $m_{BC} = -2$, product -1.

用两段的纵差除横差，乘积检验。

M1	求两个斜率
M1	应用乘积条件
A1	垂直结论

(b) · 3 marks

$D = (-2, 7)$; area 30 square units.

D=A+C−B；AB=$\sqrt{20}$，BC=$\sqrt{45}$，矩形面积=$\sqrt{900}$=30。

M1	正确第四点
M1	求两垂直边长度
A1	面积

Question 20

P1-C05-R20 · 5 marks

(a) · 2 marks

$P = (2, 1)$.

y=2x−3代入x+2y=4给5x=10。

M1	正确消元
A1	两坐标

(b) · 3 marks

$x = 2$.

AB中点为(2,3)，与P同横坐标，所以直线竖直，不能用有限斜率表达。

M1	求中点
M1	识别竖直线
A1	方程

Question 21

P1-C05-R21 · 6 marks

(a) · 2 marks

Gradient -2; $y = -2x + 2a$.

斜率$(2a-0)/(0-a)=-2$，纵截距$2a$。

M1	处理参数斜率
A1	方程

(b) · 4 marks

$a = 5$; area 25 square units.

$4=-6+2a$给$a=5$；底a、高$2a$，面积$a^2=25$。

M1	代点求a
A1	a值
M1	底高关系
A1	面积

Question 22

P1-C05-R22 · 5 marks

(a) · 3 marks

$k = \frac{3}{2}$.

AB斜率2/3；BC斜率−3/2，故(k−6)/(5−2)=−3/2。

M1	求AB斜率及垂线斜率
M1	正确参数方程
A1	k

(b) · 2 marks

$\frac{3\sqrt{13}}{2}$.

坐标差3和−9/2，长度√(9+81/4)=√117/2。

M1	正确平方距离
A1	最简根式

Question 23

P1-C05-R23 · 6 marks

(a) · 3 marks

$Q=(4,-3)$; $x=4$.

中间高度2=(7+yQ)/2给yQ=−3，横坐标不变；连接线竖直。

M1	使用反射中点关系
A1	Q
B1	直线

(b) · 3 marks

$(5,0)$.

过Q斜率3，方程y=3x−15；令y=0得x=5。

M1	正确平行直线方程
M1	令y=0
A1	交点

Question 24

P1-C05-R24 · 6 marks

(a) · 3 marks

$V = 4200 - 400t$.

初值b=4200，年变化(1800−4200)/6=−400。

M1	斜率
B1	初值
A1	模型

(b) · 3 marks

10.5 years; it would predict negative monetary values.

令V=0解t=10.5；更大t使右式负，不能代表该机器通常意义上的价值。

M1	解零值年龄
A1	年龄
B1	具体局限解释

Question 25

P1-C05-R25 · 6 marks

(a) · 2 marks

$y = \frac{3}{2}x - \frac{7}{2}$.

垂线斜率3/2，代入P求截距。

M1	正确垂直斜率
A1	方程

(b) · 4 marks

$F = (\frac{45}{13}, \frac{22}{13})$; distance $\frac{3\sqrt{13}}{13}$.

联立两线得x=45/13,y=22/13；PF坐标差6/13和9/13，长度√117/13。

M1	联立求垂足
A1	垂足
M1	两点距离
A1	精确最短距离

Question 26

P1-C05-R26 · 6 marks

(a) · 3 marks

$P = (\frac{17}{5}, \frac{17}{5})$.

AB为2x+3y=17；令y=x，得5x=17。

M1	求AB方程
M1	与y=x联立
A1	P

(b) · 3 marks

$2:3$.

P横坐标距A为12/5，距B为18/5；共线段长度之比等于相应横差之比。

M1	比较同线段的横差
M1	正确两个差
A1	最简比

Question 27

P1-C05-R27 · 6 marks

(a) · 3 marks

All pass through $(1, 2)$, since $y = k + 2 - k = 2$ when $x = 1$.

写成y=k(x−1)+2，x=1时参数项消失，点与k无关。

M1	重排或寻找参数系数为零
A1	固定点
B1	独立于k的说明

(b) · 3 marks

$k = -\frac{1}{4}$, so $y = -\frac{1}{4}x + \frac{9}{4}$.

垂直要求k×4=−1；代回截距2−k。

M1	正确垂直参数
A1	k
A1	方程

Question 28

P1-C05-R28 · 6 marks

(a) · 4 marks

Median: $y = 6x - 20$; perpendicular bisector: $x = 4$.

AB中点M=(4,4)；CM斜率6。AB水平，垂直平分线过M且竖直。

M1	中点
M1	中线斜率及方程
A1	中线式
B1	垂直平分线

(b) · 2 marks

$(\frac{10}{3}, 0)$.

令6x−20=0。

M1	使用中线方程并令y=0
A1	交点

Question 29

P1-C05-R29 · 6 marks

(a) · 2 marks

$y = 2x - 1$.

AB中点为$(2, 3)$，斜率$-1/2$；垂直平分线斜率2，故$y - 3 = 2(x - 2)$。

M1	中点与垂直斜率
A1	方程

(b) Find

(i) · 2 marks

$(2, 3)$.

联立$y = 2x - 1$和$y = x + 1$，得到$x = 2, y = 3$。

M1	联立
A1	坐标

(ii) · 2 marks

$2\sqrt{5}$.

$PA = \sqrt{(2+2)^2 + (3-5)^2} = \sqrt{20} = 2\sqrt{5}$。

M1	距离公式
A1	最简根式

Question 30

P1-C05-R30 · 5 marks

(a) · 2 marks

$T = 5R - 55$.

斜率$a = (45 - 5)/(20 - 12) = 5$，代入$(12, 5)$得$b = -55$。

M1	计算斜率
A1	完整模型

(b) A second sensor uses the model $T = 4R - 30$.

(i) · 2 marks

$R = 25$ ohms; $T = 70\,^{\circ}\mathrm{C}$.

联立$5R - 55 = 4R - 30$，得$R = 25$，代回得$T = 70$。

M1	联立两个模型
A1	共同读数

(ii) · 1 marks

The common reading is outside the calibration interval.

$25 > 20$，属于外推；模型相交仅表示预测相同，不能证明超出校准范围的准确性。

B1	指明超出校准区间

Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 6 — Trigonometric ratios
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第6章 · 三角比
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

Triangle ABC has $AB = 9$ cm, $AC = 7$ cm and $\angle BAC = 60°$.

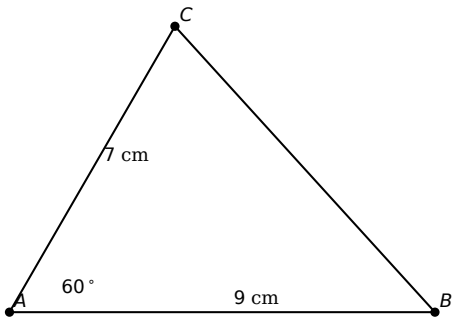


Figure 1

(a) Find the exact length of BC. (3)

(b) Find the exact area of the triangle. (2)

(Total for Question 1 is 5 marks)

2. P1-C06-R02 · P1 · Chapter 6

Triangle PQR has $PQ = 12$ m, $\angle PQR = 41^\circ$ and $\angle PRQ = 67^\circ$.

(a) Find $\angle QPR$. (2)

(b) Giving lengths to 3 significant figures, find

(i) PR, (2)

(ii) the perimeter of the triangle. (3)

(Total for Question 2 is 7 marks)

3.

In triangle ABC, $AB = 10$ cm, $BC = 8$ cm and $\angle BAC = 35°$.

(a) Use the sine rule to show that two values of $\angle ACB$ are possible. Give both values to 1 decimal place. (3)

(b) For the triangle in which $\angle ACB$ is obtuse, find AC to 3 significant figures. (3)

(Total for Question 3 is 6 marks)

4.

A triangular panel has sides 6 cm, 9 cm and 11 cm.

(a) Find its largest angle to 1 decimal place. (3)

(b) Find its area to 3 significant figures. (2)

(Total for Question 4 is 5 marks)

5.

Two paths leave a station S. A point A lies 120 m due east of S, and a point B lies 150 m from S on a bearing of 040°.

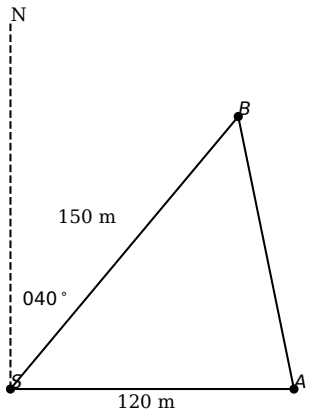


Figure 5

(a) Find $\angle ASB$. (2)

(b) Find AB and the area of triangle SAB, both to 3 significant figures. (4)

(Total for Question 5 is 6 marks)

A vertical tower stands on horizontal ground. From point A, the angle of elevation to its top is $28°$. Point B is 30 m closer to the tower on the same straight line, and its angle of elevation is $43°$.

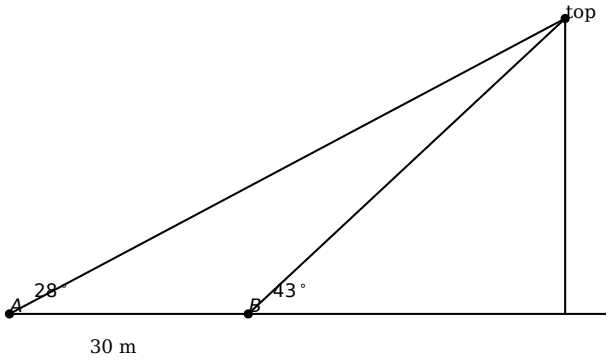


Figure 6

(a) Form an equation in d, the horizontal distance from B to the foot of the tower. (2)

(b) Find the height of the tower to 3 significant figures. (3)

(Total for Question 6 is 5 marks)

7.

In a triangle, two sides have lengths 5 cm and 8 cm and the included angle is θ. Its area is $10\sqrt{3}\ \text{cm}^2$.

(a) Find all possible values of θ in $0^\circ < \theta < 180^\circ$. (3)

(b) Given that the third side is longer than 8 cm, find its exact length. (3)

(Total for Question 7 is 6 marks)

8.

The graph $y = 3\sin x$ is considered for $0^\circ \leq x \leq 360^\circ$.

(a) Sketch the graph, marking all intercepts and all maximum and minimum points. (3)

(b) Find the values of x for which $y = \frac{3}{2}$. (2)

(Total for Question 8 is 5 marks)

A triangular support has sides $AB = AC = 13$ cm and base $BC = 10$ cm.

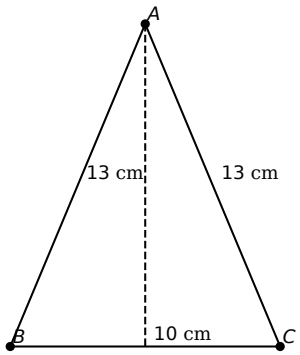


Figure 9

(a) Find the perpendicular height from A to BC and hence the area. (3)

(b) Find $\angle BAC$ to 1 decimal place. (3)

(Total for Question 9 is 6 marks)

Point B is 80 m due north of A. Point C is on a bearing of 065° from A and a bearing of 120° from B.

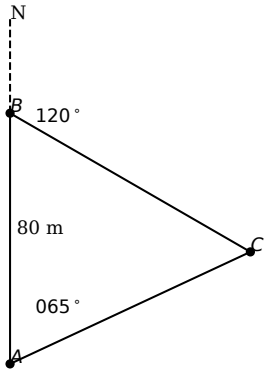


Figure 10

(a) Find all three angles in triangle ABC. (3)

(b) Find AC to 3 significant figures. (2)

(Total for Question 10 is 5 marks)

The curve is $y = 2\cos x - 1$, for $0^\circ \leq x \leq 360^\circ$.

(a) State the maximum and minimum values of y, and all values of x at which they occur. (3)

(b) Find the coordinates where the curve crosses the x-axis. (2)

(Total for Question 11 is 5 marks)

In triangle ABC, $AB = 15$ cm, $AC = 10$ cm and $\angle ABC = 30°$.

(a) Find the two possible values of $\angle ACB$. (3)

(b) For the triangle in which $\angle ACB$ is acute, find

(i) $\angle BAC$ to 1 decimal place, (2)

(ii) the area to 3 significant figures. (2)

(Total for Question 12 is 7 marks)

A ladder of length 7 m leans against a vertical wall on horizontal ground. Its angle with the ground is 64°.

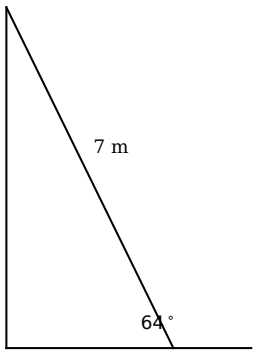


Figure 13

(a) Find the height reached on the wall to 3 significant figures. (2)

(b) The top is lowered by 1 m while the ladder remains against the wall. Find the new angle with the ground to 1 decimal place. (3)

(Total for Question 13 is 5 marks)

A parallelogram has adjacent sides 9 cm and 14 cm and an included acute angle of 38°.

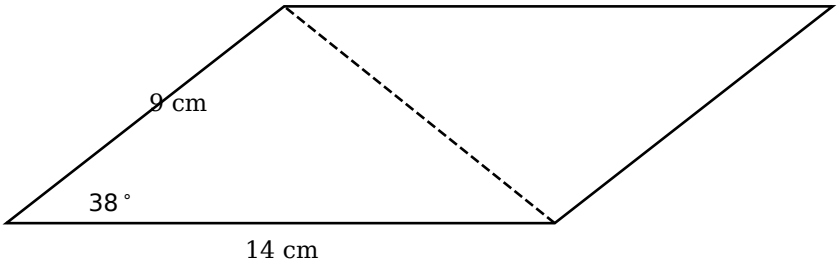


Figure 14

(a) Find its area to 3 significant figures. (2)

(b) Find the length of its shorter diagonal to 3 significant figures. (3)

(Total for Question 14 is 5 marks)

15.

Triangle ABC has $AB = 4$ cm, $AC = 7$ cm and $BC = \sqrt{37}$ cm.

(a) Find the exact value of $\cos A$ and hence find A. (3)

(b) Find the exact area of the triangle. (2)

(Total for Question 15 is 5 marks)

16.

For $0^\circ < x < 180^\circ$, $\sin x = \frac{5}{13}$ and x is obtuse.

(a) Find the exact value of $\cos x$. (2)

(b) Find the exact value of $\tan x$. (2)

(Total for Question 16 is 4 marks)

17.

The graph $y = \tan x$ is considered for $-90^\circ < x < 270^\circ$.

(a) Sketch the graph, showing all vertical asymptotes and axis intersections in this interval. (3)

(b) Find all x in the stated interval for which $y = -1$. (2)

(Total for Question 17 is 5 marks)

A triangular field has sides 120 m and 160 m enclosing an angle of $47°$. A straight fence joins the free ends of those sides.

(a) Find the length of the fence to the nearest metre. (3)

(b) A path runs along the 160 m side and has perpendicular width 2 m inside the field. For this calculation only, its area is modelled as a rectangle of dimensions 160 m by 2 m. Find

(i) the modelled path area, (1)

(ii) the percentage of the field area occupied by this modelled path, to 1 decimal place. (3)

(Total for Question 18 is 7 marks)

Two ships leave a harbour at the same time. After 2 hours, ship A has travelled 30 km on a bearing of 020° and ship B has travelled 40 km on a bearing of 110°.

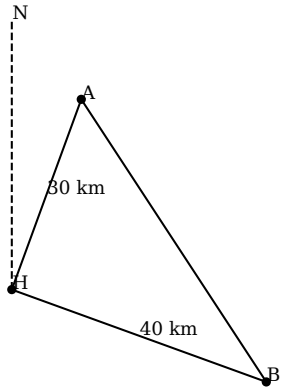


Figure 19

(a) Find their separation at this time. (3)

(b) Find the angle between the direction from A to the harbour and the direction from A to B, to 1 decimal place. (3)

(Total for Question 19 is 6 marks)

A triangle has sides x cm, $2x$ cm and $\sqrt{19}$ cm. The angle between the sides of lengths x and $2x$ is 120°, and $x > 0$.

(a) Find the exact value of x. (3)

(b) Find the exact area of the triangle. (2)

(Total for Question 20 is 5 marks)

21. P1-C06-R21 · P1 · Chapter 6

The curve is $y = 4\sin(2x)$, where angles are in degrees and $0^\circ \leq x \leq 180^\circ$.

(a) State its period and the coordinates of its maximum and minimum points on this interval. (3)

(b) Find all zeros on the interval. (2)

(Total for Question 21 is 5 marks)

Triangle ABC has area 24 cm^2, $AB = 8$ cm and $AC = 10$ cm.

(a) Find the two possible values of $\angle BAC$ to 1 decimal place. (3)

(b) Find both possible exact lengths of BC. (3)

(Total for Question 22 is 6 marks)

A mast is held by two wires attached to the same top point. The wires are fixed to the horizontal ground at opposite sides of the mast, at distances 6 m and 10 m from its foot. The shorter wire makes angle 58° with the ground.

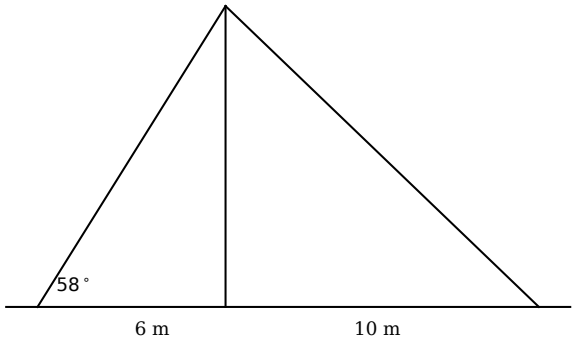


Figure 23

(a) Find the mast height to 3 significant figures. (2)

(b) For the longer wire, find

(i) its angle with the ground to 1 decimal place, (2)

(ii) its length to 3 significant figures. (2)

(Total for Question 23 is 6 marks)

For $0^\circ \leq x \leq 360^\circ$, $\cos x = -\frac{1}{2}$.

(a) Find all possible values of x. (2)

(b) For each value of x found in part (a), state

(i) the exact value of $\sin x$, (2)

(ii) the exact value of $\tan x$. (2)

(Total for Question 24 is 6 marks)

A triangle has sides 7 cm, 8 cm and 9 cm. Let θ be the angle opposite the 8 cm side.

(a) Show that $\cos\theta = \frac{11}{21}$. (2)

(b) Hence find the exact area of the triangle. (3)

(Total for Question 25 is 5 marks)

26.

The height of a point on a wheel is modelled by $h = 5 - 4\cos\theta$ metres, for $0^\circ \leq \theta \leq 360^\circ$.

(a) Find the minimum and maximum heights and state the angle at which each occurs. (3)

(b) Find all angles at which the height is 7 m. (2)

(Total for Question 26 is 5 marks)

$ABCD$ is a quadrilateral with diagonal $AC = 12$ cm. In triangle ABC, $AB = 7$ cm and $BC = 9$ cm. In triangle ACD, $AD = CD = 10$ cm. Points B and D lie on opposite sides of AC.

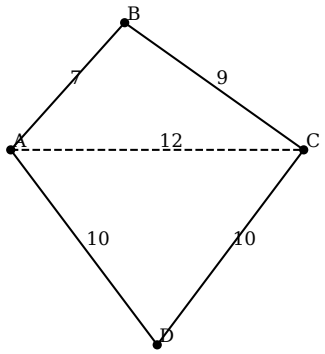


Figure 27

(a) Find $\angle ABC$ to 1 decimal place. (2)

(b) Find the area of the quadrilateral to 3 significant figures. (4)

(Total for Question 27 is 6 marks)

A hillside rises at 18° to the horizontal. A straight path on the hillside is 90 m long.

(a) Find the vertical rise and horizontal distance, each to 3 significant figures. (3)

(b) An alternative route to a platform at the same height consists of a horizontal section of length 100 m followed by a vertical climb. Find its total length to 3 significant figures. (2)

(Total for Question 28 is 5 marks)

Figure 29 shows one complete cycle of $y = a + b\cos x$, for $0^\circ \leq x \leq 360^\circ$, where $b > 0$. The maximum point is $(0, 7)$ and the minimum point is $(180, -1)$.

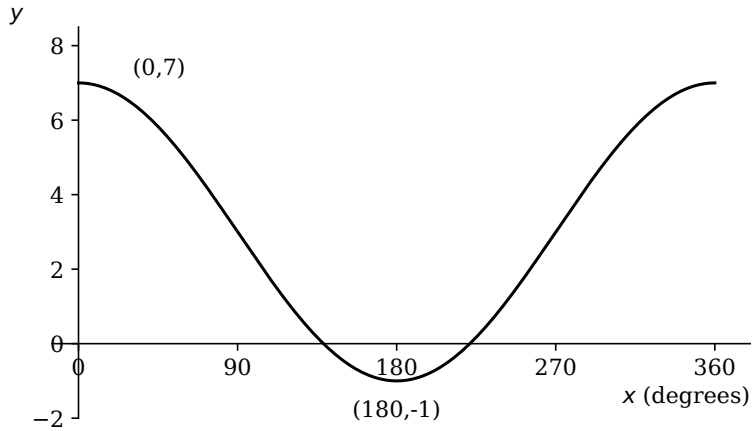


Figure 29

(a) Find a and b. (2)

(b) A horizontal line $y = 5$ cuts this curve at P and Q.

(i) Find the coordinates of P and Q. (2)

(ii) Hence solve $a + b\cos(2t) = 5$ for $0^\circ \leq t \leq 180^\circ$. (1)

(Total for Question 29 is 5 marks)

Triangle ABC has $AB = 8$ cm, $AC = 5$ cm and $\angle BAC = 120°$. Point D lies on BC and AD bisects $\angle BAC$.

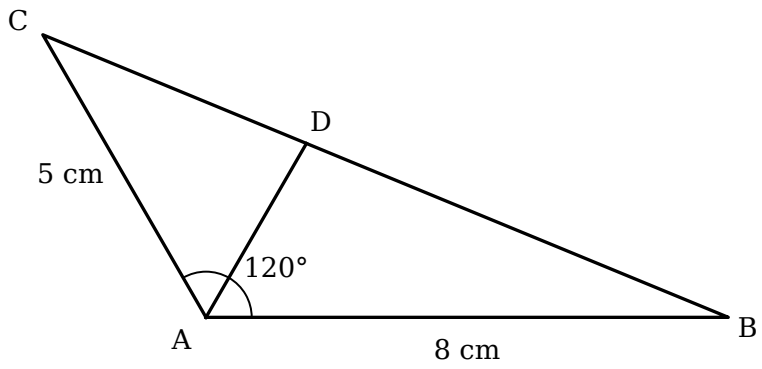


Figure 30

(a) Find the exact length of BC. (2)

(b) Let $AD = x$ cm. Use the areas of triangles ABD and ACD to find

(i) the exact value of x. (3)

(ii) the ratio $BD : DC$. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C06-R01 · 5 marks

(a) · 3 marks

$\sqrt{67}$ cm.

余弦定理$BC^2=81+49-2\times9\times7\times1/2=67$。

M1	正确余弦定理
M1	使用60度精确余弦
A1	最简根式

(b) · 2 marks

$\frac{63\sqrt{3}}{4}$ cm^2.

面积$1/2\times9\times7\times\sin60°$，代入$\sqrt{3}/2$。

M1	两边夹角面积式
A1	精确面积

Question 2

P1-C06-R02 · 7 marks

(a) · 2 marks

72°.

三角形内角和180°。

M1	内角和
A1	72度

(b) Giving lengths to 3 significant figures, find

(i) · 2 marks

$12\sin 41^\circ / \sin 67^\circ \approx 8.55$ m.

已知边PQ对67°，PR对41°，正弦定理配对应边角。

M1	正弦定理对应
A1	精度及单位

(ii) · 3 marks

$12 + 12(\sin 41^\circ + \sin 72^\circ)/\sin 67^\circ \approx 33.0$ m.

另边QR对72°，用未舍入的三边相加。

M1	正确求第三边
M1	用未舍入值求周长
A1	周长

Question 3

P1-C06-R03 · 6 marks

(a) · 3 marks

45.8° and 134.2°.

sinC=10sin35°/8≈0.7169705，两角为arcsin及180°−arcsin，且与35°相加均小于180°。

M1	正确正弦关系
M1	找补角
A1	两角及合法检查

(b) · 3 marks

$AC \approx 2.61$ cm.

用未舍入的钝角C得B=180−35−C，再由AC=8sinB/sin35°。

M1	选钝角并求余角
M1	再用正弦定理
A1	长度

Question 4

P1-C06-R04 · 5 marks

(a) · 3 marks

92.1°.

最大角对11；$\cos\theta=(36+81-121)/(2\times6\times9)=-1/27$。

M1	对应最长边与最大角
M1	余弦定理
A1	角度

(b) · 2 marks

$27.0\ \text{cm}^2$.

面积$(1/2)\times6\times9\times\sin\theta$，使用未舍入$\theta$。

M1	正确两边夹角面积
A1	精度及单位

Question 5

P1-C06-R05 · 6 marks

(a) · 2 marks

50°.

正东方位角90°，与040°的夹角50°。

M1	从北向顺时针理解方位角
A1	夹角

(b) · 4 marks

$AB \approx 117$ m; area ≈ 6890 m^2.

用余弦定理求两端距离；面积$1/2 \times 120 \times 150 \times \sin 50^\circ$。

M1	余弦定理
A1	距离
M1	面积式
A1	面积

Question 6

P1-C06-R06 · 5 marks

(a) · 2 marks

$d\tan 43^\circ = (d+30)\tan 28^\circ$.

同一塔高分别为d tan43°与(d+30)tan28°；测量点与塔脚共线，地面水平。

M1	写两种塔高
A1	距离差使用正确

(b) · 3 marks

$h = 30\tan 28^\circ \tan 43^\circ / (\tan 43^\circ - \tan 28^\circ) \approx 37.1$ m.

先解d再乘tan43°，保留中间精度。

M1	解水平距离
M1	求塔高
A1	精度与单位

Question 7

P1-C06-R07 · 6 marks

(a) · 3 marks

$60^\circ, 120^\circ$.

面积$20\sin\theta = 10\sqrt{3}$，所以$\sin\theta = \sqrt{3}/2$，在三角形范围有两个角。

M1	面积方程
M1	两个正弦解
A1	完整角度

(b) · 3 marks

$\sqrt{129}$ cm.

$\theta = 60^\circ$时第三边$^2 = 49$，$\theta = 120^\circ$时为129；长度大于8只允许后者。

M1	分别检验两夹角或比较余弦
M1	用长度条件选支
A1	根式长度

Question 8

P1-C06-R08 · 5 marks

(a) · 3 marks

Intercepts $(0,0),(180,0),(360,0)$; maximum $(90,3)$; minimum $(270,-3)$.

正弦一个完整周期，幅值3，顺序从零上升。

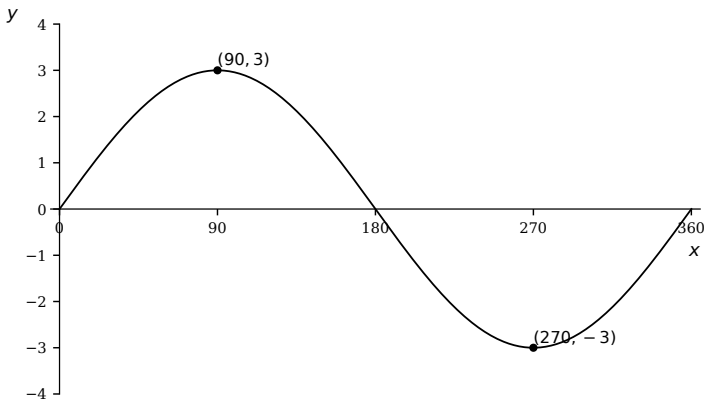
B1	三个零点
B1	两极值
A1	正确正弦形状

(b) · 2 marks

$x = 30^\circ, 150^\circ$.

sinx=1/2，第一和第二象限各一解。

M1	正确化简方程
A1	全部解



Suggested sketch

Question 9

P1-C06-R09 · 6 marks

(a) · 3 marks

Height 12 cm; area 60 cm^2.

等腰三角形高平分底边，半底5，勾股 $\sqrt{(169-25)}=12$。

M1	平分底边并勾股
A1	高
A1	面积

(b) · 3 marks

45.2°.

半顶角满足$\sin(\theta/2)=5/13$，求arcsin再乘2。

M1	正确半角直角三角关系
M1	恢复完整顶角
A1	角度

Question 10

P1-C06-R10 · 5 marks

(a) · 3 marks

$A = 65^\circ$, $B = 60^\circ$, $C = 55^\circ$.

BA从B看向正南180°，与BC的120°夹60°；最后由角和求55°。

M1	正确两处方位夹角
A1	两已知角
A1	第三角

(b) · 2 marks

$80 \sin 60^\circ / \sin 55^\circ \approx 84.6$ m.

AC对B的60°，AB对C的55°，用正弦定理。

M1	正确对应边角
A1	长度

Question 11

P1-C06-R11 · 5 marks

(a) · 3 marks

Maximum 1 at $0^\circ, 360^\circ$; minimum -3 at 180°.

cos取1时最大，取−1时最小，注意区间包含两个端点。

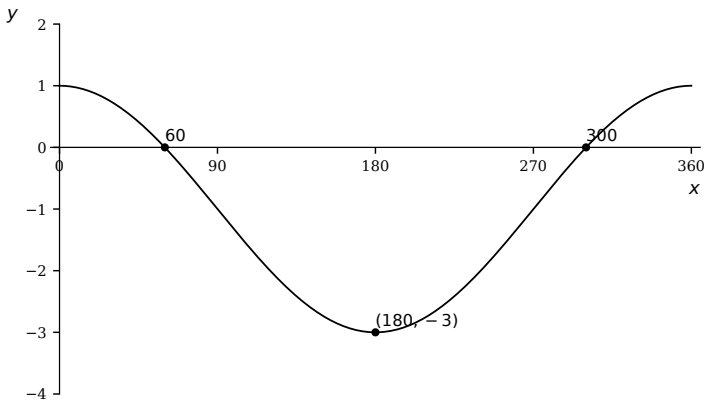
M1	用余弦范围
A1	两极值
A1	所有位置

(b) · 2 marks

$(60, 0)$ and $(300, 0)$, with x in degrees.

令cosx=1/2，第一与第四象限两个解。

M1	化简方程
A1	两个交点



Suggested sketch

Question 12

P1-C06-R12 · 7 marks

(a) · 3 marks

$\sin C = \frac{3}{4}$, so $C \approx 48.6^\circ$ or 131.4°.

正弦定理sinC/15=sin30°/10；两角均与B组成合法三角形。

M1	正弦关系
M1	补角支
A1	两角

(b) For the triangle in which $\angle ACB$ is acute, find

(i) · 2 marks

101.4°.

A=180−30−arcsin(3/4)，用未舍入角。

M1	选正确C并用内角和
A1	A角

(ii) · 2 marks

$\approx 73.5\ \text{cm}^2$.

面积1/2×15×10×sinA。

M1	正确面积式
A1	面积

Question 13

P1-C06-R13 · 5 marks

(a) · 2 marks

$7\sin 64^\circ \approx 6.29$ m.

高度是该角对边，梯长是斜边。

M1	正弦关系
A1	高度

(b) · 3 marks

$\arcsin\big((7\sin 64^\circ - 1)/7\big) \approx 49.1^\circ$.

先从原高度减1，再与不变梯长7构成正弦比；取0°到90°的角。

M1	正确新高度
M1	反求锐角
A1	角度

Question 14

P1-C06-R14 · 5 marks

(a) · 2 marks

$126\sin 38^\circ \approx 77.6\ \text{cm}^2$.

以一边为底、另一边的垂直分量为高；平行四边形面积不是三角形的一半。

M1	正确底高关系
A1	面积

(b) · 3 marks

$\sqrt{9^2+14^2-2(9)(14)\cos 38^\circ} \approx 8.86\ \text{cm}$.

较短对角线对锐角38°，使用余弦定理；另一条对补角更长。

M1	选择锐角对应短对角线
M1	余弦定理
A1	长度

Question 15

P1-C06-R15 · 5 marks

(a) · 3 marks

$\cos A = \frac{1}{2}$, $A = 60^\circ$.

余弦定理(16+49−37)/(2×4×7)=28/56。

M1	正确余弦关系
A1	精确余弦
A1	角度

(b) · 2 marks

$7\sqrt{3}\ \text{cm}^2$.

面积1/2×4×7×√3/2。

M1	正确面积式
A1	根式面积

Question 16

P1-C06-R16 · 4 marks

(a) · 2 marks

$-\frac{12}{13}$.

参考直角三角形对边5斜边13，邻边12；钝角的余弦为负。

M1	勾股得到参考边12
A1	正确负号与比值

(b) · 2 marks

$-\frac{5}{12}$.

tan为sin/cos，正除负得负。

M1	正确三角比商
A1	最简值

Question 17

P1-C06-R17 · 5 marks

(a) · 3 marks

Asymptotes $x = -90^\circ, 90^\circ, 270^\circ$ at boundaries or inside; zeros $0^\circ, 180^\circ$.

正切周期180°，奇数90°为竖直渐近线，各分支严格上升；边界虽不取值仍标渐近线。

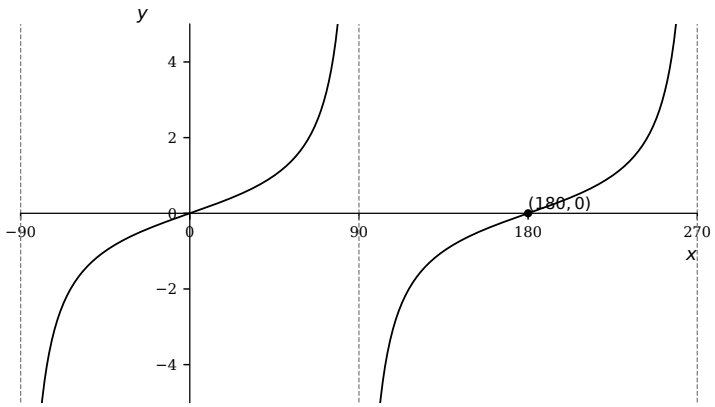
B1	渐近线
B1	两零点
A1	正确分支走向

(b) · 2 marks

$x = -45^\circ, 135^\circ$.

tan为−1的参考角45°，每180°重复，筛选开区间。

M1	使用周期并筛选
A1	全部解



Suggested sketch

Question 18

P1-C06-R18 · 7 marks

(a) · 3 marks

118 m.

余弦定理给 $\sqrt{(120^2+160^2-2\times120\times160\cos47^\circ)}$，再按米四舍五入。

M1	正确余弦式
M1	计算未舍入长度
A1	最近整数米

(b) A path runs along the 160 m side and has perpendicular width 2 m inside the field. For this calculation only, its area is modelled as a rectangle of dimensions 160 m by 2 m. Find

(i) · 1 marks

$320\ \text{m}^2$.

按明确给出的矩形近似模型计算。

B1	矩形面积

(ii) · 3 marks

$100(320)/(9600\sin47^\circ)\approx4.6\%$.

全场面积为$9600\sin47^\circ$；用近似路径面积除以全场面积再乘100。

M1	正确三角场地面积
M1	百分比关系
A1	百分比

Question 19

P1-C06-R19 · 6 marks

(a) · 3 marks

50 km.

夹角110−20=90°，勾股 $\sqrt{(30^2+40^2)}=50$。

M1	从方位得直角
M1	勾股
A1	距离

(b) · 3 marks

53.1°.

在A处夹角的对边HB=40，斜边AB=50，所以sin角=4/5；取三角形锐角。

M1	正确识别A处角的边
M1	反求角
A1	精度

Question 20

P1-C06-R20 · 5 marks

(a) · 3 marks

$x = \sqrt{19/7}$.

余弦定理$19=x^2+4x^2-4x^2\cos120^\circ=7x^2$，取正根。

M1	正确含参数余弦式
M1	使用精确余弦
A1	正根

(b) · 2 marks

$\dfrac{19\sqrt{3}}{14}\ \text{cm}^2$.

面积$x^2\sin120^\circ=(19/7)(\sqrt{3}/2)$。

M1	正确面积式
A1	精确结果

Question 21

P1-C06-R21 · 5 marks

(a) · 3 marks

Period $180°$; maximum $(45, 4)$, minimum $(135, -4)$.

输入角2x走完360°时x走180°，故周期180°；极值在2x=90°,270°。

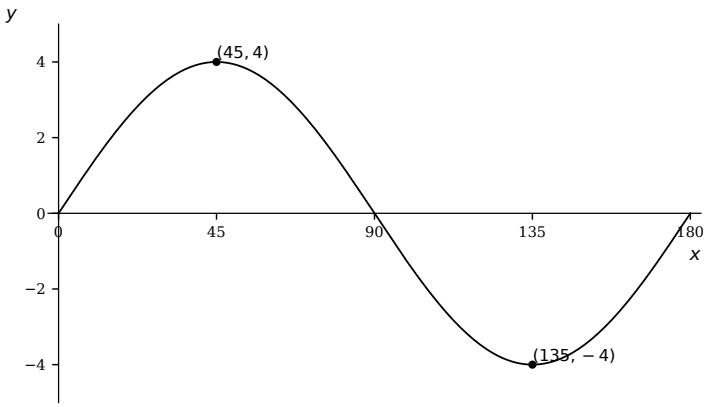
M1	正确处理周期
A1	极值位置
A1	极值坐标

(b) · 2 marks

$x = 0°, 90°, 180°$.

2x=0°,180°,360°，横坐标除2。

M1	输入角范围及零点
A1	全部x



Suggested sketch

Question 22

P1-C06-R22 · 6 marks

(a) · 3 marks

36.9° or 143.1°.

$40\sin A=24$，故$\sin A=3/5$；在0°到180°内有锐角和其补角。

M1	正确面积关系
M1	找两个角
A1	角度

(b) · 3 marks

6 cm and $2\sqrt{73}$ cm.

相应$\cos A=\pm 4/5$。余弦定理$BC^2=164-160\cos A$，分别36和292。

M1	找两余弦的正负
M1	余弦定理
A1	两精确长度

Question 23

P1-C06-R23 · 6 marks

(a) · 2 marks

$6\tan 58^\circ \approx 9.60$ m.

短线对应较近锚点6m；高度为6tan58°。

M1	正确识别短线锚点
A1	高度

(b) For the longer wire, find

(i) · 2 marks

$\arctan\left((6\tan 58^\circ)/10\right) \approx 43.8^\circ$.

同一塔高除以较远水平距离10构成正切比。

M1	正确正切关系
A1	角度

(ii) · 2 marks

$\sqrt{100 + 36\tan^2 58^\circ} \approx 13.9$ m.

水平直角边10与塔高用勾股。

M1	勾股求线长
A1	长度

Question 24

P1-C06-R24 · 6 marks

(a) · 2 marks

$120^\circ, 240^\circ$.

余弦负在第二和第三象限，参考角60°。

M1	正确参考角与象限
A1	全部解

(b) For each value of x found in part (a), state

(i) · 2 marks

$\sqrt{3}/2$ at 120°; $-\sqrt{3}/2$ at 240°.

参考角60°，正弦在第二象限正、第三负。

M1	正确参考值
A1	与两角配对

(ii) · 2 marks

$-\sqrt{3}$ at 120°; $\sqrt{3}$ at 240°.

用$\sin/\cos$或参考角符号规则。

M1	正确比值及符号
A1	两值

Question 25

P1-C06-R25 · 5 marks

(a) · 2 marks

$\cos\theta = (7^2 + 9^2 - 8^2)/(2\cdot 7\cdot 9) = 11/21.$

分子49+81−64=66，分母126，约为11/21。

M1	正确余弦定理
A1	精确化简

(b) · 3 marks

$12\sqrt{5}\ \text{cm}^2.$

θ是三角形内角且cos为正，所以sinθ=√(441−121)/21=8√5/21；面积(1/2)×7×9×该比。

M1	由正弦余弦关系或参考直角三角形得正弦
M1	正确面积式
A1	精确面积

Question 26

P1-C06-R26 · 5 marks

(a) · 3 marks

Minimum 1 m at $0^\circ, 360^\circ$; maximum 9 m at 180°.

余弦前系数负，cos=1时最低，cos=−1时最高；包含两端点。

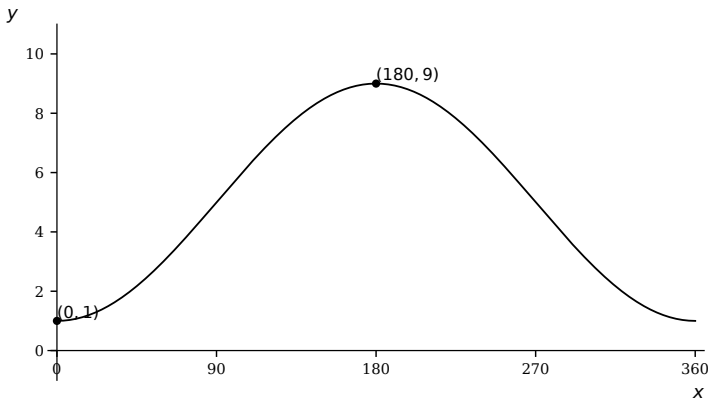
M1	正确使用负系数
A1	两高度
A1	所有角度

(b) · 2 marks

$120^\circ, 240^\circ$.

$5-4\cos\theta=7$给$\cos\theta=-1/2$。

M1	正确化简
A1	两角



Suggested sketch

Question 27

P1-C06-R27 · 6 marks

(a) · 2 marks

96.4°.

AC为角B对边，cosB=(49+81−144)/(2×7×9)=−1/9。

M1	正确余弦式
A1	角度

(b) · 4 marks

$\approx 79.3\ \text{cm}^2$.

ABC面积=(1/2)×7×9sinB=14√5≈31.305；ACD高√(100−36)=8，面积48；相加约79.3。

M1	ABC面积
M1	等腰分割求ACD面积
M1	两三角形相加
A1	总面积

Question 28

P1-C06-R28 · 5 marks

(a) · 3 marks

Rise 27.8 m; horizontal distance 85.6 m.

以路径为斜边，竖直分量$90\sin 18^\circ$，水平$90\cos 18^\circ$。

M1	两分量正确三角比
A1	竖直距离
A1	水平距离

(b) · 2 marks

128 m.

平台与原终点同高，所以竖直爬升仍为$90\sin 18^\circ$；两段总长$100+90\sin 18^\circ \approx 127.812$。

M1	给定段长相加
A1	长度

Question 29

P1-C06-R29 · 5 marks

(a) · 2 marks

$a = 3, b = 4.$

最大、最小值分别为$a + b = 7, a - b = -1$；联立得$a = 3, b = 4$。

M1	两个极值方程
A1	参数

(b) A horizontal line $y = 5$ cuts this curve at P and Q.

(i) · 2 marks

$(60, 5), (300, 5).$

$3 + 4\cos x = 5$，即$\cos x = 1/2$，区间内$x = 60^\circ, 300^\circ$。

M1	化为基本余弦方程
A1	两点

(ii) · 1 marks

$t = 30^\circ, 150^\circ.$

$2t$在原图区间内，对应$2t = 60^\circ, 300^\circ$；除以2。

B1	利用前问并正确变换区间

Question 30

P1-C06-R30 · 7 marks

(a) · 2 marks

$\sqrt{129}$ cm.

余弦定理$BC^2 = 8^2 + 5^2 - 2(8)(5)\cos 120^\circ = 129$。

M1	余弦定理
A1	精确长度

(b) Let $AD = x$ cm. Use the areas of triangles ABD and ACD to find

(i) · 3 marks

$x = 40/13$.

大三角形面积$10\sqrt{3}$。两小三角形面积之和为$\tfrac{1}{2}(8)x\sin 60^\circ + \tfrac{1}{2}(5)x\sin 60^\circ = 13x\sqrt{3}/4$；所以$x = 40/13$。不需要引用角平分线长度公式。

M1	两小面积相加
M1	与大三角形面积相等
A1	x

(ii) · 2 marks

$8:5$.

两小三角形对共同底线BC有同高，面积比等于底边比；其面积由前式给出比$8x:5x = 8:5$。

M1	等高三角形面积比
A1	比值

Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 7 — Radians
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第7章 · 弧度制
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

Figure 1 shows a footpath bounded by two concentric arcs AB and CD and two radial edges. The outer arc is twice as long as the inner arc. The two arcs have total length 18 m, and the footpath has area $27\ \text{m}^2$. Let the inner radius be r m and the outer radius be R m. The common angle is θ radians, with $0 < \theta < \pi$.

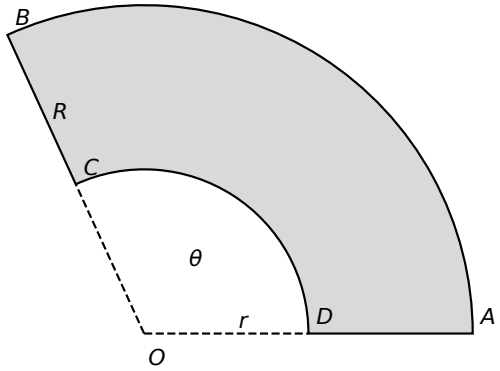


Figure 1

(a) Show that the width $R - r$ of the footpath is 3 m. (3)

(b) Hence find

(i) the inner radius r and the outer radius R, (2)

(ii) the value of θ. (2)

(c) Both radial edges also need edging. Find the total length of edging required. (2)

(Total for Question 1 is 9 marks)

Figure 2 shows a sector OAB of radius 6 cm and angle θ radians, where $0 < \theta < \pi$. Point C lies on OB with $OC = 4$ cm. Triangle OAC is removed from the sector, leaving a panel bounded by the minor arc AB and the straight segments BC and CA. The panel has perimeter 14 cm.

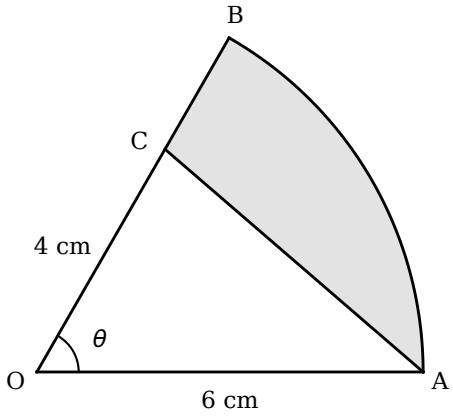


Figure 2

(a) Show that $CA = 12 - 6\theta$ cm, and hence show that $\cos\theta = \dfrac{52 - (12 - 6\theta)^2}{48}$. (3)

(b) A second panel has the same lengths OA, OB, OC but has $\theta = 1.2$; its perimeter is not fixed at 14 cm. Find

(i) the perimeter of the panel to 3 significant figures. (2)

(ii) the area of the panel to 3 significant figures. (2)

(Total for Question 2 is 7 marks)

A flower bed is a sector OAB of radius 8 m and angle 1.6 radians. Its curved fence AB is to be replaced by the straight chord AB, leaving the triangular bed OAB. Figure 3 shows the area that will be lost.

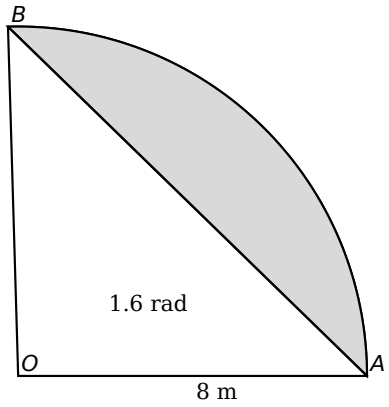


Figure 3

(a) Find the reduction in fence length, to 3 significant figures. (3)

(b) Find the area lost, to 3 significant figures. (3)

(c) The owner claims that more than one third of the original area will be lost. Determine whether the claim is correct. (2)

(Total for Question 3 is 8 marks)

Two straight roads OA and OB meet at an angle of 0.9 radians, with $OA = 5$ km and $OB = 8$ km. A circular road with centre O and radius 5 km joins A to the point C on OB.

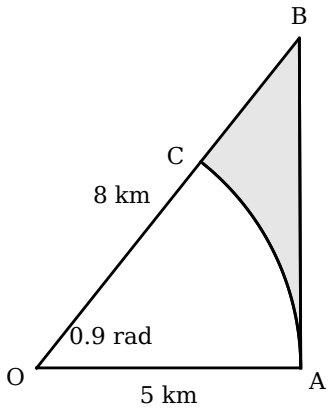


Figure 4

(a) Find, to 3 significant figures, the length of the direct road AB. (2)

(b) The alternative route from A to B follows the minor arc AC and then the straight road CB.

(i) Find the additional distance travelled using this route, to 3 significant figures. (2)

(ii) Find the area bounded by this alternative route and the direct road, to 3 significant figures. (2)

(Total for Question 4 is 6 marks)

Two circular garden beds each have radius 5 m. Their centres O_1, O_2 are 6 m apart. The circles intersect at A, B. The segments AB and O_1O_2 are perpendicular and bisect each other at M. Let $\alpha = \angle AO_1M$. Figure 5 shows the common area of the two beds.

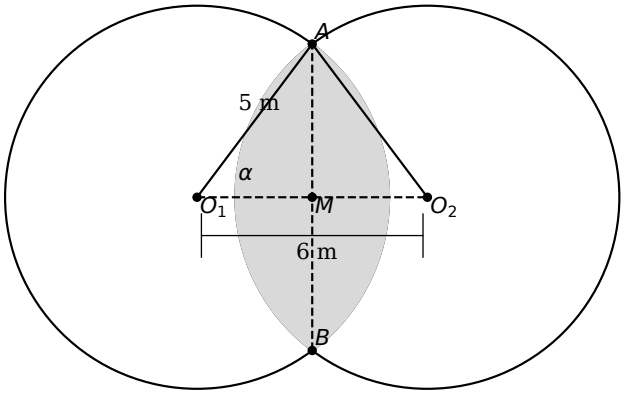


Figure 5

(a) Find AB and give an exact expression for α. (3)

(b) Find the common area, giving your answer in terms of α. (3)

(c) Find the perimeter of the common area, to 3 significant figures. (2)

(Total for Question 5 is 8 marks)

6.

A wheel of radius 0.35 m rolls in a straight line without slipping. A mark P is initially at the point of contact with the ground. The wheel makes 4 complete turns and then turns through a further 1.4 radians.

(a) Find the distance travelled by the centre of the wheel, to 3 significant figures. (3)

(b) For the final position of P, find

(i) its height above the ground, to 3 significant figures. (2)

(ii) the fraction of one full revolution represented by the final 1.4-radian turn. Give your answer to 3 significant figures. (1)

(Total for Question 6 is 6 marks)

A circular plate has centre O and radius 10 cm. Chords AB and AC lie on opposite sides of radius OA, with $\angle AOB = 0.8$ radians and $\angle AOC = 1.1$ radians. Points A, B, C lie on the minor arc from B to C.

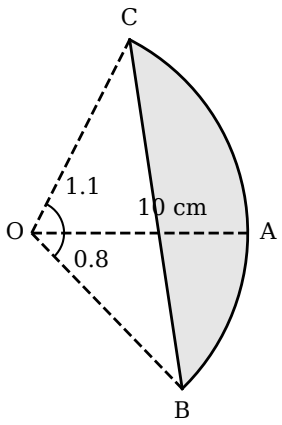


Figure 7

(a) Find the length of this arc from B through A to C. (2)

(b) A thin piece is cut off along chord BC; the piece is bounded by the minor arc BAC and chord BC. Find, to 3 significant figures,

(i) the length of BC. (2)

(ii) the area of the piece removed. (2)

(Total for Question 7 is 6 marks)

Two sector-shaped panels have the same arc length, 12 cm. Their radii are r cm and $(r + 3)$ cm. The sum of their areas is $66\ \text{cm}^2$.

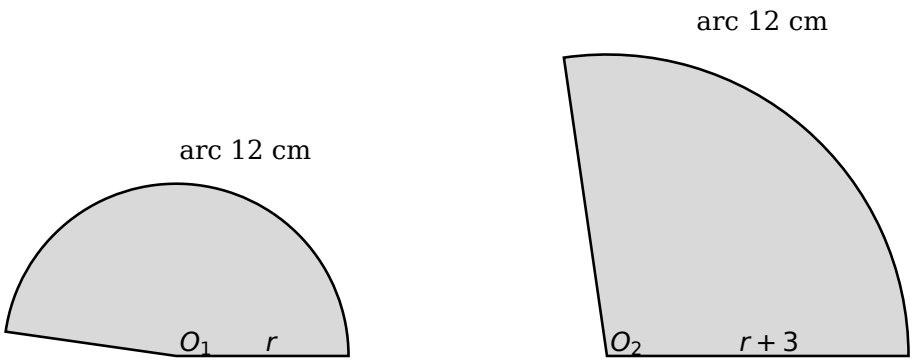


Figure 8

(a) Show that the area of the larger panel exceeds that of the smaller by $18\ \text{cm}^2$, regardless of r. (3)

(b) Find both radii and both angles. (3)

(c) Which panel has the greater perimeter? State by how much. (2)

(Total for Question 8 is 8 marks)

A windscreen wiper sweeps an annular sector of angle 2.2 radians. The distances of the ends of the rubber blade from the pivot are 0.18 m and 0.60 m. Figure 9 shows the swept region.

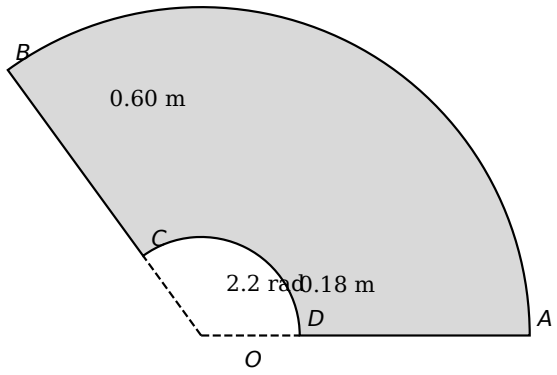


Figure 9

(a) Find the swept area to 3 significant figures. (2)

(b) Find the complete perimeter of the swept region. (3)

(c) The sweep angle may be adjusted to θ, where $0 < \theta < \pi$. The straight-line distance between the two extreme positions of the outer tip must not exceed 1.00 m.

(i) Find the greatest permitted value of θ, to 3 significant figures. (3)

(ii) State, with a reason, whether the original sweep angle meets this requirement. (1)

(Total for Question 9 is 9 marks)

A circular sheet is cut along a chord AB of length 10 cm. The chord subtends an angle of $\pi/3$ at the centre. The larger piece is kept, as shown shaded in Figure 10.

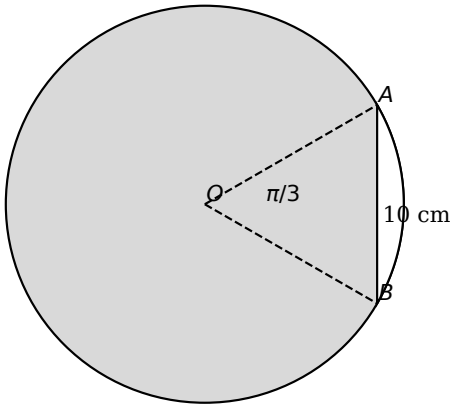


Figure 10

(a) Find the radius of the sheet. (2)

(b) Find the exact area of the piece kept. (4)

(c) Find the exact perimeter of the piece kept. (2)

(Total for Question 10 is 8 marks)

A sign consists of a rectangle of width $2r$ cm and height r cm, surmounted by a semicircle of radius r cm. Its outside perimeter is 20 cm.

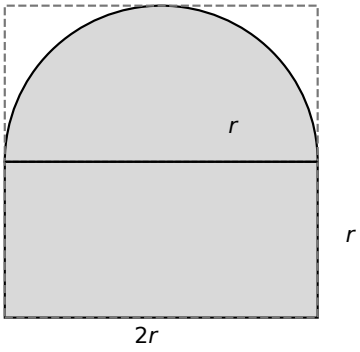


Figure 11

(a) Find r exactly. (2)

(b) Show that the area of the sign is $200/(4+\pi)$ cm^2. (3)

(c) A rectangular board of the same width and full height is used to make the sign. Find the fraction of the board that is cut away. (2)

(Total for Question 11 is 7 marks)

Triangle ABC has $AB = 8$ cm, $AC = 11$ cm and $BC = \sqrt{73}$ cm. A sector of radius 5 cm, centred at A, is removed from the corner. Its arc meets AB, AC at D, E. Figure 12 shows the remaining panel. Let $\alpha = \angle BAC$.

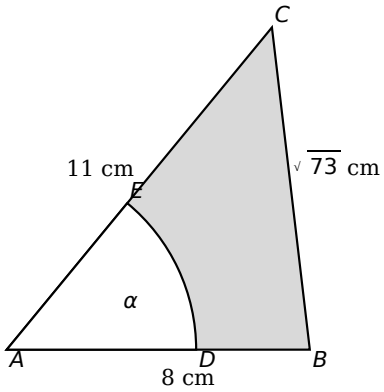


Figure 12

(a) Show that $\cos\alpha = 7/11$, and find $\sin\alpha$ exactly. (3)

(b) Find the remaining area, giving your answer in terms of α. (2)

(c) Find the perimeter of the remaining panel, to 3 significant figures. (3)

(Total for Question 12 is 8 marks)

An arm pivots at O. Starting with length 3 m, its tip moves anticlockwise through an angle θ. The arm is then extended by 2 m without turning. With its new length, it turns through a further angle θ in the same direction. The total tip path is 14 m. Assume $0 < \theta < \pi/2$.

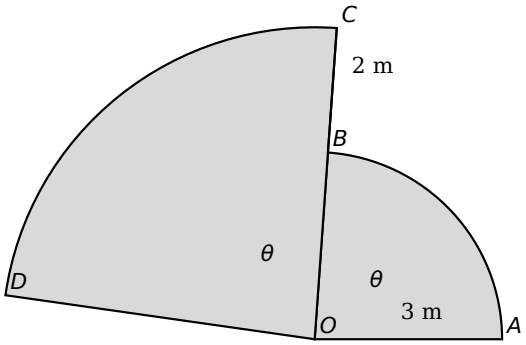


Figure 13

(a) Find θ. (3)

(b) Find the straight-line distance between the initial and final tip positions, to 3 significant figures. (3)

(c) Find the total area swept out during the two turns, excluding the zero-area extension. (2)

(Total for Question 13 is 8 marks)

A C-shaped mask is made from an annulus with outer radius 12 cm and inner radius 6 cm by removing an annular sector of angle ϕ, where $0 < \phi < \pi$. The remaining area is 81π cm^2.

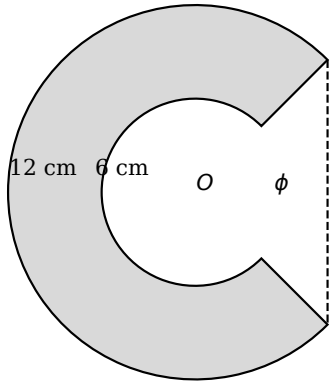


Figure 14

(a) Find ϕ exactly. (3)

(b) Find the exact perimeter of the mask. (3)

(c) A straight support joins the two outer ends across the gap. Find its length exactly. (2)

(Total for Question 14 is 8 marks)

A road of width 4 m consists of two annular sectors joined end to end along a complete radial edge. Each sector has inner radius 8 m, outer radius 12 m and angle $\pi/3$. The first bends left and the second bends right, so that each long side contains one inner arc and one outer arc. Figure 15 shows the plan.

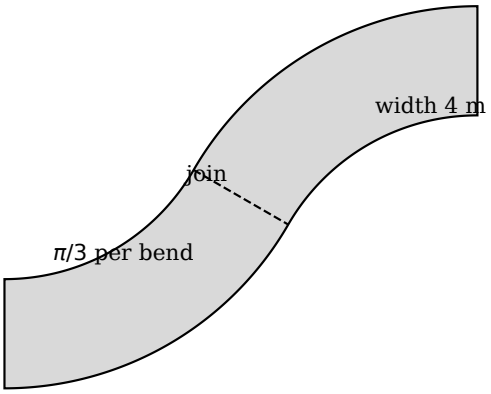


Figure 15

(a) Find the length of each curved long side exactly. (2)

(b) Find the total road area exactly. (2)

(c) Edging is required around the whole outside boundary, including the two open ends. Find the length required. (3)

(Total for Question 15 is 7 marks)

A circular window has radius 13 cm. A straight horizontal bar cuts off a minor segment whose greatest height above the bar is 8 cm. Let α be half the angle subtended by the bar at the centre, as shown in Figure 16.

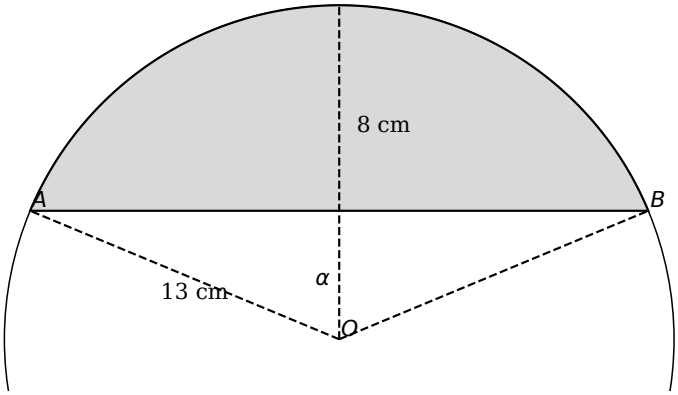


Figure 16

(a) Find the length of the bar and an exact expression for α. (3)

(b) Find the exact area of the segment, in terms of α. (3)

(c) Find the perimeter of the segment to 3 significant figures. (2)

(Total for Question 16 is 8 marks)

A quarter-disc of radius r is divided by the chord joining its two arc endpoints. Figure 17 shows the right-angled triangle next to the centre and the remaining segment.

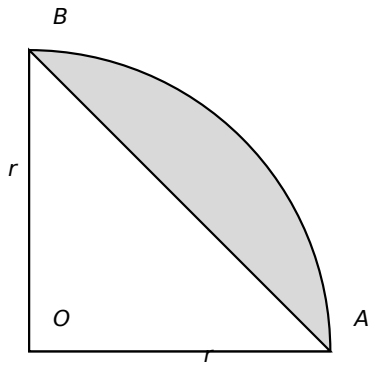


Figure 17

(a) Show that the triangle occupies the fraction $2/\pi$ of the quarter-disc, independently of r. (3)

(b) The triangle has area $18\ \text{cm}^2$. Find the exact segment area. (2)

(c) Use the result of part (a) to explain why no positive value of r makes the triangle occupy 70% of the quarter-disc. (2)

(Total for Question 17 is 7 marks)

A sector has radius r cm and angle θ radians, where $0 < \theta < \pi$. Its perimeter is 30 cm. The length of its arc exceeds the length of its chord by 2 cm.

(a) Show that $r = \dfrac{30}{2 + \theta}$ and hence show that $30\theta - 60\sin(\theta/2) = 4 + 2\theta$. (3)

(b) A trial angle is $\theta = 1.5$. Using the perimeter constraint,

(i) find the arc length minus the chord length, to 3 significant figures. (2)

(ii) state whether this trial angle satisfies the required difference of 2 cm. (1)

(Total for Question 18 is 6 marks)

19.

A clock has a minute hand of length 12 cm. At 12:00 the hand points vertically upwards. The hand moves clockwise at a constant rate.

(a) Find, in radians, the angle through which the hand turns in 18 minutes. Give your answer in terms of π. (2)

(b) Between 12:00 and 12:18, find

(i) the exact distance travelled by the tip, and the magnitude of its straight-line displacement to 3 significant figures. (3)

(ii) the area swept out by the hand, in exact form. (2)

(Total for Question 19 is 7 marks)

Three gates A, B, C lie consecutively on a minor circular arc, with common centre O. The arcs AB and BC have lengths 8 m and 12 m, respectively, and $\angle AOC = 2$ radians. A new straight path is proposed from A to C.

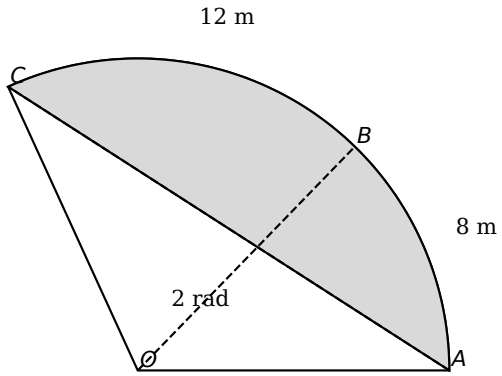


Figure 20

(a) Find the radius and the angles AOB and BOC. (3)

(b) Find the distance saved by using the new path rather than the curved route through B. (2)

(c) Find the area between the new path and the old route, to 3 significant figures. (3)

(Total for Question 20 is 8 marks)

A wedge of angle θ radians is removed from a circular sheet of radius 5 cm, with $0 < \theta < \pi$. The remaining major sector must have area at least 60 cm^2 and perimeter no greater than 35 cm.

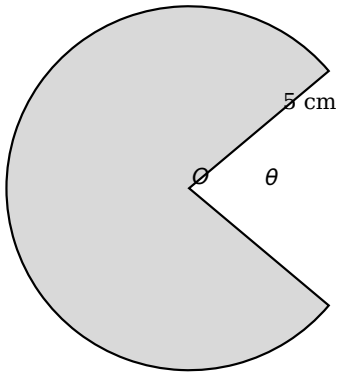


Figure 21

(a) Obtain one inequality in θ from each design requirement. (4)

(b) Give the full set of permitted values of θ. (2)

(c) Find the smallest possible straight-line gap between the two outer ends of the cut, to 3 significant figures. (3)

(Total for Question 21 is 9 marks)

Two concentric circles have centre O and outer radius 17 cm. A chord AB of the outer circle has length 30 cm. The perpendicular from O meets the midpoint of AB and has length equal to the inner radius r. The radii OA, OB cut the inner circle at D, C. Figure 22 shows the annular sector $ABCD$.

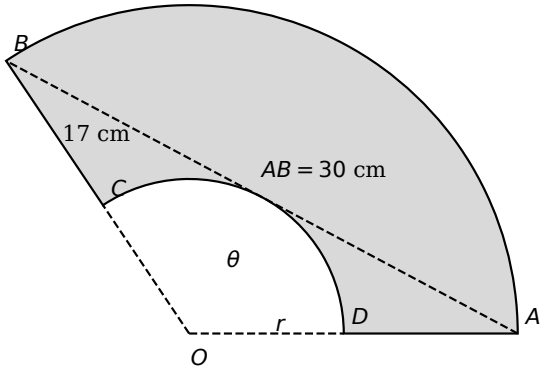


Figure 22

(a) Find r and the common angle θ. (3)

(b) Find the area of the annular sector in terms of θ. (2)

(c) Find the complete perimeter of this annular sector, to 3 significant figures. (3)

(Total for Question 22 is 8 marks)

A wire 18 cm long is bent into a circular arc. Joining its ends to the circle centre forms a sector of area $36\ \text{cm}^2$. The sector is the one bounded by the wire itself, as shown in Figure 23.

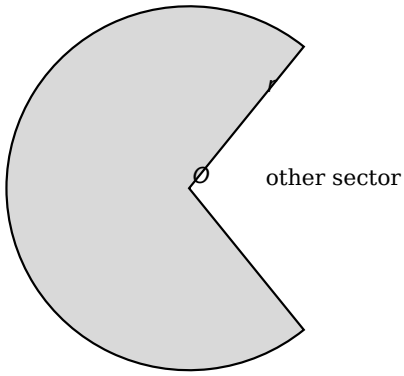


Figure 23

(a) Find the radius and the angle of this sector. (3)

(b) Explain whether this is a minor or major sector, and find its complete perimeter. (3)

(c) Find the exact area of the other sector determined by the same two radii. (2)

(Total for Question 23 is 8 marks)

A sector-shaped garden of radius r m is divided into two smaller sectors by one radial fence. Their angles are θ and 2θ. All the outer boundary and this internal fence use 54 m of fencing. The garden has area 108 m^2 and total angle less than π.

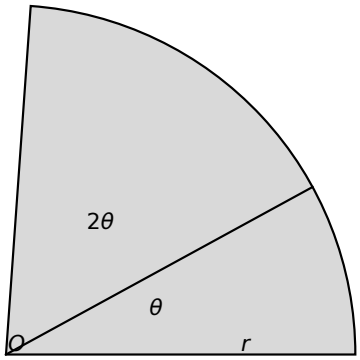


Figure 24

(a) Show that $r^2 - 18r + 72 = 0$. (3)

(b) Find the radius and both smaller angles, explaining which root must be rejected. (3)

(c) Find how much fencing is around the outside boundary only. (2)

(Total for Question 24 is 8 marks)

A rectangle of width 12 cm stands on the diameter of a semicircle of radius 10 cm. Its upper vertices A, B lie on the arc and its base is centred on O, the circle centre. The semicircle outside the rectangle consists of a cap above AB and two side regions. Figure 25 shows these regions.

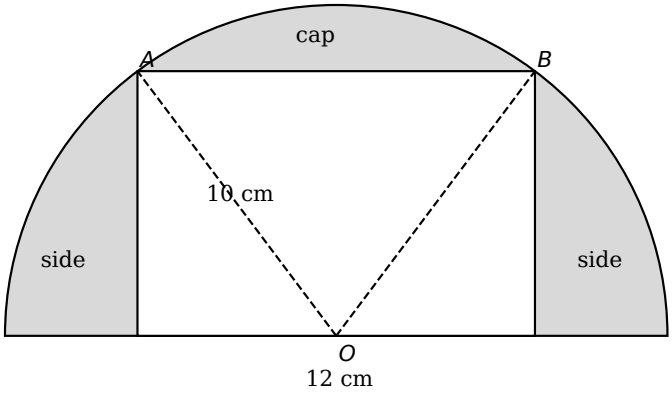


Figure 25

(a) Find the rectangle height, and show that $\angle AOB = 2\arcsin(3/5)$. (3)

(b) Let $\alpha = \arcsin(3/5)$.

(i) Find the area of the cap in terms of α. (3)

(ii) Find the combined area of the two side regions, and hence state which is larger: the cap or the side regions together. Give both areas to 3 significant figures. (3)

(Total for Question 25 is 9 marks)

A sector of radius r cm and angle θ radians has area $54\ \text{cm}^2$. Its arc is 3 times as long as one radius, and $0 < \theta < 2\pi$.

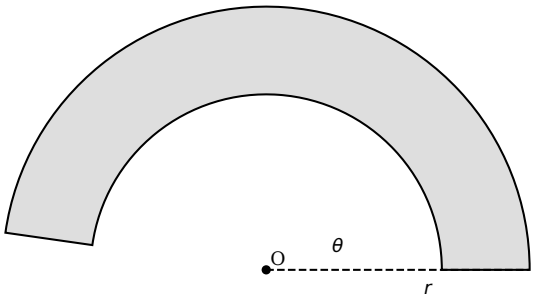


Figure 26

(a) Find r and θ. (3)

(b) A smaller concentric sector with the same angle is removed, leaving an annular sector of area $30\ \text{cm}^2$. Find

(i) the inner radius, (3)

(ii) the complete perimeter of the remaining region. (2)

(Total for Question 26 is 8 marks)

A circular window has radius 10 cm. A horizontal bar of length 12 cm forms a chord. The perpendicular from the centre bisects the chord. The smaller region above the bar is removed.

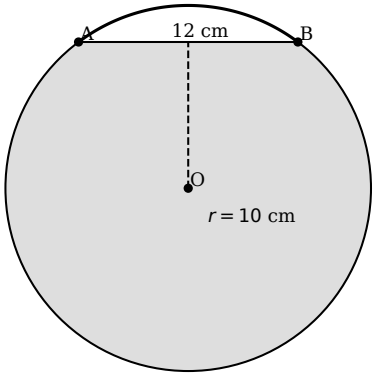


Figure 27

(a) Find the distance from the centre to the bar and the angle subtended by the bar at the centre, in radians to 3 significant figures. (4)

(b) For the remaining larger piece, find

(i) the exact area in terms of $\alpha = \arcsin(3/5)$, (3)

(ii) the complete perimeter in terms of α. (3)

(Total for Question 27 is 10 marks)

A swinging arm of length 2.4 m moves from angle $-\phi$ to angle ϕ relative to the downward vertical, where $0 < \phi < \pi/2$. The tip rises 0.6 m from its lowest position at each end of the swing.

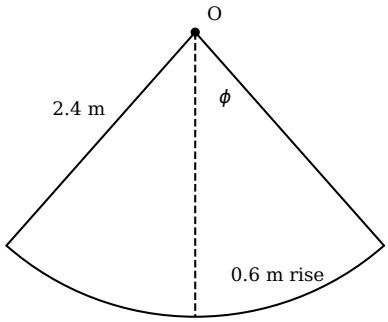


Figure 28

(a) Show that $\cos\phi = 3/4$ and find ϕ to 3 decimal places. (3)

(b) During one complete swing from the left-hand extreme to the right-hand extreme, find, to 3 significant figures,

(i) the distance travelled by the tip, (2)

(ii) the straight-line distance between the two extreme positions. (2)

(Total for Question 28 is 7 marks)

A rectangular sheet has dimensions 18 cm by 10 cm. Two semicircles of radius 5 cm are cut out, one from each short edge, with their diameters on those edges. Their interiors do not overlap.

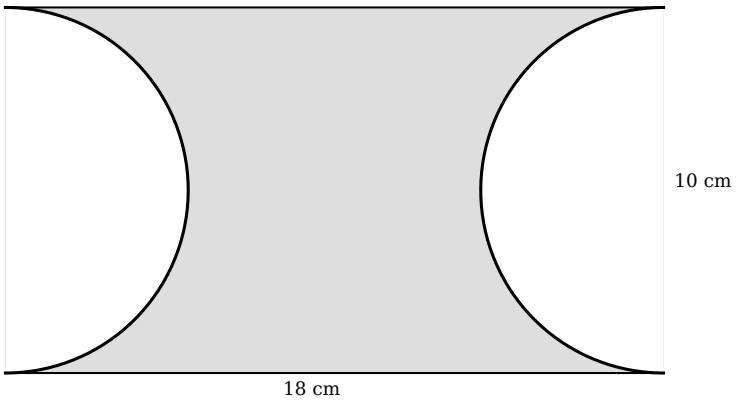


Figure 29

(a) Find the exact area and complete perimeter of the remaining sheet. (4)

(b) Each semicircular cut is replaced by a sector with the same radius and with its arc length reduced by 20%. For this comparison, consider one cut only. Find

(i) the new sector angle, (2)

(ii) the decrease in the area of the cut. (2)

(Total for Question 29 is 8 marks)

A sector-shaped sign has radius r cm and angle θ radians, where $0 < \theta < \pi$. A straight chord joins its arc endpoints. The arc length is 16 cm and the chord length is $8\sqrt{3}$ cm.

(a) Show that $\sin(\theta/2)/(\theta/2) = \sqrt{3}/2$. (3)

(b) A designer proposes $\theta = \pi/3$.

(i) Use the arc length to find the radius for this proposal. (2)

(ii) Determine whether this proposal also gives the required chord length. Give numerical evidence. (3)

(Total for Question 30 is 8 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C07-R01 · 9 marks

(a) · 3 marks

$R - r = 3$.

两弧总长 $(R+r)\theta = 18$；面积 $\frac{1}{2}(R-r)(R+r)\theta = 27$。用前一关系代入得 $9(R-r) = 27$。

M1	建立两弧总长关系
M1	把面积写成半径和与差的乘积
A1	推出宽度3

(b) Hence find

(i) · 2 marks

$r = 3$ m, $R = 6$ m.

同角弧长之比等于半径比，R=2r，结合(a)半径差3。

M1	使用弧长比和(a)
A1	两半径

(ii) · 2 marks

$\theta = 2$ radians.

两弧总长9θ=18。

M1	代入两半径求角
A1	弧度值

(c) · 2 marks

24 m.

边界包括两弧18m与两条各3m的直边，总长 $18 + 2 \times 3 = 24$。

M1	计入两条径向直边
A1	完整边界长度及单位

Question 2

P1-C07-R02 · 7 marks

(a) · 3 marks

$CA = 12 - 6\theta$; stated relation.

边界含$BC = 2$、弧$AB = 6\theta$及CA，没有已剪去的OA。因此$CA = 14 - 2 - 6\theta = 12 - 6\theta$。在$OAC$中余弦定理给$CA^2 = 36 + 16 - 48\cos\theta$，整理即得。

M1	正确弧长与周长
M1	余弦定理
A1	关系式

(b) A second panel has the same lengths OA, OB, OC but has $\theta = 1.2$; its perimeter is not fixed at 14 cm. Find

(i) · 2 marks

15.1 cm.

此设计$CA = \sqrt{52 - 48\cos 1.2} = 5.882\ldots$；周长仅为$BC + \text{arc } AB + CA = 2 + 7.2 + 5.882\ldots = 15.082\ldots$，即$15.1$ cm。

M1	以指定角度重新计算斜边
A1	周长

(ii) · 2 marks

10.4 cm^2.

面板面积为扇形减去OAC：$18(1.2) - 12\sin 1.2 = 10.4155\ldots$。

M1	扇形减三角形
A1	面积

Question 3

P1-C07-R03 · 8 marks

(a) · 3 marks

$12.8 - 16\sin 0.8 \approx 1.32$ m.

原弧长为$8\times 1.6=12.8$；将等腰三角形二等分，弦长为$16\sin 0.8$。相减才是节省长度。

M1	弧长
M1	半角求弦长
A1	差值及精度

(b) · 3 marks

$32(1.6 - \sin 1.6) \approx 19.2$ m^2.

失去的是弓形，面积为扇形 $\tfrac{1}{2}r^2\theta$ 减去三角形 $\tfrac{1}{2}r^2\sin\theta$。

M1	扇形减三角形
A1	正确精确式
A1	面积及精度

(c) · 2 marks

Correct: the lost fraction is $(1.6 - \sin 1.6)/1.6 \approx 0.375$.

用损失面积除以原扇形面积51.2，约37.5%，大于$33\tfrac{1}{3}\%$；不能把长度缩短的比例当作面积比例。

M1	用原面积作分母
A1	有数值依据的判断

Question 4

P1-C07-R04 · 6 marks

(a) · 2 marks

6.27 km.

余弦定理$AB = \sqrt{25 + 64 - 80\cos 0.9} = 6.266673\ldots$。

M1	余弦定理并使用弧度
A1	长度

(b) The alternative route from A to B follows the minor arc AC and then the straight road CB.

(i) · 2 marks

1.23 km.

弧AC长$5(0.9) = 4.5$，$CB = 3$，路线共7.5；比直达多$7.5 - 6.266673\ldots = 1.23$。

M1	弧长加直线并比较
A1	路程差

(ii) · 2 marks

4.42 km^2.

该区域为三角形OAB减扇形OAC，面积$20\sin 0.9 - \frac{1}{2}(25)(0.9) = 4.4165\ldots$。

M1	三角形与扇形面积差
A1	面积

Question 5

P1-C07-R05 · 8 marks

(a) · 3 marks

$AB = 8$ m, $\alpha = \arccos(3/5)$.

公共弦在两圆心中点处，三角形O₁MA直角，O₁M=3、斜边5，得AM=4。弦长2AM=8，cosα=3/5。

M1	二等分对称图形及勾股
A1	弦长
A1	弧度角表达式

(b) · 3 marks

$50\alpha - 24$ m^2.

重叠区由两个相同弓形组成；每个扇形角2α，面积25α，减去底8、高3的三角形面积12，再乘2。

M1	识别两个弓形
M1	扇形与三角形的面积
A1	合并正确

(c) · 2 marks

$20\alpha \approx 18.5$ m.

边界是两条角2α、半径5的圆弧；公共弦AB在内部，不应计入。

M1	只计两圆弧
A1	周长及精度

Question 6

P1-C07-R06 · 6 marks

(a) · 3 marks

9.29 m.

总转角$8\pi + 1.4$弧度；无滑动距离$r\theta = 0.35(8\pi + 1.4) = 9.286\ldots$。

M1	完整转数化为弧度
M1	使用无滑动弧长
A1	距离

(b) For the final position of P, find

(i) · 2 marks

0.291 m.

完整转数不改变标记的相对位置。相对最低点转过1.4，高度$0.35(1 - \cos 1.4) = 0.2905\ldots$。

M1	以半径竖直投影求高度
A1	数值

(ii) · 1 marks

0.223.

一周是2π弧度，比例$1.4/(2\pi) = 0.2228\ldots$。

B1	弧度与整周比例

Question 7

P1-C07-R07 · 6 marks

(a) · 2 marks

19 cm.

总圆心角$0.8+1.1=1.9$弧度，弧长$10(1.9)=19$。

M1	正确相加圆心角
A1	弧长

(b) A thin piece is cut off along chord BC; the piece is bounded by the minor arc BAC and chord BC. Find, to 3 significant figures,

(i) · 2 marks

16.3 cm.

等腰三角形BOC平分后$BC=20\sin(1.9/2)=16.268\ldots$。

M1	弦长与半角
A1	数值

(ii) · 2 marks

$47.7\ \text{cm}^2$.

弓形面积为扇形减三角形：$50(1.9-\sin 1.9)=47.685\ldots$。

M1	扇形减三角形
A1	面积

Question 8

P1-C07-R08 · 8 marks

(a) · 3 marks

Difference $= \frac{1}{2}(12)(r+3) - \frac{1}{2}(12)r = 18$.

由弧长l=rθ，扇形面积可写½rl；同弧长下两面积差为½l乘半径差。

M1	推出A=rl/2
M1	代入同弧长与半径差
A1	完整证明

(b) · 3 marks

Radii $4, 7$ cm; angles $3, 12/7$ radians, respectively.

两面积分别6r和6(r+3)，和66给r=4；各角为12除以对应半径。

M1	用面积和求r
A1	两半径
A1	两弧度角

(c) · 2 marks

The larger-radius panel, by 6 cm.

两弧相等，周长差仅来自两条半径，总差2×3=6。

B1	正确选择
B1	长度差与依据

Question 9

P1-C07-R09 · 9 marks

(a) · 2 marks

$0.360\ \text{m}^2$.

有效区域是两扇形之差：½(0.60²−0.18²)×2.2=0.36036。

M1	环形扇形面积
A1	数值及单位

(b) · 3 marks

$2.556\ \text{m}$.

两弧长合为(0.60+0.18)×2.2=1.716，两条直边各0.42，总长2.556。

M1	两弧
M1	计入两端的刀片长
A1	正确周长

(c) The sweep angle may be adjusted to θ, where $0 < \theta < \pi$. The straight-line distance between the two extreme positions of the outer tip must not exceed $1.00\ \text{m}$.

(i) · 3 marks

$\theta = 2\arcsin(5/6) \approx 1.97$ radians.

外端弦长1.20sin(θ/2)在此范围随θ增加；令弦长为1求边界角。

M1	正确弦长条件
M1	使用指定范围求边界
A1	最大角

(ii) · 1 marks

No, since $2.2 > 2\arcsin(5/6)$.

比较原角与(i)未舍入边界，原角更大，弦长超限。

B1	有数值比较支持的判断

Question 10

P1-C07-R10 · 8 marks

(a) · 2 marks

10 cm.

半弦5、半角π/6，5=r sinπ/6=r/2。

M1	二等分弦所对三角形
A1	半径

(b) · 4 marks

$250\pi/3 + 25\sqrt{3}$ cm^2.

小弓形面积为 $50(\pi/3 - \sin\pi/3) = 50\pi/3 - 25\sqrt{3}$；从整圆100π中扣除。

M1	计算小弓形
A1	小弓形精确式
M1	从完整圆扣除
A1	所留面积

(c) · 2 marks

$50\pi/3 + 10$ cm.

保留的是大弧，其角为2π−π/3=5π/3；边界还含弦10，故答案如此。

M1	选择大弧角
A1	曲边加切口的完整周长

Question 11

P1-C07-R11 · 7 marks

(a) · 2 marks

$r = 20/(4+\pi)$ cm.

外边包含底边2r、两侧共2r和半圆弧πr；半圆直径是内部接缝，不计入。

M1	正确外周长式
A1	精确r

(b) · 3 marks

$A = 2r^2 + \pi r^2/2 = \tfrac{1}{2}(4+\pi)r^2 = 200/(4+\pi)$.

将矩形与半圆面积相加，先保留因式再代r，约去一个(4+π)，避免过早小数化。

M1	面积组合
M1	提取公共因式并代r
A1	推出目标

(c) · 2 marks

$(4-\pi)/8$.

整板宽2r、高2r，面积$4r^2$；切去部分为上半个$2r\times r$矩形减半圆，即$(2-\pi/2)r^2$；除以$4r^2$。

M1	整板与切除面积正确
A1	约分后的比例

Question 12

P1-C07-R12 · 8 marks

(a) · 3 marks

$\cos\alpha = 7/11$, $\sin\alpha = 6\sqrt{2}/11$.

余弦定理给(64+121−73)/(2×8×11)=7/11。三角形内角正弦为正，用直角三角形或平方关系得6√2/11。

M1	余弦定理
A1	正确余弦
A1	正弦的精确正值

(b) · 2 marks

$24\sqrt{2} - 25\alpha/2$ cm^2.

三角形面积½×8×11×sinα=24√2，扣除半径5、角α的扇形。

M1	三角形与扇形面积
A1	正确相减结果

(c) · 3 marks

$9 + \sqrt{73} + 5\arccos(7/11) \approx 21.9$ cm.

剩余两直边为8−5和11−5，另有BC及圆弧5α，总和9+√73+5α；α须用弧度。

M1	剩余直边及BC
M1	加入新圆弧而非两半径
A1	周长及精度

Question 13

P1-C07-R13 · 8 marks

(a) · 3 marks

$\theta = 1.5$ radians.

路径由长度3θ的弧、长度2的直线、长度5θ的弧组成，故8θ+2=14。

M1	区分转动弧与伸长直线
A1	正确方程
A1	角度

(b) · 3 marks

$\sqrt{34 - 30\cos 3} \approx 7.98$ m.

两极端位置离O为3和5，夹角为2θ=3；余弦定理得距离平方9+25−30cos3。

M1	使用总角2θ而非θ
M1	余弦定理
A1	距离及精度

(c) · 2 marks

25.5 m^2.

两转动扇形占不同方向角区间，内部不重叠；面积½×9×1.5+½×25×1.5=25.5。

M1	两半径对应两扇形
A1	总面积及单位

Question 14

P1-C07-R14 · 8 marks

(a) · 3 marks

$\phi = \pi/2$.

完整圆环面积108π；剩余角为2π−φ，所以54(2π−φ)=81π，解得φ=π/2。

M1	完整圆环或剩余角面积
M1	建立方程
A1	精确角

(b) · 3 marks

$27\pi + 12$ cm.

两圆弧的剩余角都是3π/2，弧长共18×3π/2=27π，两条切口长度各6。

M1	使用剩余角而非切除角
M1	加两条切口
A1	完整周长

(c) · 2 marks

$12\sqrt{2}$ cm.

支撑是半径12、所对小角π/2的弦，长2×12×sinπ/4=12√2；不是缺口外弧长。

M1	弦长公式或勾股
A1	精确长度

Question 15

P1-C07-R15 · 7 marks

(a) · 2 marks

$20\pi/3$ m.

每条长边各含$8\times\pi/3$和$12\times\pi/3$，不是把两外弧归到同一边。

M1	每侧配一内弧一外弧
A1	两边共同长度

(b) · 2 marks

$80\pi/3$ m^2.

一段环形扇形面积$\tfrac{1}{2}(144-64)\pi/3=40\pi/3$，共两段，不重叠。

M1	一段环形扇形面积
A1	合计面积

(c) · 3 marks

$40\pi/3+8$ m.

两条长边共$40\pi/3$，只有首末两条宽4的径向边属于外界；中间接缝不计入。

M1	两曲边合计
M1	区分两端与内部接缝
A1	完整边界长度

Question 16

P1-C07-R16 · 8 marks

(a) · 3 marks

Bar length 24 cm; $\alpha = \arccos(5/13)$.

圆心到横杆距离为13−8=5；半弦长 $\sqrt{(169-25)}=12$，整杆24；cosα=5/13。

M1	由弓高找圆心至弦距离
M1	勾股求整弦
A1	长度与角表达式

(b) · 3 marks

$169\alpha - 60$ cm^2.

扇形角2α，其面积169α；中心三角形底24、高5，面积60，作差。

M1	扇形面积
M1	由底高求三角形
A1	正确弓形面积

(c) · 2 marks

$26\arccos(5/13) + 24 \approx 54.6$ cm.

边界由横杆24和角2α、半径13的弧组成，代弧度角后取3位有效数字。

M1	弧加弦
A1	数值与精度

Question 17

P1-C07-R17 · 7 marks

(a) · 3 marks

$\dfrac{r^2/2}{\pi r^2/4} = 2/\pi$.

三角形的两直角边都是r，面积$r^2/2$；四分之一圆面积$\pi r^2/4$，约去r^2说明与大小无关。

M1	两部分面积正确
M1	用整个四分之一圆作分母
A1	比例与独立性

(b) · 2 marks

$9\pi - 18$ cm^2.

$r^2/2=18$给$r^2=36$；扇形面积9π，扣18。

M1	由三角形面积求r平方
A1	精确弓形面积

(c) · 2 marks

Impossible: the fraction is fixed at $2/\pi \approx 63.7\%$ for every $r > 0$.

改变半径让两面积同倍增长，不改变占比；63.7%与70%不相等。

B1	固定比例或同倍变化理由
B1	正确设计判断

Question 18

P1-C07-R18 · 6 marks

(a) · 3 marks

Stated formulae.

周长$r\theta + 2r = 30$。弦长$2r\sin(\theta/2)$，由$r\theta - 2r\sin(\theta/2) = 2$代入$r$并整理。

M1	周长表达式
M1	弦长及差值方程
A1	整理正确

(b) A trial angle is $\theta = 1.5$. Using the perimeter constraint,

(i) · 2 marks

1.17 cm.

$r = 30/3.5 = 60/7$，差值$(60/7)[1.5 - 2\sin 0.75] = 1.1719\ldots$，即$1.17$ cm。

M1	同时更新半径并算弧弦差
A1	数值

(ii) · 1 marks

No; the difference is less than 2 cm.

试算差$1.17 < 2$，故不符合要求。这里只检验给定试值，不要求求解三角方程。

B1	依据计算比较

Question 19

P1-C07-R19 · 7 marks

(a) · 2 marks

$3\pi/5$.

每分钟转$2\pi/60$，18分钟转$18(2\pi/60)=3\pi/5$。

M1	时间比例化为弧度
A1	精确角

(b) Between 12:00 and 12:18, find

(i) · 3 marks

$36\pi/5$ cm; 19.4 cm.

走过弧长$12(3\pi/5)=36\pi/5$；位移是弦$24\sin(3\pi/10)=19.416\ldots$。

M1	弧长
M1	弦长
A1	两个结果与单位

(ii) · 2 marks

$216\pi/5\ \text{cm}^2$.

扇形面积$\tfrac12(12)^2(3\pi/5)=216\pi/5$。

M1	弧度制扇形公式
A1	精确面积

Question 20

P1-C07-R20 · 8 marks

(a) · 3 marks

Radius 10 m; angles $0.8, 1.2$ radians.

总弧长20=2r，故r=10，再各以弧长除以r求角。

M1	两相邻弧合成长角对应弧
A1	半径
A1	两角

(b) · 2 marks

$20 - 20\sin 1 \approx 3.17$ m.

新路是AC弦，长$2r\sin(2/2)=20\sin1$；从原20m路线减去。

M1	半角求AC弦
A1	节省长度及精度

(c) · 3 marks

$100 - 50\sin 2 \approx 54.5$ m^2.

两路线间的带状弓形面积为角2、半径10的扇形100，减中心三角形$50\sin2$。

M1	路线间区域识别为弓形
M1	扇形减三角形
A1	面积与单位

Question 21

P1-C07-R21 · 9 marks

(a) · 4 marks

$\theta \le 2\pi - 24/5$ and $\theta \ge 2\pi - 5$.

剩余角$2\pi-\theta$，面积$25(2\pi-\theta)/2 \geqslant 60$；周长$5(2\pi-\theta)+10 \leqslant 35$，分别解。

M1	面积不等式
A1	面积约束
M1	周长含两半径
A1	周长约束

(b) · 2 marks

$2\pi - 5 \le \theta \le 2\pi - 24/5$.

两约束取交集，得到闭区间；两端约1.283和1.483，均在原域$(0,\pi)$内。

M1	两约束与原范围取交
A1	包含等号的区间

(c) · 3 marks

$10\sin\left((2\pi - 5)/2\right) \approx 5.98$ cm.

缺口弦长$10\sin(\theta/2)$，在$0<\theta<\pi$内随角增大；取允许的最小角$2\pi-5$。

M1	弦长公式
M1	在题设角范围选择最小θ
A1	数值及单位

Question 22

P1-C07-R22 · 8 marks

(a) · 3 marks

$r = 8$ cm; $\theta = 2\arccos(8/17)$.

半弦15、斜边17，垂距$\sqrt{(289-225)}=8$；由辅助直角三角形$\cos(\theta/2)=8/17$。

M1	勾股求垂距
A1	内半径
A1	共角表达式

(b) · 2 marks

$225\theta/2$ cm^2.

面积$\tfrac{1}{2}(17^2-8^2)\theta = 225\theta/2$。弦AB不是要从此区域中扣去的边界。

M1	两扇形面积之差
A1	正确化简

(c) · 3 marks

$25\theta + 18 \approx 72.0$ cm.

内外弧共$(17+8)\theta$，两径向边各$17-8=9$，总长$25\theta+18$。

M1	两曲边合计
M1	两直边
A1	周长及精度

Question 23

P1-C07-R23 · 8 marks

(a) · 3 marks

$r = 4$ cm; $\theta = 4.5$ radians.

面积可写½r×弧长，所以36=9r给4；18=rθ给θ4.5。

M1	用面积与弧长消去θ
A1	半径
A1	角度

(b) · 3 marks

Major, since $4.5 > \pi$; perimeter 26 cm.

θ超过π而小于2π，故为大扇形；周长为18加两半径8。

B1	大扇形及角度依据
M1	完整周长
A1	数值

(c) · 2 marks

$16\pi - 36$ cm^2.

另一扇形角2π−4.5，或从圆面积16π中减36；不能把同一弧角重复使用。

M1	用互补整圆角或面积
A1	精确面积

Question 24

P1-C07-R24 · 8 marks

(a) · 3 marks

From $3r + 3r\theta = 54$ and $\frac{1}{2}r^2(3\theta) = 108$, eliminate θ to obtain the equation.

围栏含三条径向边，总长3r，曲边总长3rθ。由面积得3rθ=216/r，代总长式乘r并除3。

M1	三条径向围栏及曲边关系
M1	面积关系并消元
A1	推出二次式

(b) · 3 marks

$r = 12$ m; $\theta = 0.5$, $2\theta = 1$ radian. Reject $r = 6$, which gives total angle $6 > \pi$.

二次式分解为(r−6)(r−12)。总角216/r²；r=6给6弧度，违背总角小于π；r=12给1.5，故分角0.5、1。

M1	两个候选半径
M1	检查总角限制
A1	正确半径和两角

(c) · 2 marks

42 m.

从总围栏54减一条内部半径12，得42；也可算曲边18加两外半径24。

M1	排除内部围栏
A1	外周长度

Question 25

P1-C07-R25 · 9 marks

(a) · 3 marks

Height 8 cm; central angle $2\arcsin(3/5)$.

半宽6、斜边10，勾股得高8；以垂直对称线二等分中心三角形，半角正弦6/10=3/5。

M1	辅助直角三角形
A1	高度
A1	夹角表达式

(b) Let $\alpha = \arcsin(3/5)$.

(i) · 3 marks

$100\alpha - 48$ cm^2.

扇形角2α给面积100α；中心三角形底12、高8，面积48，相减。

M1	正确扇形面积
M1	正确中心三角形面积
A1	弓形精确表达式

(ii) · 3 marks

Sides $50\pi - 48 - 100\alpha \approx 44.7$ cm^2; cap 16.4 cm^2; sides are larger.

总空余面积50π−96；扣帽形后两侧为50π−48−100α。代α约0.643501比较，不可将整个空余区当作帽形。

M1	半圆减矩形再减帽形
A1	两个数值
B1	比较结论

Question 26

P1-C07-R26 · 8 marks

(a) · 3 marks

$\theta = 3, r = 6$.

rθ=3r且r>0给θ=3；面积$3r^2/2=54$，取正根6。

M1	弧长比例条件
M1	面积求正半径
A1	两值

(b) A smaller concentric sector with the same angle is removed, leaving an annular sector of area $30\ \text{cm}^2$. Find

(i) · 3 marks

4 cm.

去除面积54−30=24，因此$3s^2/2=24$，s=4。

M1	正确去除面积
M1	扇形关系
A1	内半径

(ii) · 2 marks

34 cm.

两弧长3(6+4)=30，两径向边各2，合34。

M1	计入两弧和两直边
A1	周长

Question 27

P1-C07-R27 · 10 marks

(a) · 4 marks

Distance 8 cm; angle $2\arcsin(3/5) \approx 1.29$ radians.

半弦6与半径10构成直角三角形；勾股得中心距8，半角正弦3/5。

M1	半弦直角三角形
A1	距离
M1	半角关系
A1	弧度角

(b) For the remaining larger piece, find

(i) · 3 marks

$100\pi - 100\alpha + 48$ cm^2.

原圆100π减小弓形(100α−48)。

M1	小弓形面积
M1	整圆减去小弓形
A1	精确面积

(ii) · 3 marks

$20\pi - 20\alpha + 12$ cm.

保留大弧角2π−2α，弧长10倍，再加弦12。

M1	使用大弧而非小弧
M1	加直弦
A1	周长

Question 28

P1-C07-R28 · 7 marks

(a) · 3 marks

$2.4(1-\cos\phi)=0.6;\ \phi=\arccos(3/4)\approx 0.723.$

端点竖直投影为2.4cosφ，升高是长度减去投影；按指定角度范围取唯一角。

M1	高度投影关系
A1	精确余弦
A1	弧度值

(b) During one complete swing from the left-hand extreme to the right-hand extreme, find, to 3 significant figures,

(i) · 2 marks

$4.8\arccos(3/4)\approx 3.47$ m.

转过总角2φ，弧长2.4×2φ。

M1	使用总角2phi
A1	路程

(ii) · 2 marks

$1.2\sqrt{7}\approx 3.17$ m.

两端对称，弦长2×2.4sinφ；sinφ=√7/4。

M1	弦长或对称投影
A1	距离

Question 29

P1-C07-R29 · 8 marks

(a) · 4 marks

Area $(180-25\pi)\ \text{cm}^2$; perimeter $(36+10\pi)\ \text{cm}$.

两半圆合为整圆去除，面积减25π；两条短直边由两半圆弧替代，保留两条长边36。

M1	两半圆面积合并
A1	面积
M1	正确识别边界
A1	周长

(b) Each semicircular cut is replaced by a sector with the same radius and with its arc length reduced by 20%. For this comparison, consider one cut only. Find

(i) · 2 marks

$4\pi/5$ radians.

原半圆弧角π，固定半径下弧长和角成比例，取0.8π。

M1	固定半径弧角比例
A1	角度

(ii) · 2 marks

$5\pi/2\ \text{cm}^2$.

一半圆面积25π/2；同半径角减少20%，面积减少其20%=5π/2。

M1	固定半径面积比例
A1	面积变化

Question 30

P1-C07-R30 · 8 marks

(a) · 3 marks

$r\theta = 16$, $2r\sin(\theta/2) = 8\sqrt{3}$; divide to obtain the result.

分别列弧长与弦长关系，消去r；不要求用超出阶段范围的方法求这个超越方程。

M1	弧长关系
M1	弦长关系
A1	消去半径目标式

(b) A designer proposes $\theta = \pi/3$.

(i) · 2 marks

$r = 48/\pi$ cm.

仅使用rθ=16并代入提议角。

M1	代弧长式
A1	半径

(ii) · 3 marks

No; proposed chord $48/\pi \approx 15.3$ cm, required $8\sqrt{3} \approx 13.9$ cm.

θ/2=π/6时2r sin(π/6)=r，分别比较未舍入值，明显不等。

M1	计算提议弦长
A1	两比较值
B1	不满足结论

Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 8 — Differentiation
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第8章 · 微分
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C08-R01 · P1 · Chapter 8

The curve C has equation $y = 2x^3 - 5x^2 + 4x - 7$.

(a) Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$. (4)

(b) Find the gradient of C at the point where $x = -1$. (2)

(Total for Question 1 is 6 marks)

C is the curve $y = x^2 + 3x - 2$. Point P on C has $x = 2$.

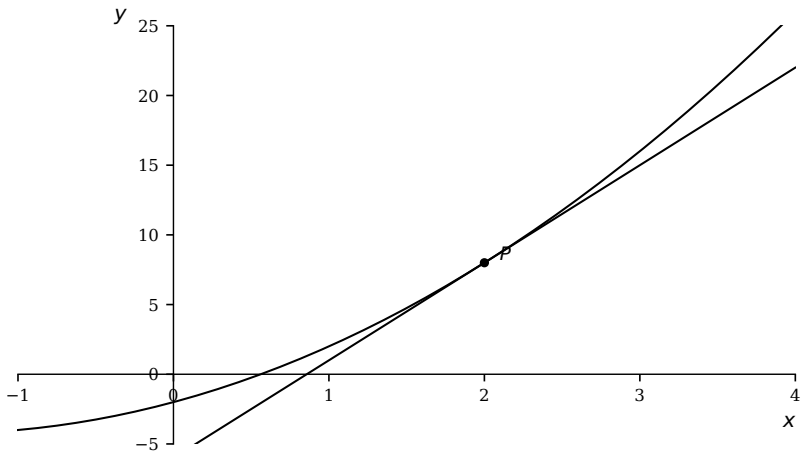


Figure 2

(a) Find an equation of the tangent to C at P. (3)

(b) Find where this tangent meets the x-axis and the y-axis. (2)

(Total for Question 2 is 5 marks)

3.

The curve C has equation $y = 4\sqrt{x} + \dfrac{3}{x}$ for $x > 0$.

(a) Find dy/dx, using powers of x. (2)

(b) Find the equation of the normal at the point where $x = 1$. (3)

(Total for Question 3 is 5 marks)

4.

A curve has equation $y = ax^3 + bx^2 + 1$. At the point $(1, 4)$, its tangent has gradient 5.

(a) Form two equations in a, b and find a, b. (4)

(b) Find the value of d^2y/dx^2 at $x = 2$. (2)

(Total for Question 4 is 6 marks)

$y = \dfrac{3x^3 - 2x + 5}{x}$, where $x \neq 0$.

(a) Express y as a sum of powers of x and hence find dy/dx. (3)

(b) Find the gradient at $x = -1$ and the value of y there. (2)

(Total for Question 5 is 5 marks)

6.

The curve C is $y = x^3 - 6x + 2$.

(a) Find the exact x-coordinates of the points where the tangent is parallel to the x-axis. (3)

(b) Find the exact y-coordinates of these points. (2)

(Total for Question 6 is 5 marks)

7.

At the point $P = (2, 5)$ on a curve, the gradient is $-\frac{1}{2}$.

(a) Find equations of the tangent and normal at P. (4)

(b) Find the area of the triangle formed by the tangent and the coordinate axes. (3)

(Total for Question 7 is 7 marks)

8. P1-C08-R08 · P1 · Chapter 8

C has equation $y = (x - 2)(x^2 + x + 1)$.

(a) Expand the equation and find dy/dx. (3)

(b) Find all points on C where the tangent has gradient 7. (3)

(Total for Question 8 is 6 marks)

A road profile is modelled by $h = 0.02x^2 + 0.3x + 4$ metres for $0 \leq x \leq 20$, where x is horizontal distance in metres.

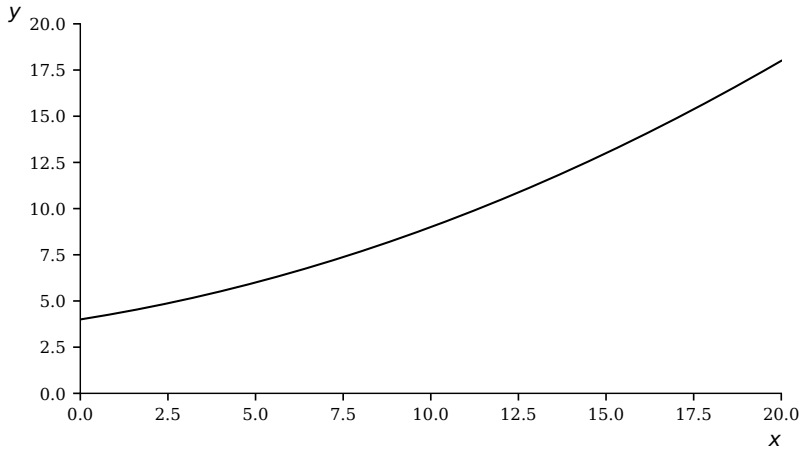


Figure 9

(a) Find dh/dx and its value at $x = 10$. (3)

(b) The gradient must not exceed 0.9. Find the part of the modelled road satisfying this condition. (3)

(Total for Question 9 is 6 marks)

The curve C has equation $y = 5x^{3/2} - 2x^{1/2}$ for $x > 0$.

(a) Find dy/dx. (2)

(b) At the point where $x = 4$, find

(i) the coordinates of the point and the tangent gradient, (3)

(ii) an equation of the tangent. (2)

(Total for Question 10 is 7 marks)

A curve is $y = ax^2 + bx + c$. Its tangent at $(0, 3)$ is $y = -4x + 3$, and the curve passes through $(2, 7)$.

(a) Find a, b, c. (4)

(b) Find where the curve has a tangent parallel to $y = 8x + 1$. (3)

(Total for Question 11 is 7 marks)

C is the curve $y = \dfrac{8}{x^2} + x$, with $x \neq 0$.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Find the equation of the tangent at $x = -2$. (3)

(Total for Question 12 is 7 marks)

13. P1-C08-R13 · P1 · Chapter 8

The curve C is $y = x^2 - 6x + 10$. A normal to C is parallel to $y = \frac{1}{2}x + 4$.

(a) Find the point of contact of this normal. (3)

(b) Find the equation of the normal and its x-intercept. (3)

(Total for Question 13 is 6 marks)

$f(x) = 3x^4 - 8x^3 + 2x.$

(a) Find $f''(x)$. (3)

(b) Find the values of x for which $f''(x) = 0$, and evaluate $f'(x)$ at each. (4)

(Total for Question 14 is 7 marks)

15. P1-C08-R15 · P1 · Chapter 8

The curve C is $y = x^3 + px + q$. At $P = (1, 2)$, its tangent is parallel to $y = 5x - 7$.

(a) Find p and q. (3)

(b) Find the equation of the normal at P. (3)

(Total for Question 15 is 6 marks)

16. P1-C08-R16 · P1 · Chapter 8

C is the curve $y = (2x + 1)^2$.

(a) By expanding first, find dy/dx. (3)

(b) At what point on C is the tangent perpendicular to the line $2x + 8y = 3$? (3)

(Total for Question 16 is 6 marks)

At a point P on $y = x^3 - 2x^2 + 1$, the tangent has gradient -1.

(a) Find all possible x-coordinates of P. (3)

(b) For $x = 1$, find the normal and its intersection with the y-axis. (4)

(Total for Question 17 is 7 marks)

18. P1-C08-R18 · P1 · Chapter 8

A curve is $y = 6x^{-1/2} + 2x$, for $x > 0$.

(a) Find the derivative and its value at $x = 9$. (3)

(b) Find the equation of the tangent at $x = 9$. (3)

(Total for Question 18 is 6 marks)

19. P1-C08-R19 · P1 · Chapter 8

The tangent to $C : y = x^2 + 2x$ at P passes through the origin. Let the x-coordinate of P be a.

(a) Show that $a = 0$. (3)

(b) Find the equation of this tangent and its perpendicular line through $(2, 1)$. (3)

(Total for Question 19 is 6 marks)

C is $y = 2x^3 - 3x^2 + 1$.

(a) Find the equations of the tangents at the points with $x = 0$ and $x = 1$. (3)

(b) Find the distance between these two tangents. (2)

(Total for Question 20 is 5 marks)

21. P1-C08-R21 · P1 · Chapter 8

The curve C is $y = x^4 - 4x^2$.

(a) Find dy/dx and factorise it completely. (3)

(b) Find all points on C with a horizontal tangent. (3)

(Total for Question 21 is 6 marks)

Curves C_1 and C_2 are $y = x^2 + ax$ and $y = bx - 1$. They touch at the point with $x = 1$.

(a) Find $a - b$. (2)

(b) Given also that the common tangent has gradient 3, find a, b and the point of contact. (3)

(Total for Question 22 is 5 marks)

$C : y = x + \dfrac{4}{x}$, where $x > 0$.

(a) Find the value of x for which the tangent is horizontal. (3)

(b) Find equations of the tangent and normal at this point. (3)

(Total for Question 23 is 6 marks)

The curve C is $y = \dfrac{x^2 + 3x - 4}{\sqrt{x}}$ for $x > 0$.

(a) Express y as a sum of powers of x and find dy/dx. (3)

(b) Find the equation of the tangent at the point on C with $x = 1$. (3)

(Total for Question 24 is 6 marks)

The curve C is $y = x^3 - 3x + 4$. The tangent at P is parallel to the tangent at Q, where $x_P = 2$ and $x_Q \neq 2$.

(a) Find x_Q and the coordinates of P, Q. (3)

(b) Find the y-intercepts of these two tangents. (2)

(Total for Question 25 is 5 marks)

$f(x) = ax^4 + bx^2$. It is given that $f'(1) = 12$ and $f''(1) = 28$.

(a) Find a, b. (4)

(b) Find $f(-1)$ and the gradient of the curve there. (2)

(Total for Question 26 is 6 marks)

The curve C is $y = 3x^2 - 2x + 5$.

(a) Find the point on C where the normal has gradient $-\frac{1}{4}$. (3)

(b) Find the exact area of the triangle formed by this normal and the coordinate axes. (4)

(Total for Question 27 is 7 marks)

A curve has equation $y = x^3 + kx^2$. Its tangent at the point with $x = 1$ passes through $(0, -3)$.

(a) Find k. (3)

(b) For this value of k, find the tangent and the normal at $x = 1$. (4)

(Total for Question 28 is 7 marks)

A curve C has equation $y = x^3 - 3x^2 + 4$. Point P on C has x-coordinate 1.

(a) Find the equation of the normal to C at P. (3)

(b) The normal meets the y-axis at N.

(i) Find the exact area of triangle OPN, where O is the origin. (2)

(ii) Find the equations of the tangents to C that are parallel to the x-axis. (2)

(Total for Question 29 is 7 marks)

The curve $y = f(x)$ passes through $(1, 4)$ and has gradient $f\prime(x) = 6x^2 - 8x$.

(a) Find $f(x)$. (2)

(b) The line $y = k$ is tangent to the curve.

(i) Find both possible values of k. (2)

(ii) For $k = 6$, find the x-coordinate of the other intersection of the line and curve. (2)

(Total for Question 30 is 6 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C08-R01 · 6 marks

(a) · 4 marks

$6x^2 - 10x + 4$; $12x - 10$.

逐项用幂函数法则，常数导数为0；再对一阶导求导。

M1	一阶幂法则
A1	一阶导
M1	再求导
A1	二阶导

(b) · 2 marks

20.

代入一阶导6+10+4；不是把x代回原曲线。

M1	代入正确导函数
A1	梯度20

Question 2

P1-C08-R02 · 5 marks

(a) · 3 marks

$y = 7x - 6$.

P=(2,8)，导数2x+3在2处为7；y−8=7(x−2)。

B1	正确P
M1	求切线斜率
A1	切线方程

(b) · 2 marks

$(\frac{6}{7}, 0)$ and $(0, -6)$.

对切线方程分别令y=0、x=0。

M1	使用切线而非曲线
A1	两截距

Question 3

P1-C08-R03 · 5 marks

(a) · 2 marks

$2x^{-1/2} - 3x^{-2}$.

将根式写为4x^{1/2}、分式写为3x^{-1}后逐项求导。

M1	写成幂并使用法则
A1	两项正确

(b) · 3 marks

$y = x + 6$.

点(1,7)，切线斜率2−3=−1，法线斜率1。

B1	曲线上点
M1	求切线及负倒数
A1	法线式

Question 4

P1-C08-R04 · 6 marks

(a) · 4 marks

$a+b=3,\ 3a+2b=5;\ a=-1,b=4.$

点条件代原式，梯度条件代导式；消元a=−1,b=4。

M1	正确点条件
M1	正确导数条件
M1	解方程组
A1	两系数

(b) · 2 marks

$-4.$

代系数后二阶导为−6x+8，x=2时−4。

M1	正确二阶导
A1	代值

Question 5

P1-C08-R05 · 5 marks

(a) · 3 marks

$y = 3x^2 - 2 + 5x^{-1}$; $dy/dx = 6x - 5x^{-2}$.

先逐项除x，常数−2导数为0。

M1	正确逐项除法
M1	对化简式求导
A1	完整导式

(b) · 2 marks

Gradient -11; $y = -4$.

导式代−1得−6−5；原化简式给3−2−5。

M1	分别用导式及原式
A1	两值

Question 6

P1-C08-R06 · 5 marks

(a) · 3 marks

$x = \pm\sqrt{2}$.

水平切线要求$3x^2-6=0$，保留正负。

M1	求导并令零
M1	解平方关系
A1	两根

(b) · 2 marks

$y = 2-4\sqrt{2}$ at $x=\sqrt{2}$; $y = 2+4\sqrt{2}$ at $x=-\sqrt{2}$.

在这些点$x^3=2x$，所以$y=2-4x$；配对对应两个x。

M1	正确代入或利用x平方关系
A1	两个配对纵坐标

Question 7

P1-C08-R07 · 7 marks

(a) · 4 marks

Tangent $y = -x/2 + 6$; normal $y = 2x + 1$.

切线用已给斜率；法线斜率为负倒数2，两条都通过P。

M1	切线点斜式
A1	切线
M1	法线斜率
A1	法线

(b) · 3 marks

36 square units.

切线截距$(12,0)$、$(0,6)$，面积$12\times 6/2$。

M1	求切线两截距
M1	正确面积式
A1	面积

Question 8

P1-C08-R08 · 6 marks

(a) · 3 marks

$y = x^3 - x^2 - x - 2$; $dy/dx = 3x^2 - 2x - 1$.

先展开两个因子，再逐项求导。

M1	正确展开
M1	幂法则
A1	导函数

(b) · 3 marks

$(2, 0)$ and $(-\frac{4}{3}, -\frac{130}{27})$.

$3x^2-2x-1=7$给$3x^2-2x-8=0$，$x=2$或$-4/3$；代原式求y。

M1	梯度方程
M1	解两横坐标
A1	两点

Question 9

P1-C08-R09 · 6 marks

(a) · 3 marks

$dh/dx = 0.04x + 0.3$; value 0.7.

按幂法则求导，代x=10。

M1	正确求导
A1	导式
A1	数值

(b) · 3 marks

$0 \le x \le 15$.

0.04x+0.3⩽0.9给x⩽15，再与模型0⩽x⩽20求交。

M1	建立梯度不等式
M1	使用模型定义域
A1	区间

Question 10

P1-C08-R10 · 7 marks

(a) · 2 marks

$\frac{15}{2}x^{1/2} - x^{-1/2}$.

每项系数乘原指数，指数减1。

M1	两项使用分数指数法则
A1	完整导式

(b) At the point where $x = 4$, find

(i) · 3 marks

Point $(4, 36)$; gradient $29/2$.

y=5×8−2×2=36；导数15−1/2=29/2。

B1	点坐标
M1	导式代值
A1	梯度

(ii) · 2 marks

$y = \frac{29}{2}x - 22$.

y−36=(29/2)(x−4)，展开。

M1	正确点斜式
A1	最终方程

Question 11

P1-C08-R11 · 7 marks

(a) · 4 marks

$a = 3, b = -4, c = 3.$

点(0,3)给c=3，切线斜率在0处给b=−4，再由7=4a−8+3求a=3。

B1	c
M1	用切线求b
M1	代另点求a
A1	三系数

(b) · 3 marks

$(2, 7).$

y'=6x−4=8给x=2，再用原曲线求y=7。

M1	梯度匹配
M1	回代原曲线
A1	点

Question 12

P1-C08-R12 · 7 marks

(a) · 4 marks

$1-16x^{-3}$; $48x^{-4}$.

写8x^{−2}再求导两次。

M1	一阶负幂求导
A1	一阶导
M1	再求导
A1	二阶导

(b) · 3 marks

$y=3x+6$.

点(−2,0)，导数1−16/(−8)=3，故y=3(x+2)。

B1	点
M1	求梯度
A1	方程

Question 13

P1-C08-R13 · 6 marks

(a) · 3 marks

$(2, 2)$.

法线斜率1/2意味着切线斜率−2；2x−6=−2给x=2，代回得2。

M1	法切斜率对应
M1	解横坐标
A1	接触点

(b) · 3 marks

$y = \frac{1}{2}x + 1$; $(-2, 0)$.

过(2,2)用斜率1/2，随后令y=0。

M1	正确法线点斜式
A1	方程
A1	截距

Question 14

P1-C08-R14 · 7 marks

(a) · 3 marks

$36x^2 - 48x$.

先得f'=12x^3−24x^2+2，再求导。

M1	正确一阶导
M1	再求导
A1	二阶导

(b) · 4 marks

$x = 0, 4/3$; $f'(0) = 2$, $f'(4/3) = -110/9$.

二阶导12x(3x−4)=0；代入一阶导得到2和−110/9。

M1	解二阶导零点
A1	两x
M1	代一阶导
A1	两梯度

Question 15

P1-C08-R15 · 6 marks

(a) · 3 marks

$p = 2, q = -1.$

3+p=5给p=2；1+p+q=2给q=−1。

M1	正确梯度条件
M1	正确点条件
A1	两参数

(b) · 3 marks

$y = -\frac{1}{5}x + \frac{11}{5}.$

法线斜率−1/5且过(1,2)。

M1	正确法线斜率
M1	点斜式
A1	方程

Question 16

P1-C08-R16 · 6 marks

(a) · 3 marks

$dy/dx = 8x + 4$.

展开$4x^2+4x+1$，再用幂法则。

M1	正确展开
M1	逐项求导
A1	导式

(b) · 3 marks

$(0, 1)$.

给定线斜率$-1/4$，切线应斜率4；$8x+4=4$给$x=0$。

M1	从直线确定所需切线斜率
M1	解梯度条件
A1	点

Question 17

P1-C08-R17 · 7 marks

(a) · 3 marks

$x = \frac{1}{3}, 1.$

$3x^2-4x=-1$给$(3x-1)(x-1)=0$。

M1	求导建立条件
M1	分解求根
A1	两x

(b) · 4 marks

$y = x - 1;\ (0, -1).$

点$(1,0)$，切线斜率-1，法线斜率1。

B1	点
M1	垂线斜率及过点
A1	法线
A1	纵截距

Question 18

P1-C08-R18 · 6 marks

(a) · 3 marks

$dy/dx = -3x^{-3/2} + 2$; value $17/9$.

9^{3/2}=27，故−3/27+2=17/9。

M1	正确负半次幂求导
A1	导式
A1	梯度

(b) · 3 marks

$y = \frac{17}{9}x + 3$.

原曲线y=6/3+18=20；点斜式y−20=(17/9)(x−9)。

B1	正确纵坐标
M1	点斜式
A1	方程

Question 19

P1-C08-R19 · 6 marks

(a) · 3 marks

Tangent is $y=(2a+2)x-a^2$; through the origin gives $a^2=0$.

P=(a,a²+2a)，点斜式整理后的截距为−a²，令其为0。

M1	用a写点与导数
M1	正确切线截距
A1	得a=0

(b) · 3 marks

Tangent $y=2x$; perpendicular $y=-x/2+2$.

a=0时导数2；垂线斜率−1/2，过(2,1)。

B1	切线
M1	垂线斜率及点
A1	垂线方程

Question 20

P1-C08-R20 · 5 marks

(a) · 3 marks

$y = 1$ and $y = 0$, respectively.

导数$6x^2-6x$在0和1都为0；对应纵坐标1和0，所以两条水平切线。

M1	求导并正确代入
M1	求两纵坐标
A1	两水平线

(b) · 2 marks

1.

两条水平直线的垂直距离为纵截距差$|1-0|$。

M1	使用水平线垂直距离
A1	结果

Question 21

P1-C08-R21 · 6 marks

(a) · 3 marks

$4x(x^2-2)=4x(x-\sqrt{2})(x+\sqrt{2})$.

逐项导出$4x^3-8x$，提公因式并平方差分解。

M1	正确求导
M1	提公因式
A1	完全因式分解

(b) · 3 marks

$(0,0)$ and $(\pm\sqrt{2},-4)$.

导数为0的x为0、$\pm\sqrt{2}$，分别代入原式。

M1	导式零点
M1	回代三点
A1	全部坐标

Question 22

P1-C08-R22 · 5 marks

(a) · 2 marks

$a - b = -2$.

接触点具有相同纵坐标：1+a=b−1给a−b=−2；导数相等也给2+a=b。

M1	同点条件
A1	参数差

(b) · 3 marks

$a = 1, b = 3$; contact $(1, 2)$.

切线是C2本身，b=3；差−2给a=1，代点求y=2。

M1	用共同斜率求b
M1	求a与回代
A1	参数及接触点

Question 23

P1-C08-R23 · 6 marks

(a) · 3 marks

$x = 2$.

y′=1−4/x²=0给x²=4，由定义域选2。

M1	正确求导
M1	解并使用定义域
A1	x

(b) · 3 marks

Tangent $y = 4$; normal $x = 2$.

点(2,4)，水平切线的法线是竖直线，不能写成有限斜率的y形式。

B1	正确点
A1	水平切线
A1	竖直法线

Question 24

P1-C08-R24 · 6 marks

(a) · 3 marks

$y = x^{3/2} + 3x^{1/2} - 4x^{-1/2}$; $y\prime = \frac{3}{2}x^{1/2} + \frac{3}{2}x^{-1/2} + 2x^{-3/2}$.

每项除以x^{1/2}，再按幂法则求导；最后一项负系数乘负指数变正。

M1	正确三个幂
M1	分数幂求导
A1	导函数

(b) · 3 marks

$y = 5x - 5$.

点(1,0)，梯度3/2+3/2+2=5。

B1	点
M1	导式代值
A1	方程

Question 25

P1-C08-R25 · 5 marks

(a) · 3 marks

$x_Q = -2$; $P = (2, 6)$, $Q = (-2, 2)$.

导数$3x^2-3$；在2处为9，解$3x^2-3=9$得± 2，排除2。

M1	求P梯度
M1	相等梯度方程
A1	两点

(b) · 2 marks

-12 at P; 20 at Q.

两条线斜率9；截距$y-9x$分别$6-18$与$2+18$。

M1	正确切线截距关系
A1	两个截距

Question 26

P1-C08-R26 · 6 marks

(a) · 4 marks

$a = 2, b = 2$.

4a+2b=12；12a+2b=28；相减8a=16。

M1	一阶条件
M1	二阶条件
M1	消元
A1	两参数

(b) · 2 marks

$f(-1) = 4$; gradient -12.

原式在±1同值，导式$8x^3+4x$在−1给−12。

M1	代入原式和导式
A1	两值

Question 27

P1-C08-R27 · 7 marks

(a) · 3 marks

$(1, 6)$.

切线斜率4，6x−2=4给x=1，代曲线y=6。

M1	由法线求切线斜率
M1	解条件
A1	点

(b) · 4 marks

$\frac{625}{8}$ square units.

法线y=−x/4+25/4，截距25和25/4，面积1/2×25×25/4。

M1	正确法线
M1	求两截距
M1	面积式
A1	精确面积

Question 28

P1-C08-R28 · 7 marks

(a) · 3 marks

$k = 1$.

切点纵坐标1+k、梯度3+2k，切线截距(1+k)−(3+2k)=−2−k；令等于−3。

M1	表示点与梯度
M1	建立截距条件
A1	k

(b) · 4 marks

Tangent $y = 5x - 3$; normal $y = -x/5 + 11/5$.

切点(1,2)，切线斜率5；法线斜率−1/5。

M1	代入参数求点和斜率
A1	切线
M1	法线负倒数
A1	法线

Question 29

P1-C08-R29 · 7 marks

(a) · 3 marks

$y = \frac{1}{3}x + \frac{5}{3}$.

$P = (1, 2)$，梯度$3x^2 - 6x$在1处为-3，法线斜率$1/3$。

M1	求导并求切线梯度
M1	负倒数
A1	法线方程

(b) The normal meets the y-axis at N.

(i) · 2 marks

$5/6$.

$N = (0, 5/3)$，以ON为底、P到y轴的距离1为高，面积$\frac{1}{2}(5/3)(1) = 5/6$。

M1	正确底与高
A1	面积

(ii) · 2 marks

$y = 4$ and $y = 0$.

$3x(x-2) = 0$给$x = 0, 2$，曲线上对应纵坐标为$4, 0$。

M1	导数为零
A1	两条切线

Question 30

P1-C08-R30 · 6 marks

(a) · 2 marks

$f(x) = 2x^3 - 4x^2 + 6$.

逐项积分得$2x^3 - 4x^2 + c$，代入$(1, 4)$得$c = 6$。

M1	反向使用幂法则
A1	积分常数

(b) The line $y = k$ is tangent to the curve.

(i) · 2 marks

$k = 6$ or $k = 98/27$.

水平切线要求$6x^2 - 8x = 0$，即$x = 0, 4/3$；代入f得$6, 98/27$。

M1	水平切线条件
A1	两个纵坐标

(ii) · 2 marks

$x = 2$.

$f(x) = 6$化为$2x^2(x - 2) = 0$。$x = 0$为切点，另一交点为$x = 2$。

M1	方程因式分解
A1	排除切点后的交点

Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 9 — Integration
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第9章 · 积分
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C09-R01 · P1 · Chapter 9

The curve C has gradient function $\frac{dy}{dx} = 6x^2 - 8x + 3$ and passes through $(1, 5)$.

(a) Find an equation of C. (4)

(b) Find the equation of the tangent at $(1, 5)$. (3)

(Total for Question 1 is 7 marks)

For $x > 0$, $\dfrac{dy}{dx} = 3\sqrt{x} - \dfrac{4}{x^2}$. The curve passes through $(4, 7)$.

(a) Find y in terms of x. (4)

(b) Find the point on the curve with $x = 1$. (2)

(Total for Question 2 is 6 marks)

A curve has $\frac{d^2y}{dx^2} = 12x - 6$. At $x = 1$, the curve has gradient 4 and passes through $(1, 2)$.

(a) Find dy/dx. (3)

(b) Find the equation of the curve. (3)

(Total for Question 3 is 6 marks)

Find each indefinite integral, simplifying your answer.

(a) $\int (8x^3 - 6x + 5)\, dx.$ (3)

(b) $\int \dfrac{6x^4 - 4x}{x^3}\, dx$, for $x > 0$. (2)

(Total for Question 4 is 5 marks)

5. P1-C09-R05 · P1 · Chapter 9

The derivative of $f(x)$ is $4x^3 + 2x$. It is given that $f(0) = f(1) - 2$.

(a) Show that this condition does not determine the integration constant. (3)

(b) Given also that $f(2) = 23$, find $f(x)$. (2)

(Total for Question 5 is 5 marks)

6.

The curve C has $dy/dx = (x - 1)(3x + 2)$ and passes through the origin.

(a) Find an equation of C. (4)

(b) Find the exact coordinates of the points where C has a horizontal tangent. (3)

(Total for Question 6 is 7 marks)

For $x > 0$, $dy/dx = 5x^{3/2} - 3x^{-1/2}$. At $x = 1$, the tangent to the curve is $y = 2x + 1$.

(a) Find the coordinates of the point of contact. (2)

(b) Find the equation of the curve. (4)

(Total for Question 7 is 6 marks)

8. P1-C09-R08 · P1 · Chapter 9

A curve has $dy/dx = ax + b$. Its gradient is 1 at $x = 0$ and 7 at $x = 3$. It passes through $(2, 9)$.

(a) Find a, b. (3)

(b) Find an equation of the curve and its y-intercept. (4)

(Total for Question 8 is 7 marks)

9. P1-C09-R09 · P1 · Chapter 9

The curve C has $dy/dx = 4x - 6$ and passes through $(3, 5)$.

(a) Find an equation of C. (3)

(b) Find the normal at $(3, 5)$ and where it meets the y-axis. (3)

(Total for Question 9 is 6 marks)

10. P1-C09-R10 · P1 · Chapter 9

Two curves have the same gradient function $dy/dx = 3x^2 - 2$. One passes through $(0, 4)$ and the other through $(2, 1)$.

(a) Find equations of both curves. (4)

(b) Describe the single translation mapping C_1 onto C_2. (2)

(Total for Question 10 is 6 marks)

Find the following integrals for $x > 0$.

(a) $\displaystyle\int \frac{9x^2 + 3}{\sqrt{x}}\,dx.$ (4)

(b) $\displaystyle\int (\sqrt{x} - 2)^2\,dx.$ (4)

(Total for Question 11 is 8 marks)

The curve C satisfies $dy/dx = 6x^2 + k$. It passes through $(0, 1)$ and $(1, 6)$.

(a) Find k and the equation of C. (4)

(b) Find the tangent to C at $(1, 6)$. (3)

(Total for Question 12 is 7 marks)

13.

The gradient of a curve is $dy/dx = 12x^2 - 12x$. The curve has a horizontal tangent at a point with coordinates $(1, 3)$.

(a) Find the equation of the curve. (3)

(b) Find the other point where the tangent is horizontal and state the equations of both horizontal tangents. (3)

(Total for Question 13 is 6 marks)

For $x > 0$, $dy/dx = \dfrac{2x^2 - 3}{x^2}$. The curve passes through $(3, 4)$.

(a) Find y. (4)

(b) Find all intersections of the curve with $y = 2$. (3)

(Total for Question 14 is 7 marks)

15.

A curve has $d^2y/dx^2 = 6x + 4$. Its tangent at the origin is $y = -2x$.

(a) Find dy/dx. (3)

(b) Find the equation of the curve. (3)

(Total for Question 15 is 6 marks)

$F(x) = ax^4 + bx^2 + c$ is an antiderivative of $12x^3 - 10x$. It is given that $F(1) = 0$.

(a) Find a, b, c. (4)

(b) Factorise $F(x)$ completely into factors with integer coefficients. (3)

(Total for Question 16 is 7 marks)

17.

The gradient of C at any x is $8x + 6$. At the point where $x = 1$, the normal passes through $(0, 5)$.

(a) Find the coordinates of the point on C with $x = 1$. (3)

(b) Find the equation of C. (3)

(Total for Question 17 is 6 marks)

18.

For $x > 0$, $dy/dx = \dfrac{5}{\sqrt{x}} + \dfrac{6}{x^3}$. It is given that $y = 17$ when $x = 1$.

(a) Find y in terms of x. (4)

(b) Find y when $x = 4$, giving an exact answer. (2)

(Total for Question 18 is 6 marks)

19.

A curve has $dy/dx = 2ax + b$. It passes through $(0, 2)$ and $(2, 10)$ and has gradient 3 at $x = 1$.

(a) Show that no such curve exists. (4)

(b) If the stated gradient is changed from 3 to 4, determine whether a is uniquely fixed. Give a reason. (2)

(Total for Question 19 is 6 marks)

The curve C has $dy/dx = 3x^2 - 6x + 2$ and passes through $(2, 1)$.

(a) Find the equation of C. (3)

(b) Find the equation of the tangent at the y-intercept of C. (3)

(Total for Question 20 is 6 marks)

For $x > 0$, $dy/dx = \frac{3}{2}x^{1/2} + 4$. The curve passes through $(4, 25)$.

(a) Find y. (3)

(b) Find the tangent at $x = 4$ and where this tangent meets the y-axis. (3)

(Total for Question 21 is 6 marks)

The curve C has $d^2y/dx^2 = 24x^2 - 4$. It has a horizontal tangent at $(1, 2)$.

(a) Find dy/dx. (3)

(b) Find the equation of C. (3)

(Total for Question 22 is 6 marks)

The derivative of $f(x)$ is $(2x-1)^2$.

(a) Find $f(x)$ in terms of an arbitrary constant. (4)

(b) Given that $f(0) = 2$, find

(i) $f(\frac{1}{2})$, (2)

(ii) the gradient of $y = f(x)$ when $x = \frac{1}{2}$. (2)

(Total for Question 23 is 8 marks)

The curve C has $dy/dx = 5x^4 - 4x^3$. Its y-intercept is -2.

(a) Find an equation of C. (3)

(b) Find the coordinates of the two points where $dy/dx = 0$, and state the equation of the tangent at $x = 0$. (4)

(Total for Question 24 is 7 marks)

A curve has $dy/dx = 2x - \dfrac{6}{x^2}$ for $x > 0$. Its tangent at $x = 1$ passes through $(0, 10)$.

(a) Find the coordinates of the point on the curve with $x = 1$. (3)

(b) Find an equation of the curve. (3)

(Total for Question 25 is 6 marks)

For a curve C, $dy/dx = 6x^2 + ax$ and $y(0) = 3$. At $x = 2$, the tangent is parallel to $y = 16x + 1$.

(a) Find a. (2)

(b) Find the equation of C and the equation of its tangent at $x = 2$. (4)

(Total for Question 26 is 6 marks)

Find the following indefinite integrals for $x > 0$.

(a) $\displaystyle\int x^{1/3}(x+2)\,dx.$ (4)

(b) $\displaystyle\int \frac{x^4-1}{x^2}\,dx.$ (4)

(Total for Question 27 is 8 marks)

28. P1-C09-R28 · P1 · Chapter 9

The gradient function of C is $dy/dx = 4x^3 - 6x^2$. The curve meets the x-axis at $x = 2$.

(a) Find the equation of C. (3)

(b) Find all x-intercepts of C and state which also have a horizontal tangent. (3)

(Total for Question 28 is 6 marks)

The curve $y = f(x)$ has equation $f(x) = ax^2 + b$, where a, b are constants. Its tangent at $x = 2$ has gradient 8. A function F satisfies $F\prime(x) = f(x)$, $F(0) = 1$ and $F(2) = 11$.

(a) Find a and b. (3)

(b) For these values of a, b, find

(i) the equation of the tangent to $y = f(x)$ at $x = 2$. (2)

(ii) an expression for $F(x)$. (2)

(Total for Question 29 is 7 marks)

For $x > 0$, $f(x) = \dfrac{3x^2 + 2x - 4}{\sqrt{x}}$.

(a) Find an expression for $\int f(x)\,dx$, giving each term in simplest form. (3)

(b) A function F satisfies $F\prime(x) = f(x)$ and $F(1) = 0$.

(i) Find $F(4)$ exactly. (2)

(ii) A second function G satisfies $G\prime(x) = 2f(x) + 3$ and $G(1) = 5$. Find $G(4)$ without integrating f again. (3)

(Total for Question 30 is 8 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C09-R01 · 7 marks

(a) · 4 marks

$y = 2x^3 - 4x^2 + 3x + 4.$

逐项积分得$2x^3-4x^2+3x+C$；代$(1,5)$得$C=4$。

M1	正确积分三项
M1	加常数并代点
A1	积分常数
A1	完整方程

(b) · 3 marks

$y = x + 4.$

由给定梯度函数在1处得1；过$(1,5)$写点斜式。

M1	从梯度式求斜率
M1	点斜式
A1	方程

Question 2

P1-C09-R02 · 6 marks

(a) · 4 marks

$y = 2x^{3/2} + 4x^{-1} - 10$.

将根式和分式写成幂，积分得2x^{3/2}+4/x+C，代4得7=16+1+C。

M1	分数负指数积分
M1	常数条件
A1	常数负10
A1	方程

(b) · 2 marks

$(1, -4)$.

代入已确定常数的曲线得2+4−10=−4。

M1	使用完整曲线而非梯度
A1	坐标

Question 3

P1-C09-R03 · 6 marks

(a) · 3 marks

$6x^2 - 6x + 4$.

第一次积分加常数A；在1处6−6+A=4，所以A=4。

M1	一次积分
M1	用梯度条件
A1	完整一阶导

(b) · 3 marks

$y = 2x^3 - 3x^2 + 4x - 1$.

再积分并加另一个常数B，代(1,2)给2=2−3+4+B。

M1	二次积分
M1	新增独立常数及点条件
A1	方程

Question 4

P1-C09-R04 · 5 marks

(a) · 3 marks

$2x^4 - 3x^2 + 5x + C$.

逐项加1指数，再除以新指数，常数5积分成5x。

M1	幂积分
A1	三项
B1	积分常数

(b) · 2 marks

$3x^2 + 4x^{-1} + C$.

先化简为6x−4x^{-2}，再用幂积分；不含指数−1的被积项。

M1	化简
A1	积分式

Question 5

P1-C09-R05 · 5 marks

(a) · 3 marks

$f(x) = x^4 + x^2 + C$, so $f(1) - f(0) = 2$ for every C.

积分后代入两个点，C在差中抵消，给定条件对任意C都成立。

M1	正确积分含C
M1	代入两点并相减
A1	解释不确定性

(b) · 2 marks

$f(x) = x^4 + x^2 + 3$.

23=16+4+C，故C=3。

M1	使用新增绝对值条件
A1	常数与方程

Question 6

P1-C09-R06 · 7 marks

(a) · 4 marks

$y = x^3 - \frac{1}{2}x^2 - 2x.$

先展开导式$3x^2-x-2$，积分得$x^3-x^2/2-2x+C$；过原点给$C=0$。

M1	正确展开
M1	逐项积分
M1	原点条件
A1	方程

(b) · 3 marks

$(1, -\frac{3}{2})$ and $(-\frac{2}{3}, \frac{22}{27})$.

由已给因式得$x=1$或$-2/3$；代入曲线求y。

M1	梯度零点
M1	正确回代
A1	两坐标

Question 7

P1-C09-R07 · 6 marks

(a) · 2 marks

$(1, 3)$.

切点在给定切线上，将x=1代入其方程。

M1	正确使用切线条件
A1	点

(b) · 4 marks

$y = 2x^{5/2} - 6x^{1/2} + 7$.

积分后在1处y=−4+C=3，故C=7。

M1	正确两项积分
M1	用切点求常数
A1	C
A1	曲线

Question 8

P1-C09-R08 · 7 marks

(a) · 3 marks

$a = 2, b = 1.$

第一梯度直接给b=1；第二给3a+b=7。

B1	b
M1	第二梯度条件
A1	a

(b) · 4 marks

$y = x^2 + x + 3;\ (0, 3).$

积分2x+1得x^2+x+C，代(2,9)得C=3。

M1	积分
M1	点条件
A1	曲线
B1	截距

Question 9

P1-C09-R09 · 6 marks

(a) · 3 marks

$y = 2x^2 - 6x + 5$.

积分$2x^2-6x+C$，代3时前两项抵消，C=5。

M1	积分含常数
M1	代点
A1	曲线

(b) · 3 marks

$y = -x/6 + 11/2$; intercept $(0, 11/2)$.

梯度$4\times3-6=6$，法线斜率$-1/6$，过$(3,5)$。

M1	求切线与法线斜率
M1	点斜式
A1	方程与截距

Question 10

P1-C09-R10 · 6 marks

(a) · 4 marks

$C_1: y = x^3 - 2x + 4;\ C_2: y = x^3 - 2x - 3.$

共同原函数x^3-2x，各自常数由两点独立求得4和-3。

M1	正确共同原函数
M1	第一常数
M1	第二常数
A1	两曲线

(b) · 2 marks

$\begin{pmatrix}0\\-7\end{pmatrix}$.

同一x下纵坐标差恒为-7，只需下移7。

M1	比较常数差
A1	平移向量

Question 11

P1-C09-R11 · 8 marks

(a) · 4 marks

$\frac{18}{5}x^{5/2} + 6x^{1/2} + C.$

先化成9x^{3/2}+3x^{-1/2}，分别积分。

M1	两项幂化简
M1	分数幂积分
A1	两项系数
B1	常数

(b) · 4 marks

$\frac{1}{2}x^2 - \frac{8}{3}x^{3/2} + 4x + C.$

先展开成x−4√x+4，再积分，不能把整体指数直接加1。

M1	正确展开
M1	逐项积分
A1	三项
B1	常数

Question 12

P1-C09-R12 · 7 marks

(a) · 4 marks

$k = 3;\ y = 2x^3 + 3x + 1.$

积分$2x^3+kx+C$；首点给$C=1$，第二点给$6=2+k+1$。

M1	正确积分含两个未知常数
M1	第一条件
M1	第二条件
A1	k及曲线

(b) · 3 marks

$y = 9x - 3.$

梯度$6+3=9$，点斜式$y-6=9(x-1)$。

M1	梯度代值
M1	过点
A1	方程

Question 13

P1-C09-R13 · 6 marks

(a) · 3 marks

$y = 4x^3 - 6x^2 + 5$.

积分加C，在(1,3)处4−6+C=3给C=5；已给点的横坐标确使导数为0。

M1	积分
M1	使用点条件
A1	曲线

(b) · 3 marks

Other point $(0, 5)$; tangents $y = 3$ and $y = 5$.

给定导式=12x(x−1)，另一根0，曲线y(0)=5。

M1	找另一梯度零点
A1	另一点
A1	两水平线

Question 14

P1-C09-R14 · 7 marks

(a) · 4 marks

$y = 2x + 3x^{-1} - 3$.

导式先化为2−3x^{−2}，积分2x+3/x+C；在3处y=7+C=4。

M1	正确幂化简
M1	积分
M1	点条件
A1	方程

(b) · 3 marks

$(\frac{3}{2}, 2)$ and $(1, 2)$.

2x+3/x−3=2给$2x^2-5x+3=0$，因式(2x−3)(x−1)，两根均正。

M1	化为二次方程
M1	解并检查定义域
A1	两交点

Question 15

P1-C09-R15 · 6 marks

(a) · 3 marks

$3x^2 + 4x - 2.$

积分得$3x^2+4x+A$；原点切线斜率-2给$A=-2$。

M1	积分二阶导
M1	用斜率初值
A1	一阶导

(b) · 3 marks

$y = x^3 + 2x^2 - 2x.$

再次积分得x^3+2x^2-2x+B；曲线过原点给$B=0$。

M1	再积分
M1	单独确定第二常数
A1	方程

Question 16

P1-C09-R16 · 7 marks

(a) · 4 marks

$a=3, b=-5, c=2.$

积分对应a=3,b=−5；F(1)=3−5+c=0给c=2。

M1	对应四次项
M1	对应二次项
M1	点条件
A1	三常数

(b) · 3 marks

$F(x)=(3x^2-2)(x-1)(x+1).$

先作关于x^2的二次式，$3x^4-5x^2+2=(3x^2-2)(x^2-1)$，再平方差。

M1	关于x平方分解
M1	分解平方差
A1	完全因子

Question 17

P1-C09-R17 · 6 marks

(a) · 3 marks

$\left(1, \frac{69}{14}\right)$.

切线斜率14，法线−1/14；通过(0,5)故法线y=5−x/14，x=1时69/14。

M1	求法线斜率
M1	用已知法线点
A1	曲线上点

(b) · 3 marks

$y = 4x^2 + 6x - \frac{71}{14}$.

积分$4x^2$+6x+C；在1处10+C=69/14给C=−71/14。

M1	积分
M1	使用(a)点
A1	曲线

Question 18

P1-C09-R18 · 6 marks

(a) · 4 marks

$y = 10\sqrt{x} - 3x^{-2} + 10.$

积分5x^{-1/2}得10√x；积分6x^{-3}得−3x^{-2}；初值给C=10。

M1	第一项积分
M1	第二项积分
M1	确定常数
A1	完整式

(b) · 2 marks

$\frac{477}{16}$.

y=20−3/16+10=477/16。

M1	代入完整原函数
A1	精确值

Question 19

P1-C09-R19 · 6 marks

(a) · 4 marks

Point conditions imply $2a + b = 4$, but the gradient condition requires $2a + b = 3$.

积分y=ax²+bx+C，首点C=2；第二点给4a+2b=8，即2a+b=4，与导数在1处3矛盾。

M1	积分并用首点
M1	第二点约束
M1	梯度约束
A1	明确矛盾

(b) · 2 marks

No; any a with $b = 4 - 2a$ satisfies the conditions.

改后的梯度条件与两点推出的条件完全相同，只有一条独立参数关系。

M1	比较两条件是否独立
A1	正确解释并给参数关系

Question 20

P1-C09-R20 · 6 marks

(a) · 3 marks

$y = x^3 - 3x^2 + 2x + 1.$

积分各项，x=2时不含常数部分为0，故C=1。

M1	正确积分
M1	代入条件
A1	方程

(b) · 3 marks

$y = 2x + 1.$

y截距(0,1)，梯度在0处为2。

B1	截距点
M1	梯度
A1	方程

Question 21

P1-C09-R21 · 6 marks

(a) · 3 marks

$y = x^{3/2} + 4x + 1$.

积分得x^{3/2}+4x+C；8+16+C=25，所以C=1。

M1	分数指数积分
M1	初值条件
A1	曲线

(b) · 3 marks

$y = 7x - 3$; $(0, -3)$.

梯度3+4=7，过(4,25)截距25−28=−3。

M1	正确梯度
M1	点斜式
A1	切线与截距

Question 22

P1-C09-R22 · 6 marks

(a) · 3 marks

$8x^3 - 4x - 4.$

第一次积分$8x^3-4x+A$，水平切线使$8-4+A=0$。

M1	一次积分
M1	水平梯度条件
A1	一阶导

(b) · 3 marks

$y = 2x^4 - 2x^2 - 4x + 6.$

再积分$2x^4-2x^2-4x+B$，代点$1,2$给$B=6$。

M1	二次积分
M1	点条件
A1	完整式

Question 23

P1-C09-R23 · 8 marks

(a) · 4 marks

$f(x) = \frac{4}{3}x^3 - 2x^2 + x + C.$

先展开$4x^2-4x+1$，再逐项积分。

M1	正确展开
M1	幂积分
A1	三项
B1	任意常数

(b) Given that $f(0) = 2$, find

(i) · 2 marks

$\frac{13}{6}$.

$C=2$；代$1/2$得$1/6-1/2+1/2+2=13/6$。

M1	使用常数及分数代入
A1	精确值

(ii) · 2 marks

0.

直接把$1/2$代入已给导式$(2x-1)^2$。

M1	区分函数值和导数值
A1	梯度

Question 24

P1-C09-R24 · 7 marks

(a) · 3 marks

$y = x^5 - x^4 - 2$.

逐项积分，截距直接给常数−2。

M1	正确积分
M1	截距条件
A1	曲线

(b) · 4 marks

$(0, -2)$ and $(\frac{4}{5}, -\frac{6506}{3125})$; tangent $y = -2$.

导式$x^3(5x-4)=0$；x=4/5时代值为1024/3125−256/625−2=−6506/3125。

M1	找全部导数零点
M1	精确代回
A1	两点
B1	水平切线

Question 25

P1-C09-R25 · 6 marks

(a) · 3 marks

$(1, 6)$.

切线斜率-4，且截距10，故切线$y=-4x+10$，代1得6。

M1	求切线斜率
M1	用所给截距
A1	点

(b) · 3 marks

$y = x^2 + 6/x - 1$.

积分导式得$x^2+6/x+C$，代$(1,6)$求$C=-1$。

M1	负幂积分
M1	使用切点
A1	曲线

Question 26

P1-C09-R26 · 6 marks

(a) · 2 marks

$a = -4$.

导数在2处24+2a=16，所以a=−4。

M1	正确梯度条件
A1	a

(b) · 4 marks

$C: y = 2x^3 - 2x^2 + 3$; tangent $y = 16x - 21$.

积分$6x^2-4x$后常数由y(0)=3得3；x=2时y=11，切线截距11−32=−21。

M1	积分及常数
A1	曲线
M1	求切点并过点
A1	切线

Question 27

P1-C09-R27 · 8 marks

(a) · 4 marks

$\frac{3}{7}x^{7/3}+\frac{3}{2}x^{4/3}+C.$

展开成x^{4/3}+2x^{1/3}，指数各加1并除以新指数。

M1	正确幂展开
M1	两项积分
A1	系数
B1	常数

(b) · 4 marks

$\frac{1}{3}x^{3}+x^{-1}+C.$

先化简x^2-x^{-2}，第二项积分符号变为正。

M1	正确分式化简
M1	负幂积分
A1	两项
B1	常数

Question 28

P1-C09-R28 · 6 marks

(a) · 3 marks

$y = x^4 - 2x^3$.

积分x^4-2x^3+C，代$(2,0)$得$C=0$。

M1	积分
M1	横截距条件
A1	曲线

(b) · 3 marks

Intercepts $(0,0), (2,0)$; only $(0,0)$ has horizontal tangent.

曲线$x^3(x-2)=0$；导数在0处0，在2处$32-24=8$，不为0。

M1	分解求全部截距
M1	检验导数
A1	正确分类

Question 29

P1-C09-R29 · 7 marks

(a) · 3 marks

$a = 2, b = 7/3.$

先由$f'(2) = 4a = 8$得$a = 2$。逐项积分得$F(x) = ax^3/3 + bx + c$。$F(0) = 1$给$c = 1$；$F(2) = 11$给$8a/3 + 2b = 10$，故$b = 7/3$。

M1	梯度条件
M1	由原函数两点形成方程
A1	常数

(b) For these values of a, b, find

(i) · 2 marks

$y = 8x - 17/3.$

曲线上点为$(2, 31/3)$，切线斜率已知为8，所以$y - 31/3 = 8(x - 2)$。

M1	切点坐标与点斜式
A1	方程

(ii) · 2 marks

$F(x) = \frac{2}{3}x^3 + \frac{7}{3}x + 1.$

将$a = 2, b = 7/3, c = 1$代回原函数表达式；可求导检查$F' = f$。

M1	积分幂与系数
A1	全表达式

Question 30

P1-C09-R30 · 8 marks

(a) · 3 marks

$\frac{6}{5}x^{5/2} + \frac{4}{3}x^{3/2} - 8x^{1/2} + c.$

先除以$x^{1/2}$得到$3x^{3/2} + 2x^{1/2} - 4x^{-1/2}$，再逐项使用幂法则。

M1	分解为幂
M1	正确积分幂与系数
A1	全式及常数

(b) A function F satisfies $F\prime(x) = f(x)$ and $F(1) = 0$.

(i) · 2 marks

$578/15.$

在1处原函数的不含常数部分为$6/5 + 4/3 - 8 = -82/15$，所以$c = 82/15$。代入4得$192/5 + 32/3 - 16 + 82/15 = 578/15$。

M1	利用初值确定常数
A1	精确值

(ii) · 3 marks

$1366/15.$

由导数关系可写$G(x) = 2F(x) + 3x + c$。代入$x = 1$得$c = 2$；所以$G(4) = 2(578/15) + 12 + 2 = 1366/15$。

M1	用已知原函数组成G
M1	初值确定常数
A1	$G(4)$